Writing the Methodology chapter

Chapter 3 tells your examiner exactly how you answered each research question, and why that was the right way to answer it. This guide walks through the four subsections every economics Methodology chapter needs, then the knowledge you need to write each one well.

3 Methodology

3.1 Data B1
Sampling design and questionnaire, or your secondary sources

3.2 Research questions and methods B2
The RQ–method table
Every research question needs its own method. Put them side by side in one table.

3.3 Variables B3
Operationalisation, measurement and the codebook

3.4 Model specification B4
Your econometric model, and why it fits the question

3.5 Ethics, limitations and summary B5
Putting the chapter together

Before you write: what each part of this guide does

The contents above are the chapter you will hand in. Everything else in this guide is the knowledge behind it. Work through the parts in order the first time; afterwards, use search (Ctrl K) to jump to what you need while drafting.

A Foundations
What "methodology" means and how it differs from research design and methods; what you can and cannot claim from your design.
A1 Methodology, design and methods A2 Correlation and causation

B Writing Chapter 3
The four subsections, each with a model write-up, a template to fill in and a worksheet.
B1 Data B2 RQ–method table B3 Codebook B4 Model specification B5 Putting it together

C Data

Levels of measurement, the four data structures and how each changes the way you write the model, collecting primary data, finding secondary data, and cleaning.

C1 Levels of measurement C2 Data structures C3 Primary data C4 Secondary data C5 Data cleaning

D Sampling

Populations and sampling frames, probability and non-probability designs with worked examples, and the question every survey student asks: how many respondents?

D1 Concepts D2 Probability sampling D3 Non-probability sampling D4 How many respondents?

E The instrument

Designing the questionnaire, adopting and adapting established scales, Likert versus rating scales, and establishing validity and reliability.

E1 Questionnaire design E2 Adopt and adapt E3 Likert and rating scales E4 Validity and reliability

F Practice and resources

All worksheets in one place, a glossary, a verified reading list, and the offline download.

F1 Worksheets F2 Glossary F3 Reading list F4 Downloads

Ten things examiners look for in Chapter 3

1. **A clear line from each research question to a method.** The examiner should be able to point at any RQ and find the data, variables and model that answer it.
2. **Choices that are justified, not just described.** "I used stratified sampling" is a description. "I stratified by state because consumption differs sharply between East and West Malaysia" is a justification.
3. **A sampling design you can defend**, including the population, sampling frame, selection procedure, sample size and response rate.
4. **A sample size that matches what you want to infer**: precision for a proportion, or power for a hypothesis test. A population size alone is not a justification.
5. **Every variable operationalised**: the question that measures it, its level of measurement and its coding.
6. **An instrument with a known pedigree**: which scales were adopted, which adapted, and the evidence of validity and reliability in your sample.
7. **A model written for your data structure**, with the right subscripts (i, t, it) and every symbol defined.
8. **An estimator chosen for the outcome variable** (binary, ordinal, count, continuous) and for the identification problem.
9. **Transparent data handling**: what was cleaned, recoded or dropped, and why.
10. **Honest limits**: what your design can and cannot support, stated before the results.

How to use this guide

Parts of a section — Most sections are split into *Notes*, *Worked example*, *Write-up*, *Worksheet* and *Quiz*. The tabs under each title switch between them.

Model answers	Each worksheet question has a “Show model answer” button. Write your own answer first; the answer box keeps it in this browser.
Examiner's notes	Red-pen notes in the margin point out what examiners query most often. Hide them with the pen button in the top bar.
Projector mode	For teaching: the text grows to 26px and the contents fold away. The A− and A+ buttons adjust text size at any time.
Code	Examples come in Stata and Python; the switch on any code block changes all of them. The same code is in the downloadable do-files and notebooks, which run without errors.
Offline copy	Download the whole guide as one ZIP from F4 Downloads and open it without an internet connection.

A1 Methodology, research design and methods

Three words that students use interchangeably and examiners do not. Methodology contains research design, and research design contains methods. Knowing which is which tells you what each part of Chapter 3 has to do.

About 20 minutes | Worksheet with model answers | Five-question quiz

After this section you can:

1. explain which of methodology, research design and methods is a subset of which;
2. place any element of your own study at the right level;
3. write the short philosophy-and-design paragraph that opens an economics Chapter 3.

The short answer

Methodology ⊃ research design ⊃ methods. Methodology is the outermost layer: the logic that says why a certain way of producing knowledge is valid for your problem. Inside it sits the research design, the blueprint for the whole study. Inside the design sit the methods, the specific techniques and tools you use to collect and analyse data.

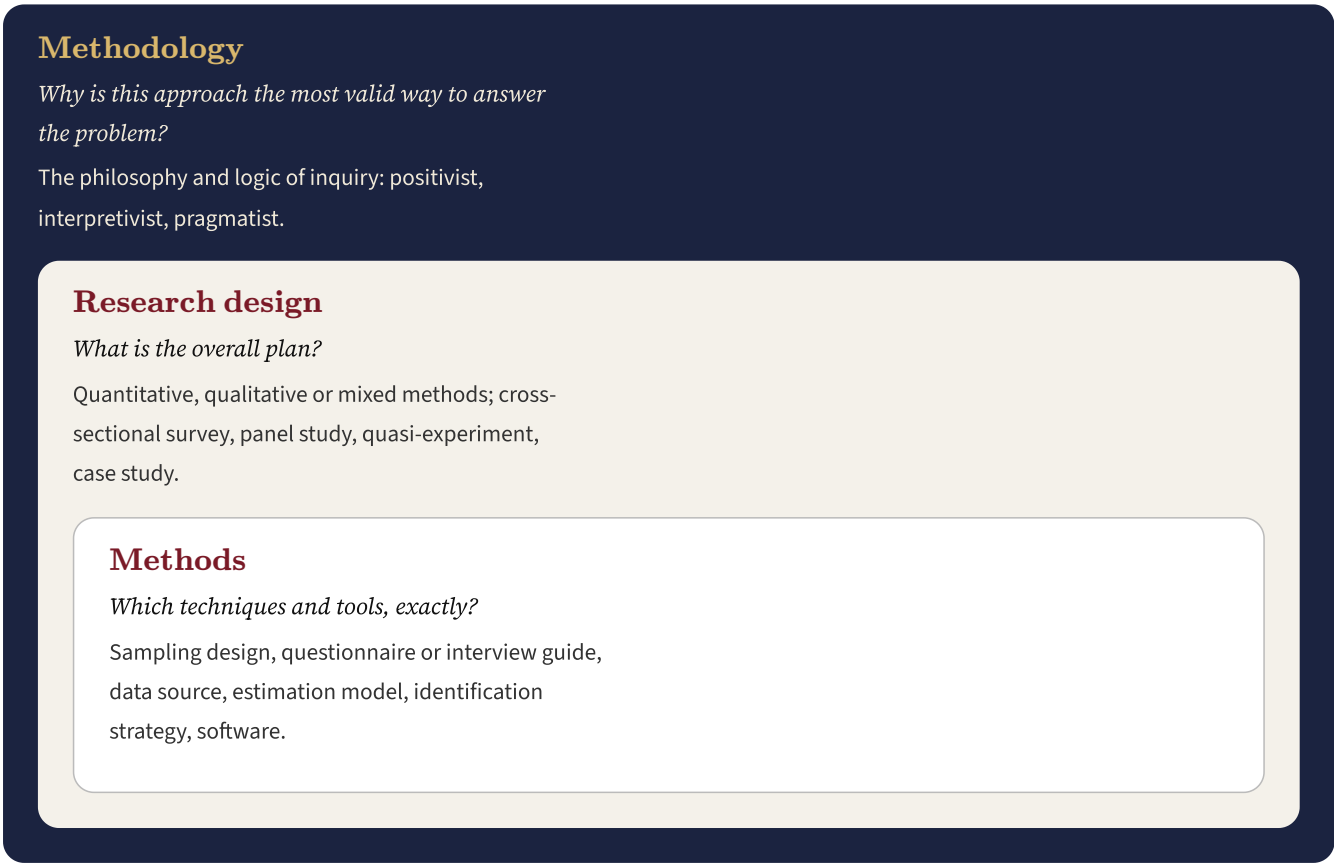


Figure A1.1. *Each layer constrains the one inside it. A positivist methodology leads to a quantitative design, which leads to methods such as random sampling and regression.*

The three levels in detail

Table A1.1. Methodology, research design and methods compared

Level	Question it answers	Examples in an economics thesis
Methodology	Why is this way of knowing valid for my problem?	Positivism: relationships between measurable variables can be observed and tested. Phenomenology: understanding lived experience through qualitative data.
Research design	What is the overall plan, the blueprint?	Quantitative cross-sectional survey; panel study of firms; difference-in-differences around a policy change; explanatory sequential mixed methods.
Methods	Which specific techniques and tools?	Stratified random sampling; a 30-item questionnaire; DOSM household survey data; ordered logit; fixed-effects panel regression; Stata 18.

The first lecture's examples of *methods* are a useful reminder of how broad this layer is. Methods include:

- the **estimation model** (OLS, logit, partial proportional odds, DiD, fuzzy RD);
- the **sampling design** (simple random, stratified, purposive, snowball);
- the **data collection technique**: primary or secondary data; questionnaire survey, interview, focus group discussion (FGD), ethnography, case study; web scraping;
- the **identification strategy** (instrumental variables, natural experiment, regression discontinuity);
- the **literature review method** (traditional narrative, bibliometric, systematic review, meta-analysis);
- even the **writing method** (a monograph or "big book" thesis versus a three-paper thesis) and the **referencing format** (APA, Harvard, Chicago, MLA).

✓ **Examiner:** *Calling Chapter 3 "Research Methodology" and then describing only methods is common. That is fine, but open with two sentences that place your methods inside a design and a methodology.*

Paradigms in one table

A paradigm is a shared set of assumptions about what exists (ontology) and how we can know it (epistemology). Your methodology is the paradigm put to work on your research problem.

Table A1.2. Three paradigms you will meet in economics and the social sciences

Paradigm	Assumes	Typical design	Typical methods	Economics example
Positivism (post-positivism)	An objective reality exists and can be measured, with error.	Quantitative: survey, panel, experiment, quasi-experiment	Probability sampling, structured questionnaires, secondary data, regression	Does financial literacy raise household saving?
Interpretivism (including phenomenology)	Reality is understood through the meanings people give it.	Qualitative: case study, ethnography, phenomenological study	Purposive sampling, in-depth interviews, FGDs, thematic analysis	How do flood victims decide whether to evacuate?
Pragmatism	Use whatever works to answer the question.	Mixed methods	A survey followed by interviews (or the reverse)	Survey on e-wallet use, then interviews explaining non-use among B40 households

Most empirical economics is positivist or post-positivist, usually without saying so. You do not need an essay on philosophy. One or two sentences that name the paradigm and connect it to your design are enough. Carter and Little (2007) show how epistemology, methodology and method should be visibly consistent with one another. Morgan (2007) makes the case for pragmatism in applied work.

? **Examiner:** *If you name a paradigm, use it. A paragraph on interpretivism followed by a regression model will be queried.*

What the Methodology chapter is not

Not a literature review — Prior studies appear here only to justify methodological choices. “Mincer (1974) uses log wages; I follow this convention” belongs in Chapter 3. “Many studies have found positive returns to schooling” belongs in Chapter 2.

Not a results chapter — No estimates, no significance tests, no findings. Those are Chapter 4. Pilot-test reliability statistics are the usual exception.

Not a tutorial — Do not teach your examiner what OLS is. State the model, justify it, and cite a textbook for standard material.

✓ **Examiner:** *In a three-paper thesis each paper carries its own data and methods section. Chapter 3 then becomes a short overview of the common data and design, or disappears entirely. Check your university's rules.*

Key takeaways

1. Methodology contains research design, which contains methods.
2. Methodology answers *why*, design answers *what plan*, methods answer *which tools*.

3. In an economics thesis, a short, consistent statement of paradigm and design is better than a long philosophical one.

References cited

Carter, S. M., & Little, M. (2007). Justifying knowledge, justifying method, taking action: Epistemologies, methodologies, and methods in qualitative research. *Qualitative Health Research, 17*(10), 1316–1328.

Crotty, M. (1998). *The foundations of social research: Meaning and perspective in the research process*. Sage.

Mincer, J. (1974). *Schooling, experience, and earnings*. National Bureau of Economic Research.

Morgan, D. L. (2007). Paradigms lost and pragmatism regained. *Journal of Mixed Methods Research, 1*(1), 48–76.

Example

Worked example: placing a study's elements at the right level

A PhD student plans a study titled *"Digital financial literacy and saving behaviour among Malaysian households."* Here is every element of the plan, sorted into the three levels.

Table A1.3. One study, three levels

Element of the plan	Level	Why
Saving behaviour is a measurable outcome that responds to measurable causes	Methodology	An assumption about what can be known and how: positivism.
A quantitative, cross-sectional household survey	Research design	The overall plan: type of data, timing, one wave.
Stratified random sample of 400 households by state and urban/rural	Method	A specific sampling technique.
Questionnaire adapting the OECD/INFE financial literacy items	Method	A specific data collection instrument.
OLS for saving rate; logit for whether the household saves regularly	Method	Specific estimation models.
Stata 18; cleaning in a do-file	Method	Tools.

Worked example: a secondary-data macro study

"Has the inflation–unemployment trade-off in Malaysia changed since 1990?"

- **Methodology:** positivist. A stable statistical relationship (a Phillips curve) can be estimated from observed data.
- **Research design:** quantitative time-series design with a structural-break analysis.
- **Methods:** annual data from DOSM and the World Bank WDI, 1990–2023; ARDL model; Bai–Perron break tests; EViews or Stata.

Notice that the design changes as soon as the question changes. Ask instead *how small firms set prices during high inflation* and the design becomes qualitative: interviews with owners, purposive sampling, thematic analysis.

Writeup

How it reads in the thesis

The opening of Chapter 3 needs only a short paragraph that places the study in a paradigm and a design, then signposts the subsections. Annotations in red explain what each sentence is doing.

3.0 Introduction

This chapter sets out how the study answers its three research questions. The study adopts a post-positivist position: household saving is treated as an observable outcome whose determinants can be measured and tested, while acknowledging that all measurement contains error.[1] Consistent with this position, it uses a quantitative, cross-sectional survey design.[2] Section 3.1 describes the sampling design and the questionnaire; Section 3.2 maps each research question to its method; Section 3.3 defines and codes the variables; and Section 3.4 specifies the econometric models. Section 3.5 discusses ethics and the limitations of the design.[3]

1 *The paradigm in one sentence, with what it means for this study.*

2 *The design follows from the paradigm, and says so.*

3 *A signpost for every subsection, so the examiner knows the map before reading.*

Model write-up. *Adapt the wording to your study; do not copy it.*

Template

This chapter sets out how the study answers its number research questions. The study adopts a paradigm position: what this means for your phenomenon. Consistent with this, it uses a quantitative / qualitative / mixed-methods, cross-sectional / longitudinal / experimental design. Section 3.1 … Section 3.4 …

Worksheet

Worksheet A1

Answer in your own words first, then compare with the model answer.

Q1 Arrange *methods*, *methodology* and *research design* from the broadest to the narrowest, and explain in two sentences why the order matters when you write Chapter 3. (4 marks)

Model answer

Methodology, then research design, then methods. The order matters because each level must be justified by the one above it: the methods must suit the design, and the design must follow from the methodology. An examiner reading Chapter 3 looks for that chain of justification.

Q2 Classify each item as methodology (M), research design (D) or method (T): (a) semi-structured interviews; (b) pragmatism; (c) explanatory sequential mixed methods; (d) the Krejcie–Morgan sample size; (e) a fixed-effects panel regression; (f) a quasi-experimental difference-in-differences design. (6 marks)

Model answer

(a) T, a data collection technique; (b) M, a paradigm; (c) D, an overall plan combining two strands; (d) T, part of the sampling method; (e) T, an estimation model; (f) D, the overall design, which is implemented through a method (the DiD regression). Accept (f) as T if the student justifies it as the estimator rather than the design.

Q3 A student's draft Chapter 3 opens with three pages on positivism, interpretivism and critical realism, then moves straight to the sampling design. What would you advise? (4 marks)

Model answer

Cut the survey of paradigms to one or two sentences that name the position the study takes and show what it implies for this study's design. Three pages of general philosophy read like a textbook. Chapter 3 is not a tutorial, and paradigms not adopted are irrelevant. Then add a sentence linking the paradigm to the design before the sampling section begins.

Q4 Which of these sentences belong in Chapter 3 and which in Chapter 2? (a) “Following Mincer (1974), earnings enter the model in logarithms.” (b) “Returns to schooling in Southeast Asia range from 6% to 12% (Author, Year).” (c) “The partial proportional odds model is used because the parallel-lines assumption is violated for two covariates (Williams, 2006).” (3 marks)

Model answer

(a) Chapter 3: it justifies a methodological choice. (b) Chapter 2: it is a substantive finding from the literature. (c) Chapter 3: it justifies the estimator with a methodological reference.

A2 Correlation, causation and what your design can claim

Every Methodology chapter makes a promise about the kind of answer the study will give. Descriptive, correlational and causal claims need different designs. The threats to a causal reading decide which econometric method you need.

About 20 minutes | Worksheet with model answers | Quiz

After this section you can:

1. say whether a research question is descriptive, correlational or causal;
2. name the five classic threats to a causal interpretation and the methods that address each;
3. write claims in Chapter 3 that match what your design can support.

Three kinds of empirical question

Table A2.1. Descriptive, correlational and causal research

Type	Research question	Methods	How the finding is worded
Descriptive	What is the pattern?	Means, proportions, trends, cross-tabulations	"The average wage is higher among university graduates than non-graduates."
Correlational (association)	Are X and Y related, conditional on controls?	Correlation; regression with observed controls	"Education is positively associated with wages, conditional on age, gender and location."
Causal	Does changing X cause Y to change?	RCT, natural experiment, credible quasi-experimental design with stated identifying assumptions	"The programme increased wages by X%, under the identifying assumptions discussed."

None of these is better than the others. A well-executed descriptive study is worth more than a weak causal one. What matters is that the claim in Chapter 1, the design in Chapter 3 and the wording in Chapter 4 all agree.

? **Examiner:** *"Impact", "effect" and "determinant" are causal words. With a single cross-sectional survey and OLS, write "associated with".*

Why a correlation may not be causal

Table A2.2. Threats to a causal interpretation and the econometric responses

Threat	Meaning	Example	Econometric solution
Omitted variable bias	An unobserved factor affects both X and Y.	Wages = f(education). Ability affects both education and wages.	Fixed effects, instrumental variables, proxy variables
Reverse causality	X affects Y, but Y also affects X.	Income = f(training). Higher income may allow more training; more training raises income.	Lagged X, instrumental variables, difference-in-differences
Selection bias	Individuals choose treatment based on characteristics related to the outcome.	Salary = f(Harvard). Brilliant students choose Harvard and later earn more anyway.	Randomised controlled trials, quasi-experimental methods
Measurement error	X or Y is measured with noise.	Self-reported income is noisy.	Instrumental variables, structural equation modelling
Simultaneity	X and Y are jointly determined.	Price and quantity are set together by supply and demand.	Two- and three-stage least squares (2SLS, 3SLS)

✓ **Examiner:** *Name the threat that worries you most and say what your design does about it. Silence reads as unawareness.*

Omitted variable bias in one line

Suppose the true model is $wage_i = \beta_0 + \beta_1 educ_i + \beta_2 ability_i + u_i$, but ability is not in your data. OLS on education alone gives

$$\operatorname{plim} \hat{\beta}_1 = \beta_1 + \beta_2 \frac{\operatorname{Cov}(educ, ability)}{\operatorname{Var}(educ)}$$

Ability raises wages ($\beta_2 > 0$) and is positively correlated with education, so $\hat{\beta}_1$ overstates the return to schooling. The sign of the bias is the product of the two signs. Being able to state it is exactly the kind of reasoning an examiner wants to see in Chapter 3.

From threat to design

The identification strategy is the part of your methods that turns a correlation into a credible causal estimate. Each standard design rests on one assumption that you must state, and test where possible.

Table A2.3. Identification strategies and their key assumptions

Design	Key assumption	How you support it in the thesis
Randomised controlled trial	Random assignment balances observed and unobserved characteristics.	Balance table of pre-treatment characteristics.
Difference-in-differences	Parallel trends: without treatment, both groups would have moved together.	Pre-treatment trends graph; event-study coefficients.
Fixed effects (panel)	The omitted factor does not change over time.	Argue why the confounder is time-invariant; Hausman test against random effects.
Instrumental variables	The instrument shifts X (relevance) and affects Y only through X (exclusion).	First-stage F statistic; argument for exclusion.
Regression discontinuity	Units just either side of the cut-off are comparable.	Density test at the cut-off; covariate balance; bandwidth robustness.

Angrist and Pischke (2010) explain why design-based identification now dominates applied economics; Imbens and Wooldridge (2009) survey the estimators and their assumptions. Both are on the reading list.

Key takeaways

1. Decide in Chapter 3 whether your study is descriptive, correlational or causal, and word every claim accordingly.
2. Name the main threat to causal interpretation and the method that addresses it.
3. Every identification strategy rests on an assumption; state it and say how you will check it.

References cited

Angrist, J. D., & Pischke, J.-S. (2010). The credibility revolution in empirical economics: How better research design is taking the con out of econometrics. *Journal of Economic Perspectives, 24*(2), 3–30.

Imbens, G. W., & Wooldridge, J. M. (2009). Recent developments in the econometrics of program evaluation. *Journal of Economic Literature, 47*(1), 5–86.

Example

Worked example: is perceived usefulness a cause of e-wallet use?

The worked survey in this guide (400 students, hypothetical) finds that a one-point higher perceived-usefulness score is associated with a 12-percentage-point higher probability of using an e-wallet (logit average marginal effect 0.119, see B4). Can Chapter 4 say that usefulness *causes* use?

Reverse causality	Students who already use e-wallets learn how useful they are. Use may raise perceived usefulness rather than the other way round.

Omitted variables	General digital confidence may raise both perceived usefulness and use. It is not measured.
Measurement error	Perceived usefulness is a four-item scale with reliability of about 0.86. Some attenuation is likely (E4).

Verdict: the design supports an association, not a causal effect. Chapter 3 should say so before any results appear. A causal question would need a different design, for example randomly assigning a short demonstration of e-wallet features and comparing later use.

Worked example: an incinerator and house prices

Kiel and McClain (1995) studied house prices near a planned incinerator. Comparing prices near and far from the site in 1981 alone mixes the incinerator's effect with the fact that the area was cheaper all along. The difference-in-differences design compares the *change* in prices near the site with the change far from it, between 1978 (before the rumour) and 1981 (during construction).

In the simulated version used in C2, the estimated effect is −11,729 (robust SE 3,112), close to the −11,500 built into the simulation. The estimate is causal only if, without the incinerator, prices near and far would have grown at the same rate. That is the parallel-trends assumption, and Chapter 3 must state it.

Writeup

How it reads in the thesis

3.4.3 Interpretation and identification

The estimates from Equations (3.1)–(3.3) are interpreted as conditional associations rather than causal effects.[1] Two features of the design prevent a causal reading. First, perceived usefulness and e-wallet use are measured at the same time, so use may shape perceptions as much as perceptions shape use. Second, unobserved digital confidence may influence both.[2] Accordingly, results in Chapter 4 are described as associations, and the policy implications in Chapter 5 are framed as hypotheses for future experimental work.[3]

1 *The claim is stated plainly and early.*

2 *The specific threats are named, not a generic "correlation is not causation".*

3 *It commits later chapters to consistent wording.*

Model write-up. *Adapt to your study.*

Worksheet

Worksheet A2

Q1 Classify each research question as descriptive, correlational or causal: (a) What share of Malaysian SMEs accept e-wallet payments? (b) Is firm size associated with e-wallet acceptance, controlling for sector? (c) Did the 2020 e-Tunai Rakyat e-wallet credit increase e-wallet adoption among recipients? (3 marks)

Model answer

(a) Descriptive: a proportion. (b) Correlational: an association conditional on a control. (c) Causal: it asks about the effect of a policy. It needs a design such as a difference-in-differences or an eligibility-based discontinuity.

Q2 A student regresses household saving on whether the household attended a financial-literacy workshop, using a cross-section. Name the most serious threat to a causal reading and the direction of the likely bias. (4 marks)

Model answer

Selection bias: households that choose to attend are probably more motivated and more financially organised, and these traits also raise saving. The bias is upward, overstating the workshop's effect. Possible fixes are random assignment of workshop invitations, or a DiD comparing saving before and after for attendees and similar non-attendees.

Q3 Rewrite this sentence so it matches a single-survey OLS design: “Financial literacy has a significant positive impact on saving, so the government should fund literacy programmes.” (3 marks)

Model answer

“Financial literacy is positively associated with saving after controlling for income, age and education. If the association reflects a causal effect, literacy programmes may raise saving, a possibility that future experimental or quasi-experimental research could test.”

Q4 For a panel of 13 states over 11 years, explain in two sentences why fixed effects address one kind of omitted variable but not another. (4 marks)

Model answer

State fixed effects remove every influence that is constant over time within a state, such as geography, culture or long-run institutions, even if unobserved. They do not remove omitted factors that change over time within a state, such as a state-specific policy introduced in 2018; those still bias the estimates unless they are measured and controlled for.

B1 3.1 Data: describing where your data come from

The first subsection of Chapter 3 tells the examiner where every number in your thesis comes from. For primary data that means the sampling design, the questionnaire and how its items become your variables. For secondary data it means the datasets, their providers, coverage and versions.

About 30 minutes | Two model write-ups | Template and worksheet

After this section you can:

1. list everything Section 3.1 must report for primary and for secondary data;
2. build the table that maps questionnaire items to variables;
3. write a complete data description an examiner can audit.

What Section 3.1 has to do

Section 3.1 has to let a reader reproduce your dataset in principle: who or what was observed, how they were chosen, how they were measured and what was obtained. Write it so that someone with your instructions could collect the same kind of data again.

If you collected primary data

Report these, in this order. Each item links to the section of this guide that explains it.

1. **Target population and unit of analysis**: who the findings are about (for example, undergraduates at public universities in northern Malaysia), and why that population (D1).
2. **Sampling frame**: the list you sampled from, how complete it is, and who it misses (D1).
3. **Sampling design**: probability or non-probability, the specific design, and why it suits the research questions (D2, D3).
4. **Sample size**, justified by precision or by power, and inflated for expected non-response (D4).
5. **The instrument**: its sections, number of items, response formats, which scales were adopted or adapted and from whom (E1, E2, E3).
6. **How questionnaire items map to your variables**, in a table (below).
7. **Pretesting and pilot**: expert review, cognitive interviews, the pilot sample and its reliability results (E4).
8. **Administration**: mode (face to face, online, telephone), language versions, fieldwork dates, who collected the data.
9. **Response**: number invited, returned and usable, giving the response rate with its denominator.
10. **Ethics**: approval reference, consent, anonymity and data protection (or put these in Section 3.5).

✓ **Examiner:** *"The response rate was 80%" means nothing without the numerator and the denominator. Write "400 usable questionnaires from 617 invitations (64.8%)".*

The item-to-variable map

This table connects Section 3.1 (the instrument) to Section 3.3 (the variables) and is the single most useful table for an examiner checking your measurement. One row per construct or variable.

Table B1.1. Example: questionnaire sections, items and the variables they measure

Section	Items	Variable (construct)	Response format	Source
A. Background	A1–A7	female, age, faculty, year of study, home state, urban hometown, monthly allowance	Categories; numeric (RM)	Researcher-developed
B. E-wallet use	B1–B3	ewallet_user (past 30 days); ewallet_days (past 7 days); spending (RM)	Yes/No; 0–7; numeric	Researcher-developed
C. Perceptions	C1–C4	Perceived usefulness (PU)	5-point agreement	Adapted from Davis (1989)
	C5–C8	Perceived security (PS)	5-point agreement	Researcher-adapted from published scales
	C9–C11	Ease of use (EOU)	5-point agreement	Adapted from Davis (1989)
D. Knowledge	D1–D3	Financial literacy score (0–3)	Multiple choice, scored correct/incorrect	Lusardi and Mitchell (2014)
E. Intentions	E1–E2	Intention to continue; satisfaction	5-point	Researcher-developed

If you use secondary data

1. **The dataset(s)**: full name, provider, edition or vintage, and date accessed. Data are revised, and results can change with the revision (Johnson et al., 2013).
2. **Coverage**: countries, states, households or firms; years; frequency (annual, quarterly, monthly).
3. **Unit of observation** and how many observations you use, after merges and exclusions.
4. **A source table**: one row per variable, giving definition, series code, source and any transformation.
5. **How the provider collected the data**, briefly: for a survey such as DOSM's Household Income and Expenditure Survey, the sampling design and sample size; for administrative data, how records are generated.
6. **Known limitations**: breaks in series, changes in definitions, missing years, top-coding.

✓ **Examiner:** *Give the series code (for example, World Bank `FP.CPI.TOTL.ZG`) so a reader can download exactly what you used.*

Key takeaways

1. Section 3.1 should let a reader reproduce your data in principle.
2. For primary data, report population, frame, design, size, instrument, item map, pilot, administration and response, with numbers.
3. For secondary data, report the dataset, provider, vintage, coverage, unit, series codes and limitations.

References cited

Davis, F. D. (1989). Perceived usefulness, perceived ease of use, and user acceptance of information technology. *MIS Quarterly, 13*(3), 319–340.

Johnson, S., Larson, W., Papageorgiou, C., & Subramanian, A. (2013). Is newer better? Penn World Table revisions and their impact on growth estimates. *Journal of Monetary Economics, 60*(2), 255–274.

Lusardi, A., & Mitchell, O. S. (2014). The economic importance of financial literacy: Theory and evidence. *Journal of Economic Literature, 52*(1), 5–44.

Example

Worked example 1: primary survey data

The running example in this guide is a hypothetical survey: *"Determinants of e-wallet use among undergraduates at a public university in northern Malaysia."* Here are the design decisions that Section 3.1 has to report, with the numbers.

Table B1.2. Design decisions for the e-wallet survey

Decision	What was done	Justification
Target population	Full-time undergraduates enrolled in Semester 1, 2025/26 (N ≈ 10,000)	The research questions concern young adults' first-hand payment choices.
Sampling frame	Registrar's enrolment list	Covers every enrolled student; excludes students on leave.
Design	Stratified random sampling by faculty, proportionate allocation	E-wallet use is expected to differ by faculty; stratifying guarantees every faculty is represented.
Target sample	370 completed questionnaires	Krejcie and Morgan (1970) for N = 10,000: ±5 points at 95% confidence. Power for the logit model checked separately (D4).
Invitations	617	370 ÷ 0.60 expected response rate.
Instrument	35 items in five sections; English and Malay versions	Scales adapted from Davis (1989); back-translated (E2).
Pilot	30 students outside the sample	Checked wording, timing (12 minutes) and reliability.
Fieldwork	Online questionnaire, 3–28 November 2025, two reminders	
Response	414 returned; 6 duplicates and 8 incomplete removed; 400 usable	Usable response rate 400 / 617 = 64.8% (cleaning in C5).

Worked example 2: secondary macroeconomic data

"Has the inflation–unemployment trade-off in Malaysia changed since 1990?" The source table below is the heart of a secondary-data Section 3.1.

Table B1.3. Variables, definitions and sources (annual, 1990–2023)

Variable	Definition	Source and series code	Transformation
inflation	Inflation, consumer prices (annual %)	World Bank, World Development Indicators, `FP.CPI.TOTL.ZG`	None
unemp	Unemployment, % of labour force (modelled ILO estimate)	World Bank WDI, `SL.UEM.TOTL.ZS`; cross-checked with DOSM Labour Force Survey	None
growth	GDP growth (annual %)	World Bank WDI, `NY.GDP.MKTP.KD.ZG`	None
exports	Exports of goods and services (% of GDP)	World Bank WDI, `NE.EXP.GNFS.ZS`	First difference

Series codes are the World Bank's own identifiers; a reader can paste them into the WDI database search to retrieve the identical series. Record the date you downloaded them.

Writeup

Model write-up: primary data

3.1 Data

3.1.1 Population and sampling design

The target population comprises the approximately 10,000 full-time undergraduates enrolled at the university in Semester 1 of 2025/26; the sampling frame is the registrar's enrolment list for that semester.[1] Because e-wallet use is expected to vary across faculties, a stratified random sample was drawn with faculties as strata and proportionate allocation.[2] The target of 370 completed questionnaires follows Krejcie and Morgan (1970) for a population of 10,000 at 95% confidence and a ±5-percentage-point margin of error; an a priori power analysis confirmed that 370 observations give 80% power to detect an odds ratio of 1.5 for the main predictor in the logit model (Section 3.4).[3] Anticipating a 60% response rate, 617 students were invited.

3.1.2 Questionnaire

The self-administered questionnaire contains 35 items in five sections (Table 3.1). Perceived usefulness and ease of use were adapted from Davis (1989); financial literacy was measured with the three questions of Lusardi and Mitchell (2014).[4] The English instrument was translated into Malay and back-translated by a second bilingual reader. A pilot with 30 students outside the sample took 12 minutes on average; Cronbach's alpha exceeded 0.75 for every scale.

3.1.3 Data collection and response

The questionnaire was administered online from 3 to 28 November 2025, with two reminders. Of 414 returned questionnaires, six were duplicate submissions and eight were abandoned before half the scale items, leaving 400 usable responses, a usable response rate of 64.8% (400/617).[5]

1 *Population and frame are separate, and both are named.*

2 *The design is justified by a feature of the problem.*

3 *Sample size has a precision justification and a power check.*

4 *Every scale has a source. The full item map goes in a table.*

5 *Response is reported with numerator, denominator and reasons for exclusion.*

Model write-up. *Hypothetical study; adapt the structure, not the words.*

Model write-up: secondary data

3.1 Data

The study uses annual data for Malaysia from 1990 to 2023 (34 observations). Inflation, unemployment, GDP growth and exports are taken from the World Bank's World Development Indicators (accessed 14 January 2026); Table 3.1 gives each series' definition and code.[1] The WDI unemployment rate is a modelled ILO estimate; for the years in which DOSM's Labour Force Survey reports a national rate, the two series differ by less than 0.3 percentage points, and results using the DOSM series are reported as a robustness check.[2]

1 *Provider, access date and a pointer to the source table.*

2 *A known limitation of one series, checked against an alternative source.*

Model write-up. *The comparison with DOSM is illustrative; report the difference you actually find.*

Template: primary data

The target population comprises `who, where, when, approximately how many`. The sampling frame is `list`, which `covers / excludes …`. A `design` sample was drawn because `feature of the problem`. The target sample of `n` follows `precision or power justification`; anticipating a `%` response rate, `number` were invited. The questionnaire contains `k` items in `m` sections (Table 3.1); `scales` were adapted from `sources`. A pilot with `n` respondents … Data were collected `mode, dates`. Of `returned`, `excluded and why`, leaving `usable` (response rate `usable / invited`).

Worksheet

Worksheet B1

Q1 A draft says: “Data were collected from 300 respondents using a questionnaire. The response rate was good.” List five pieces of information missing from it. (5 marks)

Model answer

Any five of: the target population; the sampling frame; the sampling design and its justification; how the sample size was determined; the number invited (so the response rate can be computed); the questionnaire's structure and the sources of its scales; the pilot; the mode and dates of administration; exclusions from returned to usable; ethics approval.

Q2 Build the item-to-variable map (Section, Items, Variable, Format, Source) for a study of household saving with three constructs: saving behaviour (3 items), financial literacy (5 questions), and present bias (4 agreement items adapted from a published scale). (6 marks)

Model answer

Section	Items	Variable	Format	Source
B	B1–B3	Saving behaviour (saves monthly; amount; saving rate)	Yes/No; RM; %	Researcher-developed
C	C1–C5	Financial literacy score, 0–5	Multiple choice, scored correct/incorrect	Name the published questions used
D	D1–D4	Present bias (mean of four items)	5-point agreement	Adapted from the named published scale

Marks for: one row per construct, item numbers, measurement format, and a source for every adopted or adapted scale.

Q3 You use DOSM's Household Income and Expenditure Survey microdata. What should Section 3.1 say about how DOSM collected the data, and why? (4 marks)

Model answer

Briefly: the survey year, its sampling design (a stratified multistage household sample) and achieved sample size, the mode of interview, and the reference period for income and expenditure. Cite the official methodology report. This tells the examiner how representative your data are and whether you need to use the survey weights and design variables in estimation.

B2 3.2 The RQ–method table

One table that pairs every research question with the method that answers it. It is the fastest way to show an examiner that your design is complete, and the fastest way for you to find a research question that has no method.

About 20 minutes | Four real examples | Template and worksheet

After this section you can:

1. build an RQ–method table with the right columns for your study;
2. match each question to a method suited to its outcome and data structure;
3. spot a question the design cannot answer before the examiner does.

Why the table matters

Chapter 1 promises research questions; Chapter 4 delivers answers. The RQ–method table in Chapter 3 is the bridge. An examiner reads it first and checks that every question has data, variables and a model, and that each method suits the question it is paired with.

Which columns to include

Table B2.1. Columns for an RQ–method table

Column	When to include it
Research question (and sub-questions)	Always. Use exactly the numbering and wording of Chapter 1.
Method or model	Always. Name the estimator (“ordered logit”), not just “regression”.
Outcome variable(s)	Recommended. The outcome's measurement level decides the model.
Main explanatory variable(s)	Recommended.
Data source and unit	When different questions use different data or units.
Hypothesis	When the thesis is hypothesis-testing.
Data collection method; theory	Optional. Useful in mixed-methods designs.
Section where answered	Optional. A pointer to the Chapter 4 section helps long theses.

Rules for a good table

1. **One row per question or sub-question.** If a row needs two unrelated methods, the question probably needs splitting.
2. **The method must fit the outcome.** “What are their intentions?” needs summary statistics. “How intense are their intentions?” on an ordered scale needs an ordinal model. “Did the policy change outcomes?” needs a design such as DiD (see B4).
3. **The method must fit the data structure.** A panel question needs a panel estimator; a single cross-section cannot answer a question about change over time (C2).
4. **Descriptive questions get descriptive methods.** Not every row needs a regression.

5. **Wording must match Chapter 1 exactly.** Examiners notice when RQ2 has quietly changed.

? **Examiner:** *If a method appears in 3.4 but in no row of this table, I will ask which question it answers.*

Four examples

Example 1: non-return intentions of international students

Table B2.2. RQ, sub-RQ and analysis method

RQ	Sub-RQ	Analysis method
RQ1: What is the extent of non-return intentions of international students studying in New Zealand?	RQ1a: What are their non-return intentions?	Summary statistics
	RQ1b: How intense or definite are their non-return intentions?	Partial proportional odds model
	RQ1c: What are the key determinants of their non-return intentions?	Partial proportional odds model
RQ2: To what extent do intended return timeframes differ?	RQ2a: What are the students' intended return timeframes?	Summary statistics
	RQ2b: How do the characteristics differ between students who intend to return immediately, delay their return, and not return at all?	Multinomial logit model

The table may also include columns for data collection method, hypothesis, data source or theory.

Example 2: the 2015 refugee crisis and immigrants' life satisfaction

Table B2.3. RQ and method

Research question	Method
(1) Does the 2015 refugee crisis have any negative impact on resident immigrants' life satisfaction?	Difference-in-differences model
(2) To what extent can resident immigrants' education levels insulate against the impact of the crisis on their life satisfaction?	DiD model by subsample
(3) Does the type of host country play a role in whether resident immigrants' life satisfaction is affected by the crisis?	DiD model by subsample

Example 3: naturalisation and labour-market outcomes

Table B2.4. RQ and method, with the stage of a fuzzy RD design for each question

Research question	Method
(1) Are naturalised immigrants really better off than non-naturalised immigrants in income and employment chances? If yes, by how much?	Summary statistics; fuzzy RD model: second-stage equation
(2) Does the naturalisation eligibility rule itself have any significant impact on the likelihood of naturalisation and on labour-market outcomes?	Fuzzy RD model: first-stage equation; fuzzy RD model: reduced-form equation

Example 4: digital financial literacy and saving (five columns)

Table B2.5. A fuller RQ–method table

Research question	Outcome variable(s)	Main explanatory variable(s)	Data source and unit	Method or model
RQ1: What is the relationship between digital financial literacy and saving behaviour?	Monthly saving amount; saving rate; binary indicator for saving regularly	Digital financial literacy score	Household survey; individual respondent	Descriptive statistics; OLS; logit for binary saving
RQ2: Does exposure to a financial education programme improve saving behaviour?	Pre/post saving rate	Treatment exposure; post period; treatment × post	Programme and comparison groups; repeated observations	Difference-in-differences
RQ3: Through what mechanisms does the programme influence behaviour?	Perceived control; budgeting confidence; reported habits	Programme exposure; qualitative themes	Survey plus interviews	Mediation-style descriptive analysis; thematic analysis

Notice how Example 4's columns make the design auditable: RQ2 needs repeated observations of treated and comparison groups, and the table shows they exist.

Key takeaways

1. Every research question and sub-question gets a row and a named method.
2. The outcome's measurement level and the data structure decide the method.
3. Keep RQ wording identical to Chapter 1, and make sure every model in 3.4 appears in the table.

Example

Worked example: building the table for the e-wallet study

Start from the research questions, identify each outcome and its measurement level, then choose the model. The models below are estimated in `06_model_specification.do` and its notebook (B4).

Table B2.6. RQ–method table for the e-wallet survey (hypothetical)

Research question	Outcome (level)	Main explanatory variables	Data	Method
RQ1: What share of undergraduates use e-wallets, and how often?	Use in past 30 days (binary); days used in past week (0–7)	—	Survey of 400 undergraduates; individual	Proportions; means; cross-tabulation by faculty
RQ2: Which perceptions and characteristics are associated with being an e-wallet user?	ewallet_user (binary)	PU, PS, EOU, financial literacy		Logit with average marginal effects; LPM as robustness check
RQ3: Among users, what is associated with more frequent use?	ewallet_days (0–7, ratio)	PU, PS, EOU, allowance		OLS with robust standard errors, users only
RQ4: What is associated with intending to keep using e-wallets?	intention (1–5, ordinal)	PU, PS, EOU		Ordered logit; partial proportional odds if parallel lines fail

✓ **Examiner:** *RQ3 analyses users only. Say so in the table, and acknowledge in 3.4 that this conditions on a choice.*

Writeup

How it reads in the thesis

3.2 Research questions and methods

Table 3.2 links each research question to its outcome variables, data and method of analysis.[1] RQ1 is descriptive and is answered with proportions and means. RQ2 to RQ4 concern associations between perceptions and behaviour, and are answered with regression models chosen for the measurement level of each outcome: a logit model for the binary indicator of use, OLS for the number of days of use, and an ordered logit for the five-point intention scale.[2] The models are specified in Section 3.4.[3]

1 *Introduce the table; do not repeat it row by row.*

2 *Summarise the logic: descriptive versus inferential questions, and why each model fits.*

3 *Point forward to the specification.*

Model write-up. *The table itself carries the detail.*

Template

Template. Copy into your thesis and fill in (also in the ZIP as a Word file)

Research question	Outcome (measurement level)	Main explanatory variable(s)	Data source and unit	Method or model
RQ1 as in Chapter 1	variable (level)	variables	dataset; unit	estimator
RQ2				
RQ3				

Worksheet

Worksheet B2

Q1 For each question, name a suitable method and the measurement level of the outcome: (a) What proportion of B40 households have an emergency fund? (b) What factors are associated with the number of children a household has? (c) Which characteristics distinguish households that save in ASB, in a bank account, or not at all? (d) Did the 2022 increase in the minimum wage change working hours? (8 marks)

Model answer

(a) Binary outcome; proportion with a confidence interval, using survey weights. (b) Count outcome (ratio level, non-negative integers); Poisson or negative binomial regression. (c) Nominal outcome with three unordered categories; multinomial logit. (d) Ratio-level outcome, causal policy question; difference-in-differences comparing affected and less-affected workers before and after 2022, or a panel with individual fixed effects.

Q2 What is wrong with this row? "RQ2: How has household debt changed between 2016 and 2022? | Method: OLS regression | Data: 2022 household survey." (4 marks)

Model answer

The question is about change over time, but the data are a single cross-section from 2022, so they cannot show change. The design needs at least two time points: pooled cross-sections from both survey years (comparing distributions or estimating with a year dummy) or a panel. "OLS regression" is also too vague; the table should name the outcome and the specification.

Q3 Draft a three-row RQ–method table for your own thesis (or for the topic "remittances and household education spending in Malaysia"). (6 marks)

Model answer

A possible answer: RQ1 “What share of households receive remittances, and how much?” uses a household survey with proportions and means. RQ2 “Are remittances associated with higher education spending?” takes the education budget share as the outcome (ratio level), remittance receipt as the main explanatory variable, and OLS with controls; a fractional response model if many zeros. RQ3 “Does the association differ for daughters and sons?” adds an interaction of remittances and the share of daughters. Marks for wording, levels and named estimators.

B3 3.3 Variables: operationalisation and the codebook

A concept in your theory becomes a variable in your data through operationalisation: a definition, a question or source, a level of measurement and a coding rule. The codebook (or data dictionary) records all of this in one table that anyone can check.

About 25 minutes | Published example plus three more | Stata and Python code

After this section you can:

1. operationalise a concept into a measurable, coded variable;
2. build a variable operationalisation table for Section 3.3;
3. name, label and code variables so the data and the thesis agree.

From concept to variable

Operationalisation answers the question: *what exactly will I observe, and how will I record it?* "Financial literacy" is a concept. "The number of correct answers to three questions on interest, inflation and diversification, scored 0 to 3" is its operationalisation.

For each variable, Section 3.3 should state:

1. the **variable name** exactly as it appears in your dataset and tables;
2. the **operational definition**, in words;
3. the **questionnaire item or data source** it comes from;
4. the **level of measurement** (nominal, ordinal, interval, ratio; see C1);
5. the **coding**: values, units, what 1 means for a dummy, how a composite is built;
6. its **role and expected sign**, with a source for the expectation where possible.

A published example

Soon, Kamaruddin and Anuar (2018) studied why flood victims did or did not evacuate during the December 2014 floods. Their Table II is a compact operationalisation table: the variable, the exact questionnaire item, and the coding.

Table B3.1. Questionnaire items and variable operationalisation. Reproduced from Soon et al. (2018, Table II, p. 138)

Variables	Questionnaire items	Variable operationalisation
Dependent variable		
Evacuee	Did you evacuate?	1 if evacuated; 0 otherwise
Explanatory variables of interest		
First-time victim	Is the December 2014 flood your first experience as a flood victim?	1 if first-time victim; 0 otherwise
Notified to evacuate	Did you get any evacuation notices from the authority?	1 if notified; 0 otherwise
Programme existence	Were there any flood awareness programmes held at your community prior to the flood?	1 if yes; 0 otherwise
Joined programme	Did you participate in any of those programmes?	1 if participated; 0 otherwise
Distance to centre	How far away would you say your home is from the designated evacuation centre?	Estimates given in kilometres

Notice three good habits. Each dummy says what 1 means. The question wording is quoted, so the reader can judge what was really measured. The continuous variable states its unit.

✓ **Examiner:** *"Distance to centre" is the respondent's estimate, not a measured distance. Good operationalisation tables make that visible, and so does the limitations section.*

A fuller codebook

Table B3.2. Variable operationalisation and codebook for a saving study

Variable name	Operational definition	Measurement level	Coding	Expected sign and source
`income`	Household income: total monthly household income in RM	Ratio	Continuous; top-coded if necessary	Positive effect on saving; survey item Q12
`female`	Gender: respondent identifies as female	Nominal (binary)	1 = female, 0 = otherwise	Ambiguous; demographic control; Q3
`finlit_score`	Financial literacy: sum of correct answers to 10 financial-literacy questions	Interval-like index	0–10; higher = more literate	Positive; constructed from Q20–Q29
`save_regular`	Saving behaviour: respondent saves money every month	Nominal (binary)	1 = yes, 0 = no	Dependent variable; Q35
`att_risk`	Risk attitude: agreement with “I am willing to take financial risks”	Ordinal	1 = strongly disagree to 5 = strongly agree	Control; Q40

Naming and coding rules that save you later

Short, lowercase names `finlit_score`, not `Financial Literacy Score (Q20-Q29)`. Put the long description in the variable label.

Name dummies after the category coded 1 `female` (1 = female), never `gender`. Then the coefficient reads directly as “female versus male”.

One coding per concept If agreement items run 1 = strongly disagree to 5 = strongly agree, keep that direction everywhere and reverse-code negatively worded items before building scales.

Composites say how they are built “PU = mean of PU1–PU4 (1–5)”, plus the reliability evidence (E4).

Transformations are visible `ln_allow` = natural log of monthly allowance; state why (skewness, elasticity interpretation).

Missing values are documented Which codes (−9, 999) meant missing, and what you did with them (C5).

Key takeaways

1. Every variable needs a name, definition, source item, level, coding and expected role.
2. Quote the question wording; say what 1 means; state units.

3. The names in the codebook, the dataset and the regression tables must be identical.

References cited

Soon, J.-J., Kamaruddin, R., & Anuar, A. R. (2018). Flood victims' evacuation decisions: A semi-nonparametric estimation. *International Journal of Emergency Services, 7*(2), 134–146. https://doi.org/10.1108/IJES-05-2017-0031

Example

Worked example: the codebook for the e-wallet survey

This is the codebook for the clean dataset produced in C5 (`ewallet_survey_clean` , 400 respondents, hypothetical). Names match the dataset exactly.

Table B3.3. Codebook, e-wallet survey

Variable	Definition	Item	Level	Coding	Role
`ewallet_user`	Used an e-wallet in the past 30 days	B1	Nominal (binary)	1 = yes, 0 = no	Outcome, RQ2
`ewallet_days`	Days an e-wallet was used in the past 7 days	B2	Ratio (count)	0–7	Outcome, RQ3
`intention`	“I intend to keep using e-wallets”	E1	Ordinal	1 = strongly disagree … 5 = strongly agree	Outcome, RQ4
`PU`	Perceived usefulness	C1–C4	Interval-like composite	Mean of four 1–5 items	Explanatory (+)
`PS`	Perceived security	C5–C8	Interval-like composite	Mean of four 1–5 items	Explanatory (+)
`EOU`	Ease of use	C9–C11	Interval-like composite	Mean of three 1–5 items	Explanatory (+)
`finlit`	Financial literacy: correct answers to three questions	D1–D3	Interval-like index	0–3	Explanatory (+)
`female`	Respondent is female	A1	Nominal (binary)	1 = female, 0 = male	Control
`ln_allow`	Natural log of monthly allowance (RM)	A7	Ratio (log)	ln(RM)	Control (+)
`urban`	Hometown is urban	A6	Nominal (binary)	1 = urban, 0 = rural	Control (+)

Labelling the variables in code

Variable and value labels make Stata's output and your tables self-explanatory. These lines come from `03_data_cleaning.do` and its notebook.

`03_data_cleaning.do`

```
label variable female   "Female (1 = yes)"
label variable PU       "Perceived usefulness, mean of PU1-PU4 (1-5)"
label variable finlit   "Financial literacy score, correct answers (0-3)"
label define yesno 0 "No" 1 "Yes"
label values female urban ewallet_user yesno

* the codebook as Stata sees it
describe ewallet_user ewallet_days intention PU PS EOU finlit female ln_allow urban
codebook female ewallet_user, compact


# a data dictionary you can export to Word or Excel for the appendix
codebook = pd.DataFrame({
    "variable": ["ewallet_user", "PU", "finlit", "female"],
    "definition": ["Used an e-wallet in past 30 days", "Perceived usefulness, mean of PU1-PU4",
                   "Financial literacy, correct answers", "Respondent is female"],
    "level": ["nominal (binary)", "interval-like", "interval-like", "nominal (binary)"],
    "coding": ["1 = yes, 0 = no", "1-5", "0-3", "1 = female, 0 = male"],
})
codebook.to_excel("codebook.xlsx", index=False)
clean[codebook["variable"]].describe().T
```

Writeup

How it reads in the thesis

3.3 Variables and measurement

Table 3.3 defines every variable used in the analysis, the questionnaire item it comes from, its level of measurement and its coding.[1] The three outcome variables differ in measurement level: e-wallet use is binary, days of use is a count from 0 to 7, and intention to continue is ordinal. This difference determines the choice of model in Section 3.4.[2] Perceived usefulness, perceived security and ease of use are each computed as the mean of their items, after confirming internal consistency (Cronbach's α = 0.86, 0.86 and 0.77) and discriminant validity (all HTMT ratios below 0.51; Section 3.1.4).[3] Monthly allowance enters in logarithms because its distribution is strongly right-skewed.

1 *Point to the table; do not list variables in prose.*

2 *Connect measurement level to model choice.*

3 *Explain how composites are built and give their reliability.*

Model write-up. *Reliability figures are from the worked example in E4.*

Worksheet

Worksheet B3

Q1 Operationalise “household vulnerability to floods” into two different variables, one binary and one continuous. Give the questionnaire item and the coding for each. (4 marks)

Model answer

Binary: `flooded_before` : “Has your home been flooded in the past ten years?” 1 = yes, 0 = no. Continuous: `water_depth` : “During the December 2014 flood, how deep was the water inside your home, in centimetres?” recorded in cm (ratio). Other good answers include distance to the river in km, or the number of flood events experienced (a count).

Q2 Find four problems in this codebook row: “Gender | Gender of respondent | Scale | 1, 2 | Positive”. (4 marks)

Model answer

(1) The name should be a dummy named after the category coded 1, such as `female` . (2) “Scale” is not a measurement level; gender is nominal (binary). (3) The coding “1, 2” does not say which value is which, and should be 0/1 for a dummy. (4) “Positive” is meaningless without saying which category the sign refers to and why; give a source or call it a control.

Q3 Your PU scale has four items, and item PU3 is negatively worded (“E-wallets make paying more complicated”). What must you do before computing `PU` , and how would you record it in the codebook? (3 marks)

Model answer

Reverse-code PU3 so that higher values mean higher usefulness: `PU3r = 6 - PU3` on a 1–5 scale. Then compute PU as the mean of PU1, PU2, PU3r and PU4. Record in the codebook “PU3 reverse-coded (6 − PU3) before averaging”, and check that PU3r correlates positively with the other items.

B4 3.4 Model specification

Section 3.4 states the econometric model that answers each research question: the estimating equation with every symbol defined, the estimator and why it suits the outcome and the identification problem, how inference is done, and how robustness will be checked. If your method is unusual, this is where you review the methodological literature that justifies it.

About 40 minutes | Equation templates for 10 models

Stata and Python estimation

After this section you can:

1. write an estimating equation with correct subscripts, error term and defined symbols;
2. choose an estimator from the outcome's measurement level, the data structure and the identification problem;
3. write a methodological justification for a less familiar estimator.

What Section 3.4 contains

1. **The estimating equation(s)**, numbered (3.1), (3.2)…, one per research question or model. Use subscripts that match the data structure: i for units, t for time, it for panels (C2).
2. **A definition for every symbol**: outcome, regressors (pointing to the codebook in 3.3), parameters, the error term, fixed effects.
3. **The estimator and its justification**: why this estimator suits the outcome variable and your identification problem.
4. **The key assumptions** and the diagnostic tests you will run.
5. **Inference**: robust, clustered or bootstrapped standard errors, and why.
6. **What will be reported**: coefficients, odds ratios, marginal effects, elasticities.
7. **Robustness checks**: alternative estimators, samples, variable definitions.
8. **A short methodological review** when the method is less standard (semi-nonparametric, partial proportional odds, fuzzy RD, spatial models), citing methods papers rather than textbook definitions.

Choosing the model from the outcome

Table B4.1. Outcome variable, model and what to report

Outcome variable	Example	Model	Report
Continuous (interval or ratio)	Saving rate; log wage	OLS	Coefficients; elasticities for logs
Binary	Evacuated or not; uses an e-wallet	Logit, probit; linear probability model as a check	Average marginal effects; odds ratios
Ordinal	Five-point intention; income band	Ordered logit/probit; partial proportional odds if parallel lines fail	Marginal effects on each category's probability
Nominal, more than two categories	Bank, ASB, or no saving	Multinomial logit	Relative-risk ratios; marginal effects
Count	Number of children; days of use	Poisson, negative binomial (or OLS when counts are spread out)	Incidence-rate ratios; marginal effects
Censored at zero	Charitable giving with many zeros	Tobit; two-part model	Effects on the probability and on the amount
Fraction 0–1	Budget share for education	Fractional logit	Average marginal effects

✓ **Examiner:** *With a binary outcome, report marginal effects, not just logit coefficients. Readers cannot interpret a log-odds of 0.874.*

Equation templates

Copy the form that matches your design, and replace the generic x with your variables.

Cross-section, continuous outcome (OLS)

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \cdots + \beta_k x_{ki} + u_i \tag{3.1}$$

Binary outcome (logit)

$$\Pr(y_i = 1 \mid \mathbf{x}_i) = \Lambda(\mathbf{x}_i'\boldsymbol{\beta}) = \frac{\exp(\mathbf{x}_i'\boldsymbol{\beta})}{1 + \exp(\mathbf{x}_i'\boldsymbol{\beta})} \tag{3.2}$$

Ordinal outcome (ordered logit)

$$y_i^* = \mathbf{x}_i'\boldsymbol{\beta} + \varepsilon_i, \qquad y_i = j \text{ if } \kappa_{j-1} < y_i^* \le \kappa_j,\ j = 1, \ldots, J \tag{3.3}$$

In the partial proportional odds model (Williams, 2006), some coefficients are allowed to differ across the cut-points j: $\Pr(y_i > j) = \Lambda(\alpha_j + \mathbf{x}_{1i}'\boldsymbol{\beta}_1 + \mathbf{x}_{2i}'\boldsymbol{\beta}_{2j})$.

Nominal outcome (multinomial logit)

$$\Pr(y_i = m \mid \mathbf{x}_i) = \frac{\exp(\mathbf{x}_i'\boldsymbol{\beta}_m)}{\sum_{h=1}^{M}\exp(\mathbf{x}_i'\boldsymbol{\beta}_h)}, \quad \boldsymbol{\beta}_1 = \mathbf{0} \tag{3.4}$$

Time series (one unit, many periods)

$$y_t = \beta_0 + \beta_1 x_t + \beta_2 y_{t-1} + u_t \tag{3.5}$$

Pooled cross-sections, difference-in-differences

$$y_{it} = \beta_0 + \delta_0\, Post_t + \beta_1\, Treat_i + \delta_1 (Post_t \times Treat_i) + \mathbf{x}_{it}'\boldsymbol{\gamma} + u_{it} \tag{3.6}$$

Panel, fixed effects

$$y_{it} = \mathbf{x}_{it}'\boldsymbol{\beta} + a_i + \lambda_t + u_{it} \tag{3.7}$$

Instrumental variables (two-stage least squares)

$$x_i = \pi_0 + \pi_1 z_i + \mathbf{w}_i'\boldsymbol{\pi} + v_i, \qquad y_i = \beta_0 + \beta_1 \hat{x}_i + \mathbf{w}_i'\boldsymbol{\gamma} + u_i \tag{3.8}$$

Regression discontinuity

$$y_i = \alpha + \tau D_i + f(r_i - c) + D_i\, g(r_i - c) + u_i, \quad D_i = \mathbf{1}[r_i \geq c] \tag{3.9}$$

In a fuzzy RD, crossing the cut-off c raises the probability of treatment without determining it, and D_i instruments actual treatment, giving a first stage, a reduced form and a second stage (see B2, Example 3).

? **Examiner:** *Why is there no error term in Equation (3.4)? Because a probability model has none. Know which of your equations should have one.*

Inference: which standard errors?

Heteroskedasticity-robust — The default for cross-sections (`vce(robust)` in Stata; `cov_type="HC1"` in Python).

Clustered — When observations are grouped (students in classes, households in villages, years within a state). Cluster at the level of treatment assignment or sampling (`vce(cluster id)`). With fewer than about 30 clusters, say how you deal with it (for example, a wild cluster bootstrap).

Newey–West (HAC)	For time series with serial correlation (`newey`).
Survey design	With complex survey data (strata, clusters, weights), use the design (`svyset` then `svy:`).

When the method needs a methodological review

A standard estimator needs only a sentence and a textbook citation. A less familiar one needs a short review of the methods literature: what problem the method solves, why the usual alternative fails here, and evidence that the method works. Two examples:

- **Partial proportional odds.** The ordered logit assumes each regressor's effect is the same at every cut-point (parallel lines). When a Brant test rejects this for some regressors, Williams (2006) shows how to relax it only for those, keeping the model parsimonious.
- **Semi-nonparametric binary choice.** Logit and probit fix the error distribution. Gallant and Nychka (1987) proposed approximating an unknown density with a Hermite series, so the error distribution is estimated rather than assumed. Soon et al. (2018) use this approach for flood evacuation decisions.

Key takeaways

1. Number every equation, define every symbol, and match subscripts to the data structure.
2. The outcome's measurement level chooses the family of model; the identification problem chooses the design.
3. State the standard errors and what you will report; review the methods literature for non-standard estimators.

References cited

Gallant, A. R., & Nychka, D. W. (1987). Semi-nonparametric maximum likelihood estimation. *Econometrica, 55*(2), 363–390.

Soon, J.-J., Kamaruddin, R., & Anuar, A. R. (2018). Flood victims' evacuation decisions: A semi-nonparametric estimation. *International Journal of Emergency Services, 7*(2), 134–146.

Williams, R. (2006). Generalized ordered logit/partial proportional odds models for ordinal dependent variables. *The Stata Journal, 6*(1), 58–82.

Example

Worked example: three outcomes, three models

The e-wallet survey (400 students, hypothetical) has three outcomes at three measurement levels, so Section 3.4 specifies three models. Here they are estimated on the clean data, using `06_model_specification.do` and its notebook.

06_model_specification.do

```
use "$data/ewallet_survey_clean.dta", clear
global X "PU PS EOU finlit female ln_allow urban"

* RQ2: binary outcome. LPM for comparison, logit as the main model
regress ewallet_user $X, vce(robust)
logit ewallet_user $X, vce(robust) nolog
margins, dydx(*)                                    // average marginal effects

* RQ3: days of use in the past week, users only
regress ewallet_days $X i.year_study if ewallet_user == 1, vce(robust)

* RQ4: ordinal outcome
ologit intention $X, vce(robust) nolog
margins, dydx(PU) predict(outcome(5))               // effect on Pr(strongly agree)


import statsmodels.formula.api as smf
from statsmodels.miscmodels.ordinal_model import OrderedModel

df = load("ewallet_survey_clean.csv")
X = "PU + PS + EOU + finlit + female + ln_allow + urban"

lpm = smf.ols(f"ewallet_user ~ {X}", df).fit(cov_type="HC1")
logit = smf.logit(f"ewallet_user ~ {X}", df).fit(disp=False, cov_type="HC1")
print(logit.get_margeff(at="overall").summary_frame())        # average marginal effects

ols_days = smf.ols(f"ewallet_days ~ {X} + C(year_study)", df[df["ewallet_user"] == 1]).fit(cov_type="HC1")

d3 = df.dropna(subset=["intention", "PU", "PS", "EOU", "finlit", "female", "ln_allow", "urban"])
ologit = OrderedModel(d3["intention"].astype(int), d3[["PU", "PS", "EOU", "finlit", "female", "ln_allow", "urban
                      distr="logit").fit(method="bfgs", disp=False)
```

Table B4.2. Selected estimates (robust standard errors in parentheses; N = 394 after missing allowance)

	LPM: user	Logit: user	Logit AME	Ordered logit: intention
PU	0.128 (0.021)	0.874 (0.153)	0.119 (0.019)	1.563 (0.133)
PS	0.088 (0.020)	0.618 (0.148)	0.084 (0.018)	0.923 (0.119)
EOU	0.046 (0.022)	0.320 (0.150)	0.044 (0.020)	0.705 (0.121)
finlit	0.010 (0.019)	0.092 (0.139)	0.013 (0.019)	0.006 (0.097)

Source: `06_model_specification.log`. Data are simulated for teaching.

Reading the output

- The LPM coefficient on PU (0.128) and the logit average marginal effect (0.119) tell the same story: one more point of perceived usefulness goes with a 12–13 percentage-point higher probability of using an e-wallet. When the two agree, the result does not hinge on the functional form.
- The logit coefficient (0.874) is a change in log-odds; $e^{0.874} = 2.40$ is the odds ratio.
- Financial literacy has no detectable association with use once perceptions are controlled for. That is a substantive finding for Chapter 4, so the model is not “wrong”.

The ordered logit's effect of PU on the probability of answering “strongly agree” is 0.162: one more point of PU goes with a 16-point higher chance of the top category.

Writeup

How it reads in the thesis

3.4 Model specification

3.4.1 E-wallet use (RQ2)

Because e-wallet use is binary, its association with perceptions is estimated with a logit model:

$$\Pr(user_i = 1 \mid \mathbf{x}_i) = \Lambda(\beta_0 + \beta_1 PU_i + \beta_2 PS_i + \beta_3 EOU_i + \beta_4 finlit_i + \mathbf{w}_i'\boldsymbol{\gamma}) \qquad (3.1)$$

where $\Lambda(\cdot)$ is the logistic distribution function, PU_i, PS_i and EOU_i are respondent i's perceived usefulness, perceived security and ease-of-use scores, $finlit_i$ is the financial literacy score, and $\mathbf{w}_i$ contains gender, log allowance and urban origin (Table 3.3).[1] Results are reported as average marginal effects, and a linear probability model is estimated as a check on functional form.[2] Standard errors are robust to heteroskedasticity; because respondents were sampled individually within faculties rather than in clusters, clustering is not required.[3]

3.4.2 Intention to continue (RQ4)

Intention is measured on a five-point ordered scale, so it is modelled with an ordered logit (Equation 3.3). The parallel-lines assumption is tested with a Brant test; if it fails for a regressor, the partial proportional odds model (Williams, 2006) relaxes the assumption for that regressor only.[4]

1 *Every symbol defined, pointing to the codebook rather than repeating it.*

2 *What will be reported, plus a functional-form check.*

3 *The choice of standard errors is justified by the sampling design.*

4 *The model's key assumption, its test, and the fallback, with a methods citation.*

Model write-up. *Adapt to your study.*

Worksheet

Worksheet B4

Q1 Write the estimating equation, with subscripts, for: household monthly saving (RM) regressed on financial literacy, income, age and number of children, using a 2024 household survey. Then rewrite it for a panel of the same households observed in 2020, 2022 and 2024 with household fixed effects. (6 marks)

Model answer

Cross-section: $saving_i = \beta_0 + \beta_1 finlit_i + \beta_2 income_i + \beta_3 age_i + \beta_4 children_i + u_i$.

Panel: $saving_{it} = \beta_1 finlit_{it} + \beta_2 income_{it} + \beta_3 children_{it} + a_i + \lambda_t + u_{it}, t \in \{2020, 2022, 2024\}$. Age drops out (or is collinear with λ_t and a_i) because everyone ages two years between waves. The intercept is absorbed by the fixed effects.

Q2 A student plans to estimate "the number of times a household used BNM's online banking in the past month" with OLS. Comment on the choice and suggest an alternative. (4 marks)

Model answer

The outcome is a non-negative count, probably with many zeros and a long right tail. OLS can predict negative counts and gives poor inference here. Poisson regression with robust standard errors is the natural choice (consistent if the mean is correctly specified); a negative binomial model handles overdispersion. Report incidence-rate ratios or marginal effects. OLS can still be reported as a robustness check.

Q3 Write a three-sentence methodological justification for using clustered standard errors in a study of 1,200 students sampled by randomly selecting 40 tutorial classes. (3 marks)

Model answer

"Students were sampled in clusters (tutorial classes), and students in the same class share instructors and peer influences, so their errors are likely to be correlated. Ignoring this correlation would understate standard errors and overstate significance. Standard errors are therefore clustered at the class level, the level at which sampling took place; with 40 clusters, conventional cluster-robust inference is adequate."

B5 3.5 Ethics, limitations and putting the chapter together

The last section closes the chapter: research ethics, the limits of the design, and a short summary that hands over to the results. This page also gives the whole chapter's skeleton, a checklist to run before you submit, and the queries examiners raise most often.

About 25 minutes | Interactive submission checklist

Chapter skeleton with suggested lengths

The skeleton of an economics Chapter 3

Table B5.1. A typical structure. Lengths are a guide for a monograph-style thesis, not a rule

Section	Contents	Typical length
3.0 Introduction	Paradigm and design in two or three sentences; map of the chapter (A1)	½ page
3.1 Data	Population, frame, sampling, sample size, instrument, item map, pilot, administration, response, or the secondary sources (B1)	4–8 pages
3.2 Research questions and methods	The RQ–method table and a paragraph introducing it (B2)	1–2 pages
3.3 Variables	Operationalisation table; composites, reliability and validity; transformations (B3, E4)	3–5 pages
3.4 Model specification	Equations, estimators, assumptions, inference, robustness; methodological review where needed (B4)	4–8 pages
3.5 Ethics and limitations	Approval, consent, confidentiality, data protection; limits of the design	1–2 pages
3.6 Summary	What was done, in a paragraph; bridge to Chapter 4	½ page

Data preparation and cleaning (C5) fit at the end of 3.1 or as a short 3.3.x. Universities differ; follow your faculty's thesis guide where it is stricter.

Writing the ethics section

- **Approval**: the committee and the reference number, for example “approved by the university's Research Ethics Committee (ref. …)”.
- **Informed consent**: how participants were told the purpose, that taking part was voluntary, and that they could withdraw; how consent was recorded.
- **Confidentiality and anonymity**: whether names were collected, how identifiers were separated from answers.
- **Data protection**: storage, access, retention period; in Malaysia, handling personal data in line with the Personal Data Protection Act 2010.
- **Vulnerable participants and sensitive topics**: extra safeguards, for example for flood victims, minors or low-income households.

- **Secondary data**: the licence or terms of use, and any restricted-access agreement.

Writing limitations without undermining the thesis

State limitations of *design*, not of effort. Name each limitation, its likely consequence, and what you did about it.

Table B5.2. Limitation, consequence, response

Limitation	Likely consequence	Response in the thesis
Single university, convenience recruitment within faculties	Findings may not generalise to all Malaysian students	Restrict claims to the target population; compare sample profile with enrolment data
Cross-sectional, self-reported data	Associations, not causal effects; common-method bias	Associational wording; procedural remedies; Harman test reported with caveats
Self-reported income	Measurement error, attenuation	Income bands checked against allowance norms; log transformation

? **Examiner:** *"Time and money were limited" is not a methodological limitation. Tell me what the design cannot show.*

Checklist before you submit Chapter 3

Tick each item as you check your draft. Ticks are kept in this browser only.

- [ ] Every research question in Chapter 1 appears, word for word, in the RQ–method table.
- [ ] Every model in 3.4 answers a question in the table, and every question has a model.
- [ ] The population, sampling frame and sampling design are stated and justified.
- [ ] The sample size is justified by precision or power, not by population size alone.
- [ ] The response rate is reported with numerator and denominator.
- [ ] Every adopted or adapted scale has a source, and adaptations are documented.
- [ ] Reliability and validity evidence is reported for my own sample.
- [ ] Every variable has a name, definition, item, level and coding in one table.
- [ ] Equations are numbered, subscripts match the data structure, and every symbol is defined.
- [ ] The estimator suits the outcome's measurement level, and its key assumption is stated and tested.
- [ ] The standard errors are stated and justified.
- [ ] Data cleaning decisions and the number of observations dropped are reported.
- [ ] Causal language appears only where the design supports it.
- [ ] Ethics approval, consent and data protection are described.

- [ ] Limitations name consequences and responses.

0 of 15 checked.

The queries examiners raise most often

“Why this sample size?”	Answer with a precision or power calculation (D4).
“Is the sample representative?”	Compare the sample's profile with the population on known characteristics; be honest about non-probability recruitment (D3).
“Where did these items come from?”	The adoption and adaptation table (E2).
“Is a Likert item interval data?”	Explain how you treat single items and composite scales (E3).
“Alpha is 0.95; is that good?”	Possibly redundant items; report omega and validity evidence too (E4).
“Is this effect causal?”	State the identifying assumption, or reword as an association (A2).

Writeup

Model write-ups: ethics, limitations, summary

3.5 Ethics and limitations

The study was approved by the university's Research Ethics Committee (reference to be inserted). Participants read an information sheet explaining the study's purpose, that participation was voluntary and anonymous, and that they could stop at any point; continuing to the questions recorded consent.[1] No names or student numbers were collected, and the data are stored on a password-protected university drive for five years.[2]

Three limitations qualify the findings. First, the sample comes from one university, so results describe its undergraduates rather than Malaysian students generally. Second, all variables are self-reported in one questionnaire at one time, so estimated relationships are associations and may be inflated by common-method variance; Section 3.1.2 describes the procedural remedies used. Third, allowance is self-reported and may contain error, which would bias its coefficient towards zero.[3]

3.6 Summary

This chapter described a stratified random survey of 400 undergraduates, the measurement of perceived usefulness, security and ease of use with adapted scales of acceptable reliability and validity, and three regression models matched to the measurement level of each outcome. Chapter 4 presents the results in the order of the research questions.

1 *Approval and consent procedure, concretely.*

2 *Anonymity and data protection.*

3 *Each limitation is followed by its consequence.*

Model write-up. *Hypothetical study.*

Worksheet

Worksheet B5

Q1 Rewrite this limitation so that it helps rather than hurts the thesis: “Due to time constraints, the sample size was small and the results may not be accurate.” (4 marks)

Model answer

“With 150 respondents, the study has 80% power to detect medium associations ($f^2 = 0.15$ with five predictors needs 92 observations) but not small ones, so non-significant estimates for weaker predictors should not be read as evidence of no association. Confidence intervals are reported so readers can judge precision.” It names the consequence, quantifies it and says what was done.

Q2 Run the 15-item checklist on a draft Chapter 3 (your own or a classmate's). Which three items failed? Write one sentence for each saying how you will fix it. (6 marks)

Model answer

Answers vary. The most common failures in drafts are item 4 (sample size justified only by Krejcie–Morgan or by “rule of thumb”), item 7 (reliability quoted from the original study rather than the student's own sample) and item 13 (causal verbs such as “impact” with a cross-sectional design). Fixes: add a power calculation; report alpha, CR and AVE from your data; reword to “associated with”.

C1 Levels of measurement: nominal, ordinal, interval, ratio

A variable's level of measurement decides which statistics are meaningful and which model suits it as an outcome. Stevens (1946) set out the four levels, often remembered as NOIR.

About 15 minutes | Sorting game | Worksheet with model answers

After this section you can:

1. assign any variable to its level of measurement;
2. say which statistics each level permits;
3. explain what the mean of a dummy variable means.

Nominal

Values are categories with no inherent order. Examples: gender, ethnicity, industry code, country of birth, marital status. The only meaningful comparison is equal or not equal (=, ≠). Permissible statistics: mode, frequencies, proportions, the chi-square test.

Means are meaningless for nominal codes. A “mean gender” is nonsense, even though Stata will obediently compute one if male = 1 and female = 2.

Ordinal

Values have a meaningful order, but the distances between them are not equal, or not known. Examples: income category (B40, M40, T20), a Likert item, academic performance bands (excellent, good, pass). Median and percentiles are meaningful; differences are not. Moving from “agree” to “strongly agree” need not be the same step as from “disagree” to “neutral”.

Interval

Values have order *and* equal intervals, but no true zero: zero is just a point on the scale, not the absence of the attribute. Examples: temperature in Celsius (0 °C is not “no temperature”), shoe size, calendar year. Differences are meaningful; ratios are not. 20 °C is not “twice as hot” as 10 °C.

Ratio

Interval level plus a true zero that means the absence of the attribute. Examples: income, age, distance, weight, price, GDP, the unemployment rate, number of children, years of schooling. All arithmetic is meaningful: RM 4,000 is twice RM 2,000.

Table C1.1. What each level permits

Level	Order?	Equal intervals?	True zero?	Central tendency	Typical tests	As an outcome, model with…
Nominal	No	No	No	Mode	Chi-square	Logit/probit (binary); multinomial logit
Ordinal	Yes	No	No	Median	Mann–Whitney, Spearman	Ordered logit/probit
Interval	Yes	Yes	No	Mean	t-test, Pearson, regression	OLS
Ratio	Yes	Yes	Yes	Mean (geometric mean too)	All of the above	OLS; logs for elasticities; count models for counts

✓ **Examiner:** *Composite scores built from several Likert items are usually treated as interval-like. Say that you do this, and why (E3).*

The mean of a dummy variable

If gender is coded as a dummy with female = 0 and male = 1, and Stata reports a mean of 0.57, what does it represent? It is the **proportion of observations coded 1**: 57% of the sample is male. This is why dummies are coded 0/1 and named after the category coded 1 (`male` , or better `female` with female = 1). The mean then reads directly as a share, and the regression coefficient as the difference from the 0 category.

$$\bar{d} = \frac{1}{n}\sum_{i=1}^{n} d_i = \frac{\#\{d_i = 1\}}{n}$$

Key takeaways

1. Nominal: categories. Ordinal: ordered categories. Interval: equal steps, no true zero. Ratio: equal steps and a true zero.
2. The level limits the statistics and decides the model for an outcome.
3. The mean of a 0/1 dummy is the proportion coded 1.

References cited

Stevens, S. S. (1946). On the theory of scales of measurement. *Science, 103*(2684), 677–680.

Example

Sort the variables

Choose the level of measurement for each variable. Feedback appears as you choose.

Home state (Kedah, Selangor, Sabah…)
Income group: B40, M40, T20
Monthly household income (RM)
Temperature in Celsius
“I trust e-wallets”: strongly disagree … strongly agree
Number of children
Industry (MSIC code)
Unemployment rate (%)
Degree classification (first, second upper, second lower, third)
Year of birth

N = nominal, O = ordinal, I = interval, R = ratio.

Worksheet

Worksheet C1

Q1 A dummy `male` (1 = male, 0 = female) has mean 0.57 in a sample of 400. How many women are in the sample, and what does a regression coefficient of −250 on `male` in a monthly-saving equation mean? (3 marks)

> **Model answer**
>
> 57% are men, so 0.43 × 400 = 172 are women. The coefficient means that men save RM 250 a month less than women on average, holding the other regressors constant.

Q2 A student codes ethnicity as 1 = Malay, 2 = Chinese, 3 = Indian, 4 = Others and enters it as one variable in OLS. What is wrong, and what should they do? (4 marks)

> **Model answer**
>
> Ethnicity is nominal, so a single 1–4 variable imposes an order and equal spacing that do not exist. The coefficient would be meaningless. Create dummies for each group and omit one as the reference category (`i.ethnic` in Stata; `C(ethnic)` in Python). Each coefficient is then the difference from the reference group.

Q3 Income can be collected as an exact amount (ratio) or in bands such as B40/M40/T20 (ordinal). Give one advantage of each. (2 marks)

Model answer

Exact amount: more information, allows means, logs and elasticities. Bands: higher response rates and less recall error on a sensitive question; enough for many descriptive purposes. A common compromise asks the exact amount and offers bands to those who decline.

C2 Data structures and how the model is written

Cross-section, time series, pooled cross-sections and panel data answer different questions, and each changes how you write the econometric model: which subscripts appear, which terms you add, and which estimator you use.

About 30 minutes | Four example datasets | Stata and Python estimation

After this section you can:

1. recognise the four data structures from a dataset's layout;
2. match a research question to the structure that can answer it;
3. write the model with the subscripts and terms each structure requires.

Cross-sectional data: many units, one point in time

Each row is a different unit (individual, household, firm or country), all observed at roughly the same time.

Table C2.1. Cross-section: students in one semester (first 8 of 22 rows)

Student_ID	Study_Hours	Attendance	Sleep_Hours	PT_Job	HH_Income_k	Final_Score	Gende
STU001	18	100	8	1	75	96.7	Female
STU002	16	78	5	0	34	77.0	Female
STU003	16	88	9	1	102	95.0	Female
STU004	20	91	9	1	128	98.0	Male
STU005	7	64	7	1	33	62.6	Female
STU006	29	61	7	1	43	100.0	Male
STU007	25	63	5	1	139	100.0	Male
STU008	7	90	8	0	129	84.6	Male

The model carries the unit subscript i:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \cdots + \beta_k x_k + u$$

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + \cdots + \beta_k x_{ki} + u_i$$

$$Wage_i = \beta_0 + \beta_1 Educ_i + \beta_2 Exp_i + \beta_3 Age_i + \beta_4 Male_i + \beta_5 CGPA_i + u_i$$

Time-series data: one unit, many periods

Each row is a different period for the same unit, usually a country.

Table C2.2. Time series: one economy, 2000–2023 (selected years; values as in the lecture notes)

Year	GDP (USDb)	Unemployment	Inflation	Export (USDb)	Import (USDb)
2000	102.15	3.00	1.53	112.37	94.35
2001	101.05	3.53	1.42	102.44	86.25
2008	241.95	3.32	5.44	229.66	178.11
2009	211.85	3.66	0.58	184.90	143.89
2020	337.46	4.54	−1.14	207.80	186.31
2023	399.71	3.87	2.49	274.13	253.74

The subscript is t. One city or one country over time:

$$y_t = \beta_0 + \beta_1 x_{1t} + \beta_2 x_{2t} + \beta_3 x_{3t} + \cdots + \beta_k x_{kt} + u_t$$

$$Crimerate_t = \beta_0 + \beta_1 Unemp_t + \beta_2 PrisonPop_t + \beta_3 YoungMales_t + u_t$$

$$GDP_t = \beta_0 + \beta_1 EmpRate_t + \beta_2 InfRate_t + \beta_3 NetExp_t + u_t$$

Time series bring their own issues: lags (y_{t-1}), trends, serial correlation (Newey–West standard errors), and non-stationarity (unit-root tests; cointegration or differencing).

✓ **Examiner:** *With 24 annual observations, every extra lag costs you degrees of freedom. Justify the lag length.*

Pooled cross-sections: new units each period

Several cross-sections taken at different times. Different units are drawn each time, independently, from the same population. A unit occasionally appears twice by chance, but you cannot follow anyone over time.

Table C2.3. Pooled cross-section: two survey years, different respondents (selected rows)

Year	ID	Age	Income	Educ_yrs	Employed
2022	101	25	45,000	14	Yes
2022	102	34	60,000	16	Yes
2022	103	22	15,000	12	No
2023	201	28	52,000	14	Yes
2023	202	31	58,000	16	Yes
2023	203	40	75,000	16	Yes

Pooling lets you ask whether relationships changed between periods, and it supports difference-in-differences. Kiel and McClain (1995) studied house prices near a planned incinerator, with houses sold in 1978 (before) and 1981 (after). The hypothesis H_0 is that house prices fall if a house is located near the incinerator, where $near = 1$ if within 3 km and 0 otherwise.

$$Hprice_i = \beta_0 + \beta_1 near_i + u_i \qquad \text{(1981 only: mixes the effect with pre-existing differences)}$$

$$Hprice_i = \beta_0 + \beta_1\, year81_i + \beta_2\, near_i + \beta_3\,(year81_i \cdot near_i) + u_i$$

β_3 is the **coefficient of interest**: the difference-in-differences estimate of the incinerator's effect.

Panel (longitudinal) data: the same units, many periods

The same units are observed at several points in time. Each row is a unit–period pair.

Table C2.4. Panel: the same five countries, 2021–2023 (selected rows; values as in the lecture notes)

Country	Year	GDP (USDb)	Inflation	Unemp	Pop (m)	TradeBal (USDb)
USA	2021	23,315	4.7	5.3	332	−845
USA	2022	25,462	8.0	3.6	333	−948
USA	2023	27,360	4.1	3.6	334	−773
China	2021	17,734	0.9	3.9	1,412	676
China	2022	17,963	2.0	4.2	1,412	877
China	2023	17,700	0.2	5.2	1,410	823

The model carries both subscripts and, usually, unit and time effects:

$$y_{it} = \beta_0 + \beta_1 x_{1it} + \cdots + \beta_k x_{kit} + a_i + u_{it}$$

$$y_{it} = \beta_1 x_{1it} + \cdots + \beta_k x_{kit} + a_i + \lambda_t + u_{it} \qquad \text{(two-way fixed effects)}$$

$$Saving_{it} = \beta_1 FinLit_{it} + \beta_2 Income_{it} + a_i + \lambda_t + u_{it}$$

a_i captures everything about unit i that does not change over time, observed or not (geography, institutions, culture). λ_t captures shocks common to all units in year t, such as 2020. Fixed effects remove a_i by comparing each unit with itself over time; random effects assume a_i is unrelated to the regressors. The Hausman test helps choose between them.

? **Examiner:** *If a_i is in the equation, tell me how you estimate it away. If it is not, tell me why you are sure there is no unobserved heterogeneity.*

Matching the data structure to the research question

Table C2.5. Research questions and the structures that can answer them

Research question	Structure	Why
What factors correlate with household income in 2024?	Cross-section	One point in time is sufficient.
How has the inflation–unemployment relationship in Malaysia evolved?	Time series	One economy followed over time.
Did a 2020 policy change affect consumption patterns differently in 2024 compared with 2019?	Pooled cross-sections	Independent samples before and after the change.
Does retirement affect an individual's cognitive decline?	Panel	You need to observe the same people before and after retirement.

Key takeaways

1. Cross-section y_i; time series y_t; pooled cross-sections and panels y_{it}, but only panels follow the same units.
2. Pooled cross-sections support before/after comparisons and DiD; panels add unit fixed effects.
3. Write the subscripts and the extra terms (a_i, λ_t, lags) that your structure requires.

References cited

Kiel, K. A., & McClain, K. T. (1995). House prices during siting decision stages: The case of an incinerator from rumor through operation. *Journal of Environmental Economics and Management, 28*(2), 241–255.

Wooldridge, J. M. (2020). *Introductory econometrics: A modern approach* (7th ed.). Cengage. Chapters 1, 10, 13 and 14 cover the four structures.

Example

Worked example: estimating each structure

All four structures are estimated in `05_data_structures.do` and its notebook. The pooled and panel datasets are simulated, with known true values, so you can see each estimator succeed or fail.

05_data_structures.do

```
* 1. Cross-section
use "$data/ds_crosssection_students.dta", clear
generate byte female = Gender == "Female"
regress Final_Score Study_Hours Attendance Sleep_Hours PT_Job HH_Income_k female, vce(robust)

* 2. Time series: declare the time variable, then lags work
use "$data/ds_timeseries_macro.dta", clear
tsset Year
newey Inflation Unemployment L.Inflation, lag(2)

* 3. Pooled cross-sections: difference-in-differences
use "$data/pooled_incinerator_sim.dta", clear
regress price i.y81##i.nearinc age rooms area, vce(robust)

* 4. Panel: declare unit and time, then fixed effects
use "$data/panel_states_sim.dta", clear
encode state, generate(state_id)
xtset state_id year
xtreg saving_rate finlit_index income_k i.year, fe vce(cluster state_id)
```

```
import statsmodels.formula.api as smf
from linearmodels.panel import PanelOLS

cs = load("ds_crosssection_students.csv")
cs["female"] = (cs["Gender"] == "Female").astype(int)
smf.ols("Final_Score ~ Study_Hours + Attendance + Sleep_Hours + PT_Job + HH_Income_k + female", cs).fit(cov_type=

ts = load("ds_timeseries_macro.csv").set_index("Year")
ts["Inflation_lag1"] = ts["Inflation"].shift(1)
smf.ols("Inflation ~ Unemployment + Inflation_lag1", ts.dropna()).fit(cov_type="HAC", cov_kwds={"maxlags": 2})

pool = load("pooled_incinerator_sim.csv")
smf.ols("price ~ y81 * nearinc + age + rooms + area", pool).fit(cov_type="HC1")

panel = load("panel_states_sim.csv").set_index(["state", "year"])
PanelOLS.from_formula("saving_rate ~ finlit_index + income_k + EntityEffects + TimeEffects", panel).fit(
    cov_type="clustered", cluster_entity=True)
```

What the panel example shows

Table C2.6. Effect of the financial literacy index on the saving rate, 13 states, 2012–2022 (simulated; true value 0.15)

Estimator	Coefficient	Std. error	Verdict
Pooled OLS with year dummies	0.347	0.070	Biased upwards: ignores a_i, which is correlated with literacy
Two-way fixed effects (clustered)	0.149	0.029	Close to the true 0.15
Random effects (trend + 2020 dummy)	0.175	0.026	Biased; the Hausman test rejects RE ($p < 0.001$)

In the pooled incinerator data, the DiD estimate is −11,729 (robust SE 3,112) against a true −11,500. Without the 1978 comparison, the 1981 price gap would also include the area's pre-existing price disadvantage.

Worksheet C2

Q1 Identify the structure: (a) DOSM's Household Income and Expenditure Survey 2019 and 2022, with different households in each; (b) the World Bank's GDP series for ASEAN-10, 1990–2023; (c) the Malaysian Labour Force Survey for 2023 alone; (d) Malaysia's quarterly GDP, 2000Q1–2024Q4. (4 marks)

Model answer

(a) Pooled cross-sections, since different households are drawn in each round. (b) Panel: the same ten countries every year. (c) Cross-section. (d) Time series.

Q2 Write the model for this question, with correct subscripts: "Does minimum-wage coverage affect employment across Malaysia's 13 states, 2012–2022, controlling for state GDP?" Name the term that handles time-invariant state characteristics. (4 marks)

Model answer

$Emp_{st} = \beta_1 Coverage_{st} + \beta_2 \ln GDP_{st} + a_s + \lambda_t + u_{st}$, with s = state and t = year. a_s is the state fixed effect, absorbing time-invariant state characteristics; λ_t absorbs national shocks common to all states. Standard errors are clustered by state.

Q3 In the incinerator model, what would β_2 (the coefficient on *near*) measure, and why is it not the effect of the incinerator? (3 marks)

Model answer

β_2 is the price difference between houses near and far from the site in 1978, before the incinerator was rumoured. It captures pre-existing differences in the neighbourhood (the site was chosen in a cheaper area), not the incinerator. The effect is β_3, the extra change in the near group's prices between 1978 and 1981.

C3 Collecting primary data: surveys, interviews and focus groups

Primary data are collected by you, for your research questions. The three workhorses are the questionnaire survey, the interview and the focus group discussion. Each answers a different kind of question, and each has to be described in Chapter 3 in its own way.

About 25 minutes | Mode comparison table | Write-up examples

After this section you can:

1. choose between a survey, interviews and FGDs, and justify the choice;
2. choose a survey mode and anticipate its biases;
3. describe fieldwork in Chapter 3 so an examiner can judge its quality.

Which method for which question?

Table C3.1. Surveys, interviews and focus groups compared

	Questionnaire survey	Interview	Focus group discussion (FGD)
Answers	How many? How much? What is associated with what?	How and why, in a person's own words	How a group talks about an issue; shared views and disagreements
Typical sample	Hundreds to thousands, ideally probability-sampled	About 10–30, purposively chosen	3–6 groups of 6–10 participants
Instrument	Structured questionnaire, mostly closed items	Interview guide (structured, semi-structured or unstructured)	Discussion guide with a moderator
Analysis	Statistics and econometrics	Thematic or content analysis	Thematic analysis, attention to group dynamics
Main risk	Shallow answers; non-response; social desirability	Interviewer influence; small, selective samples	Dominant voices; conformity

Many economics theses combine them. Interviews or FGDs before a survey help to write better items. Interviews after a survey help to explain surprising patterns (an explanatory sequential design).

Survey modes

Table C3.2. Survey modes and their trade-offs

Mode	Strengths	Weaknesses	Good for
Face to face, on paper or tablet (PAPI or CAPI)	High response; can reach people without internet; interviewer can clarify	Expensive; interviewer effects; more social desirability	Household surveys, rural areas, low-literacy respondents
Telephone (CATI)	Fast; geographically spread samples	Falling response rates; short questionnaires only	Follow-ups; short opinion surveys
Online, self-administered (CAWI: Google Forms, Qualtrics)	Cheap; quick; no interviewer effect; automatic data entry	Coverage of people without internet; self-selection; low response; inattentive answers	Students, professionals, firms with email lists
Drop-off and collect, or mail	Respondent answers in their own time	Slow; incomplete forms; no clarification	Firms; employees in organisations

✓ **Examiner:** *A link shared in WhatsApp groups is a non-probability design, whatever the sample size. Describe it honestly (D3).*

Protecting data quality in a survey

- **Pretest and pilot** before fieldwork: cognitive interviews, then a pilot of 20–40 people outside the sample (E1).
- **Train enumerators** and give them a field manual; check a sample of their questionnaires.
- **Build checks into online forms**: valid ranges, required answers for key items, skip logic, an attention-check item.
- **Send reminders**: two or three reminders typically raise response rates noticeably.
- **Record the paradata**: start and finish times, mode, enumerator ID. They help you detect speeders and interviewer effects.

Interviews

Structured — The same questions in the same order; close to a spoken questionnaire.

Semi-structured — A guide of main questions and follow-up probes, with freedom to pursue what the participant raises. The usual choice in economics and business theses.

Unstructured — A conversation around a topic; used in exploratory and ethnographic work.

Kallio et al. (2016) give a five-phase framework for building a semi-structured interview guide. Record interviews (with consent), transcribe them, and analyse them systematically, for example with Braun and Clarke's (2006) thematic analysis. For the number of interviews, justify it by saturation (Guest et al., 2006) or by information power (Malterud et al., 2016), not by a round number.

Focus group discussions

An FGD is a moderated discussion among six to ten people who share a relevant characteristic, such as B40 mothers, SME owners in Kedah or first-time flood victims. The interaction is the point: participants react to each other. Run several groups so you are not relying on one

group's dynamics, and keep groups homogeneous on the dimension that matters (Morgan, 1996).

? **Examiner:** *Did one FGD of eight people give you "the community's view"? Say how many groups, who was in them, and how you handled dominant speakers.*

Key takeaways

1. Surveys measure and count; interviews and FGDs explain.
2. The survey mode shapes coverage, response and bias: report it and its consequences.
3. For qualitative work, justify the number of interviews or groups and describe the analysis method.

References cited

Braun, V., & Clarke, V. (2006). Using thematic analysis in psychology. *Qualitative Research in Psychology, 3*(2), 77–101.

Guest, G., Bunce, A., & Johnson, L. (2006). How many interviews are enough? An experiment with data saturation and variability. *Field Methods, 18*(1), 59–82.

Kallio, H., Pietilä, A.-M., Johnson, M., & Kangasniemi, M. (2016). Systematic methodological review: Developing a framework for a qualitative semi-structured interview guide. *Journal of Advanced Nursing, 72*(12), 2954–2965.

Malterud, K., Siersma, V. D., & Guassora, A. D. (2016). Sample size in qualitative interview studies: Guided by information power. *Qualitative Health Research, 26*(13), 1753–1760.

Morgan, D. L. (1996). Focus groups. *Annual Review of Sociology, 22*, 129–152.

Example

Worked example: a mixed-methods flood study

Research problem: *Why do some flood victims refuse to evacuate?* A design that uses all three methods:

1. **Two FGDs** (village heads; women who stayed at home during the last flood) to learn how evacuation decisions are discussed and which reasons come up: protecting property, distrust of warnings, conditions at relief centres.
2. **A face-to-face household survey** of flood-affected households, sampled from district relief lists, measuring evacuation (yes/no), prior flood experience, notification, awareness programmes and distance to the evacuation centre, as in Soon et al. (2018).
3. **Twelve semi-structured interviews** with households that did not evacuate despite being notified, to explain a pattern the survey cannot.

Chapter 3 would report each strand: its sample and how it was chosen, its instrument, its fieldwork, and how the strands connect. In this design, the FGDs inform the questionnaire items and the interviews explain survey results.

Writeup

How it reads in the thesis

3.1.3 Qualitative data collection

Twelve households that did not evacuate despite receiving an official notice were selected purposively from survey respondents to explain this pattern.[1] Semi-structured interviews followed a guide of six main questions with probes (Appendix C), lasted 35–60 minutes, were conducted in Malay, and were audio-recorded with consent and transcribed verbatim.[2] Recruitment stopped when the last three interviews produced no new reasons for staying (Guest et al., 2006).[3] Transcripts were analysed thematically following Braun and Clarke (2006).

1 *Who, how chosen, and why: tied to the survey finding.*

2 *The instrument, duration, language, recording and transcription.*

3 *A stopping rule with a citation, not just "until saturation".*

Model write-up. *Hypothetical study.*

Worksheet

Worksheet C3

Q1 For each aim, choose a survey, interviews or FGDs, and justify the choice in one sentence: (a) estimate the share of hawkers in Penang who accept DuitNow QR; (b) understand why older hawkers resist QR payments; (c) explore how a residents' association talks about a proposed toll. (6 marks)

Model answer

(a) Survey: a proportion needs a large, preferably probability sample. (b) Semi-structured interviews: reasons and experiences in the hawkers' own words, from a purposive sample of older hawkers. (c) FGD: the aim is the group's shared and contested views, which emerge through interaction.

Q2 Your online survey of SME owners received 180 responses from 1,500 emails (12%). Give two ways the low response could bias your results, and two things you can report or do about it. (4 marks)

Model answer

Bias: owners who are more digitally active, or more interested in the topic, are more likely to reply, overstating digital adoption; very small or struggling firms may be under-represented. Responses: compare respondents with the frame on known characteristics (sector, size, state) and report the comparison; weight to the frame's composition; compare early and late responders as a rough check; state the limitation and restrict generalisation.

C4 Collecting secondary data: where to find it and how to report it

Secondary data were collected by someone else, usually a statistics agency, central bank or international organisation, for their own purposes. They are cheaper and often larger and better sampled than anything a PhD student can collect. The cost is that you must understand exactly how they were produced.

About 25 minutes | Directory of 20 verified sources | How to cite a dataset

After this section you can:

1. find a suitable public dataset for an economics research question;
2. evaluate a dataset's fitness for your question before you commit to it;
3. cite and describe datasets so another researcher can retrieve the same data.

Before you commit: six questions about any dataset

1. **Who collected it, and why?** An administrative register, a household survey and a model-based estimate have very different error properties.
2. **How was it collected?** For surveys: the sampling design, sample size, mode and response rate. Most national household surveys need their weights, and often their strata and clusters, in your estimation.
3. **Does its definition match your concept?** “Unemployment” in the WDI is a modelled ILO estimate; in DOSM's Labour Force Survey it is a survey estimate. They can differ.
4. **What is the coverage?** Years, countries or states, gaps, and breaks in series when methods change.
5. **Which version?** Datasets are revised. Penn World Table results can change between versions (Johnson et al., 2013). Record the version and the date you downloaded it.
6. **Can you use it?** The licence, any registration, restricted microdata agreements, and the time they take to approve.

✓ **Examiner:** *Large surveys are not immune to error. Response rates and reporting quality in major household surveys have been falling (Meyer et al., 2015). Say what you know about your dataset's quality.*

A directory of public data sources

Every link below was checked when this guide was built. Portals reorganise; if a link moves, search for the dataset's name.

Table C4.1. Data sources for economics theses, with a Malaysian emphasis

Source	What it offers	Typical thesis use
Malaysia		
OpenDOSM (Department of Statistics Malaysia)	Official statistics with downloadable tables: CPI, labour force, GDP, trade, household income, population	State or national time series; descriptive context
DOSM main site	Publications and methodology reports; microdata requests for surveys such as HIES and the Labour Force Survey	Household microdata (by application)
data.gov.my	Malaysia's open data portal across ministries	Administrative series (education, health, transport)
Bank Negara Malaysia statistics and BNM OpenAPI	Monetary, banking, interest-rate and exchange-rate data	Monetary policy, exchange rates, credit
International macroeconomic data		
World Bank World Development Indicators	About 1,400 indicators for over 200 economies from 1960	Cross-country panels; growth, poverty, trade
IMF Data and the World Economic Outlook databases	Balance of payments, financial statistics, fiscal data, WEO forecasts	Macro and international finance
FRED (Federal Reserve Bank of St. Louis)	Hundreds of thousands of US and international time series, with an easy download and API	Time series; global variables such as oil prices and US rates
Penn World Table	Real GDP, capital, productivity and prices for about 180 countries	Growth accounting and convergence (Feenstra et al., 2015)
ILOSTAT	Labour statistics: employment, wages, informality	Labour markets across countries
OECD Data Explorer	OECD statistics on economy, education, tax and trade	Comparisons with OECD economies
ADB Key Indicators Database	Asia–Pacific economic and social indicators	Regional panels for Asia
UN Comtrade	Detailed bilateral trade by product	Trade flows; gravity models
Household, individual and firm microdata		
Global Findex	Individual-level survey of account ownership, saving, borrowing and digital payments in more than 120 economies	Financial inclusion; digital payments
World Bank Enterprise Surveys	Firm-level surveys on finance, regulation, innovation	Firm performance and constraints

Source	What it offers	Typical thesis use
IPUMS International	Harmonised census microdata from many countries	Migration, education, labour across countries
Demographic and Health Surveys	Household surveys on health, fertility and wealth in developing countries	Health and development economics
Luxembourg Income Study	Harmonised household income microdata (remote access)	Inequality and poverty comparisons
UNU-WIDER World Income Inequality Database	Gini coefficients and income shares across countries and years	Inequality panels
Values, education and human capital		
World Values Survey	Individual-level values, trust, life satisfaction, in waves since 1981	Trust, culture and economic outcomes
Asian Barometer	Political attitudes and values in East and Southeast Asia	Institutions and attitudes
OECD PISA data	Student test scores and background questionnaires for 15-year-olds; Malaysia participates	Education production functions
Barro–Lee educational attainment	Years of schooling by country, five-year intervals	Human capital in growth regressions
Our World in Data	Charts and cleaned series compiled from primary sources	Exploration. Cite the original source it draws on, not the aggregator

Citing a dataset

Treat a dataset like a publication: author (the organisation), year, title, version, "[Data set]", publisher and URL, with the access date for data that change.

World Bank. (2026). *World Development Indicators* [Data set]. https://databank.worldbank.org/source/world-development-indicators (accessed 14 January 2026).

Penn World Table: cite the version you used and its documentation paper (Feenstra et al., 2015, for version 8 onwards), and record the version number, for example PWT 10.01 or 11.0.

Combining datasets

Theses often merge sources, such as WDI controls with PWT productivity. Merge on standard identifiers (ISO 3-letter country codes and year; DOSM state codes), check how many observations matched, and report the final sample.

(example)

```
use wdi_extract.dta, clear                    // iso3 year inflation unemp growth
merge 1:1 iso3 year using pwt_extract.dta     // iso3 year rgdpna ctfp
tabulate _merge                               // 3 = matched in both
keep if _merge == 3
drop _merge
```

```
wdi = pd.read_csv("wdi_extract.csv")          # iso3, year, inflation, unemp, growth
pwt = pd.read_csv("pwt_extract.csv")          # iso3, year, rgdpna, ctfp
m = wdi.merge(pwt, on=["iso3", "year"], how="outer", indicator=True, validate="one_to_one")
print(m["_merge"].value_counts())             # both / left_only / right_only
panel = m[m["_merge"] == "both"].drop(columns="_merge")
```

? **Examiner:** *How many country-years were lost in the merge, and are the missing ones random? Report it.*

Key takeaways

1. Know who collected the data, how, and whether the definitions match your concepts.
2. Record version and access date; cite datasets like publications.
3. Report merges and the observations they cost.

References cited

Feenstra, R. C., Inklaar, R., & Timmer, M. P. (2015). The next generation of the Penn World Table. *American Economic Review, 105*(10), 3150–3182.

Johnson, S., Larson, W., Papageorgiou, C., & Subramanian, A. (2013). Is newer better? Penn World Table revisions and their impact on growth estimates. *Journal of Monetary Economics, 60*(2), 255–274.

Meyer, B. D., Mok, W. K. C., & Sullivan, J. X. (2015). Household surveys in crisis. *Journal of Economic Perspectives, 29*(4), 199–226.

Example

Worked example: choosing data for a financial-inclusion thesis

Research question: *Is mobile-money adoption associated with higher saving among adults in ASEAN economies?*

Table C4.2. Candidate datasets, judged against the question

Dataset	Fit	Problem	Decision
Global Findex microdata	Individual adults; saving and digital-payment questions; several waves	Repeated cross-sections, not a panel; about 1,000 respondents per country-wave	Main dataset; pooled cross-sections with country and wave fixed effects
WDI country indicators	Country-level controls (GDP per capita, inflation)	Aggregates only	Merge as country-year controls
DOSM HIES microdata	Rich Malaysian household data	One country only; access by application	Robustness check for Malaysia if approved in time

The design follows from the data: Findex is a set of repeated cross-sections, so the model is $save_{ict} = \beta\, mobile_{ict} + \mathbf{x}'_{ict}\boldsymbol{\gamma} + \mu_c + \lambda_t + u_{ict}$, with individual i, country c and wave t, estimated with Findex's survey weights and standard errors clustered by country.

Worksheet

Worksheet C4

Q1 Suggest a dataset for each question and state its unit and structure: (a) Did Malaysian state-level unemployment respond differently to the 2020 lockdown? (b) Does trust in institutions predict support for tax compliance across countries? (c) Has Malaysia's export structure diversified since 2000? (6 marks)

Model answer

(a) DOSM Labour Force Survey state estimates (via OpenDOSM): state-quarter or state-year panel. (b) World Values Survey: individuals in many countries across waves, so pooled cross-sections, with country effects. (c) UN Comtrade: Malaysia's exports by product and partner per year, a time series of product-level cross-sections; compute a diversification index each year.

Q2 Write a dataset citation and a one-sentence description for the Penn World Table version you would use. (3 marks)

Model answer

“Feenstra, R. C., Inklaar, R., & Timmer, M. P. (2015). The next generation of the Penn World Table. American Economic Review, 105(10), 3150–3182; data: Penn World Table version 11.0, Groningen Growth and Development Centre, https://www.rug.nl/ggdc/productivity/pwt/ (accessed date).” Description: “PWT provides PPP-adjusted real GDP, capital stocks and total factor productivity for about 180 countries; the study uses real GDP at constant national prices (rgdpna) for 1990–2019.”

C5 Data cleaning: inspecting, correcting and transforming

Raw data are never ready for analysis. Cleaning means inspecting the data, finding problems, correcting or flagging them by rules you can defend, and creating the variables your models need, all in a script that turns the raw file into the analysis file.

About 30 minutes | Full worked example: 414 raw records to 400 clean

Runnable do-file and notebook

After this section you can:

1. run a standard sequence of inspection and cleaning checks;
2. make and document cleaning decisions that an examiner will accept;
3. report data preparation briefly and completely in Chapter 3.

Why clean, and why with a script

Errors in raw data feed straight into your estimates: a keying error of RM 98,000 for a student's monthly allowance shifts the mean and can dominate a regression. A cleaning *script* (a do-file or notebook) makes every decision visible and repeatable. You can rerun it when you find a new problem, your supervisor can check it, and you can describe it accurately in Chapter 3. Editing cells by hand in Excel leaves no record.

✓ **Examiner:** *Keep the raw file untouched. Save the cleaned data under a new name, and keep the script that links the two (Christensen & Miguel, 2018).*

The cleaning checklist

Table C5.1. Data cleaning tasks, their purpose and Stata commands

Task	Purpose	Stata commands, for example
Inspect structure	Understand variables, labels, storage types, missingness	`describe`; `codebook`; `summarize`; `inspect`
Check duplicates	Detect repeated IDs or accidental duplicate records	`duplicates report id`; `duplicates list id`
Check missing values	Identify item non-response and patterns of missingness	`misstable summarize`; `tab var, missing`
Check outliers	Identify implausible values	`summ var, detail`; `graph box var`
Check coding consistency	Ensure categories match the codebook	`tab gender`; `tab education, missing`
Recode variables	Align categories and reference groups	`recode educ (1/2=0) (3/4=1), gen(high_educ)`
Generate new variables	Create dummies, logs, lags, interactions, scale scores	`egen score = rowmean(q1 q2 q3)`; `gen ln_income = ln(income)`
Save the data, original and cleaned	Preserve the raw data and create a reproducible cleaned file	`save clean_data.dta, replace`; also save the do-file used to clean the dataset

Rules for cleaning decisions

Missing-value codes Convert codes such as −9 or 999 to true missing before computing anything (`mvdecode`).

Impossible values Values outside the allowed range (a 6 on a 1–5 scale; 10 days in a week) are errors. Set them to missing unless the original form shows the true value.

Implausible values Possible but suspicious (an allowance of RM 98,000). Check the source; if it cannot be verified, set it to missing, winsorise or top-code, and say which. Do not drop real but unusual observations just because they are extreme; robust outlier rules help to flag them (Leys et al., 2013).

Inconsistent answers Skip-pattern contradictions (a non-user reporting days of use) are resolved by a stated rule, applied to everyone.

Incomplete responses Decide a threshold in advance, for example dropping respondents who answered fewer than half of the scale items.

Missing data	Report how much is missing for each variable. Listwise deletion is the default; for substantial missingness, consider multiple imputation and explain the assumption (missing at random) it requires (Schafer & Graham, 2002).

Key takeaways

1. Clean with a script, never by hand; keep raw and clean files separate.
2. Every change follows a stated rule applied to all observations.
3. Report the path from raw records to the analysis sample, with numbers.

References cited

Christensen, G., & Miguel, E. (2018). Transparency, reproducibility, and the credibility of economics research. *Journal of Economic Literature, 56*(3), 920–980.

Leys, C., Ley, C., Klein, O., Bernard, P., & Licata, L. (2013). Detecting outliers: Do not use standard deviation around the mean, use absolute deviation around the median. *Journal of Experimental Social Psychology, 49*(4), 764–766.

Schafer, J. L., & Graham, J. W. (2002). Missing data: Our view of the state of the art. *Psychological Methods, 7*(2), 147–177.

Example

Worked example: cleaning the e-wallet survey

The raw file `ewallet_survey_raw.csv` (hypothetical, in the downloads) has 414 records with the problems students meet in real surveys. Running `03_data_cleaning.do`, or its notebook, finds and fixes them.

Table C5.2. Problems found and decisions taken

Check	Found	Decision	Records left
Raw records			414
Exact duplicates	6 records submitted twice	Drop the second copy	408
Break-offs	8 respondents answered none of the 15 scale items	Drop (rule: fewer than half answered)	400
Missing-value codes	age = 999 (3); allowance = −9 (4); scale items = −9 (10 cells)	Set to missing	400
Out-of-range scale values	5 cells coded 0, 6 or 9 on 1–5 items	Set to missing	400
Implausible values	age = 150 (1); allowance RM 98,000 and RM 44,000 (2)	Set to missing (could not be verified)	400
Inconsistent text	Gender written 12 ways (“F”, “female”, “Male ”…); state names in mixed case and with spaces	Standardise; create dummy `female`	400
Impossible values	Days of use in past week = 8, 9, 10 (3)	Set to missing	400
Skip-pattern contradiction	4 non-users reported days of use	Set days to missing	400

03_data_cleaning.do

```
import delimited using "$data/ewallet_survey_raw.csv", clear varnames(1) case(preserve)

duplicates report                          // identical on every variable?
duplicates drop

egen n_missing = rowmiss(PU1-PU4 PS1-PS4 EOU1-EOU3 intention finlit1-finlit3)
drop if n_missing >= 8                     // fewer than half of the 15 scale items answered

mvdecode age, mv(999)
mvdecode allowance_rm PU1-PU4 PS1-PS4 EOU1-EOU3 intention, mv(-9)
foreach v of varlist PU1-PU4 PS1-PS4 EOU1-EOU3 intention {
    quietly replace `v' = . if !inrange(`v', 1, 5)
}
replace age = . if !inrange(age, 17, 40)
replace allowance_rm = . if allowance_rm > 10000

generate byte female = .
replace female = 1 if inlist(lower(strtrim(gender)), "f", "female")
replace female = 0 if inlist(lower(strtrim(gender)), "m", "male")
replace state = proper(strtrim(state))

replace ewallet_days = . if ewallet_days > 7
replace ewallet_days = . if ewallet_user == 0 & ewallet_days > 0

egen PU = rowmean(PU1-PU4)
generate ln_allow = ln(allowance_rm)
save "$data/ewallet_survey_clean.dta", replace


raw = load("ewallet_survey_raw.csv")
df = raw.drop_duplicates().copy()

n_missing = df[scale].isna().sum(axis=1)
df = df[n_missing < 8].copy()              # fewer than half of the 15 scale items answered

df["age"] = df["age"].replace(999, np.nan)
for v in ["allowance_rm"] + items + ["intention"]:
    df[v] = df[v].replace(-9, np.nan)
for v in items + ["intention"]:
    df.loc[~df[v].between(1, 5), v] = np.nan
df.loc[~df["age"].between(17, 40), "age"] = np.nan
df.loc[df["allowance_rm"] > 10000, "allowance_rm"] = np.nan

g = df["gender"].str.strip().str.lower()
df["female"] = np.select([g.isin(["f", "female"]), g.isin(["m", "male"])], [1, 0], default=np.nan)
df["state"] = df["state"].str.strip().str.title()

df.loc[df["ewallet_days"] > 7, "ewallet_days"] = np.nan
df.loc[(df["ewallet_user"] == 0) & (df["ewallet_days"] > 0), "ewallet_days"] = np.nan

df["PU"] = df[["PU1", "PU2", "PU3", "PU4"]].mean(axis=1)
df["ln_allow"] = np.log(df["allowance_rm"])
```

Both versions end with the same 400 records and the same means (for example, 58.25% female and 75.25% e-wallet users); the notebook checks this against Stata automatically.

Writeup

How it reads in the thesis

3.1.5 Data preparation

All cleaning was carried out in a Stata do-file that reads the raw export and writes a separate analysis file (Appendix D).[1] Of 414 submissions, six exact duplicates and eight break-offs (respondents answering fewer than half of the scale items) were removed, leaving 400.[2] Missing-value codes, out-of-range responses on the five-point items (five cells), an age of 150 and two monthly allowances above RM 10,000 that could not be verified were set to missing. Days of e-wallet use were set to missing where they exceeded seven (three cases) or contradicted a report of non-use (four cases).[3] Item non-response is below 2% for every variable, so models use complete cases; the logit sample is 394 because six allowances are missing.[4]

1 *Reproducibility: a script, raw and clean files kept separate.*

2 *The path from submissions to the analysis sample, with reasons.*

3 *Each rule, and how many cases it affected.*

4 *How remaining missing data are handled, and its effect on N.*

Model write-up. *Numbers are from the worked example.*

Worksheet

Worksheet C5

Q1 Download `ewallet_survey_raw.csv` and run the do-file or notebook. Then add one more check: list respondents whose `ewallet_spend_rm` is positive although they are non-users. How many are there, and what rule would you apply? (4 marks)

Model answer

Stata: `count if ewallet_user == 0 & ewallet_spend_rm > 0 & !missing(ewallet_spend_rm)`. In this dataset the answer is zero: spending was generated only for users. In real data, apply the same rule as for days of use (set spending to missing for contradictory non-users), or check the skip logic of the online form. The point is to write the rule before looking at results.

Q2 A student drops every observation with income more than three standard deviations above the mean. Give two objections and a better approach. (4 marks)

Model answer

Objections: (1) the mean and SD are themselves inflated by the outliers, so the rule is unstable; Leys et al. (2013) recommend the median absolute deviation instead; (2) high incomes may be genuine, and dropping them biases the sample and changes the population studied. Better: check suspicious values against the source; set verified errors to missing; for genuine extremes, use logs, top-coding or winsorising, and show results are robust with and without them.

Q3 Write the code to create `high_educ` = 1 for degree holders (education codes 3 and 4) and 0 for codes 1 and 2, keeping missing values missing, in Stata and in Python. (2 marks)

Model answer

Stata: `recode educ (1/2 = 0) (3/4 = 1), generate(high_educ)` (missing stays missing).
Python: `df["high_educ"] = df["educ"].map({1: 0, 2: 0, 3: 1, 4: 1})` (unmapped and missing become NaN).

D1 Population, sampling frame and sample

Before choosing a sampling design you must be precise about whom your findings describe, which list you can actually draw from, and the units you select. Most sampling problems that examiners find start here, not in the formula.

About 15 minutes | Worksheet with model answers

After this section you can:

1. distinguish population, target population, sampling frame, sampling unit and sample;
2. identify coverage problems in a sampling frame;
3. separate sampling error from non-sampling error.

From population to sample

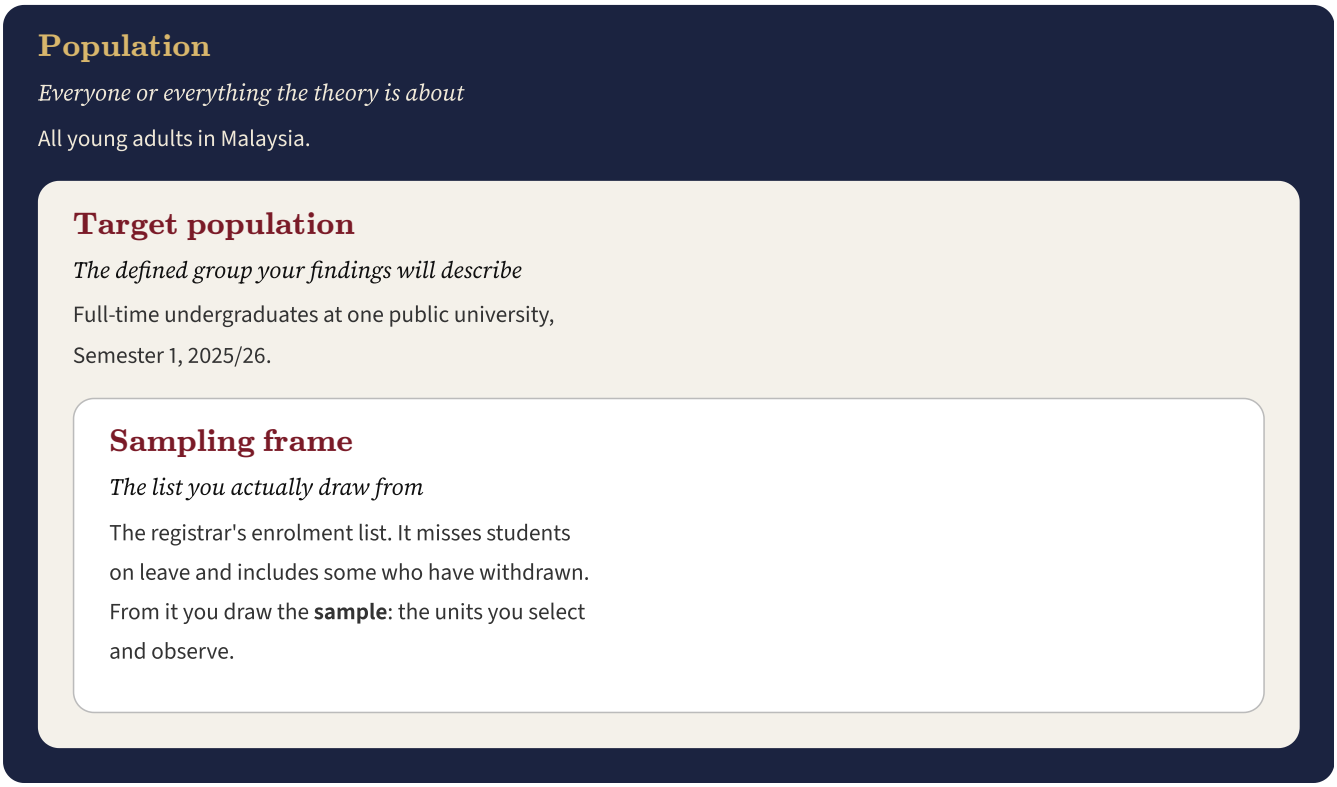


Figure D1.1. *Each layer is narrower than the one around it. Generalising from the sample to the target population needs a good frame and a good design. Going on to the wider population is an argument, not a statistical inference.*

Population	The full set of units the research question concerns.
Target population	The population defined precisely by who, where and when. Your conclusions apply here.
Sampling frame	A list or procedure that identifies the units you can select: enrolment records, the SSM company register, DOSM enumeration blocks, a membership list.
Element (unit of analysis)	The unit you measure and analyse: a student, a household, a firm.

Sampling unit	The unit you select. It can differ from the element: you may select villages, then households within them.
Sample	The units selected. The *achieved* sample is the subset who actually respond.
Census	Observing every unit in the target population, which is sensible when it is small (for example, every licensed commercial bank).

Frame problems

- **Undercoverage**: units missing from the frame. An online panel misses people without internet; a telephone directory misses mobile-only households.
- **Overcoverage**: ineligible units on the frame, such as closed firms still registered, or graduates still on an alumni list.
- **Duplicates**: units listed twice have twice the chance of selection.
- **Clustering**: one frame entry covers several elements, such as one address with several households.

? **Examiner:** *"The population is all SMEs in Malaysia." Which list did you sample from, and who is not on it?*

Two kinds of error

Table D1.1. Sampling and non-sampling error

Error	Cause	Shrinks with a larger sample?	What helps
Sampling error	Chance: observing a sample, not everyone	Yes, in proportion to $1/\sqrt{n}$	Larger probability samples; stratification
Coverage error	The frame misses or adds units	No	A better frame; weighting to known totals
Non-response error	Those who respond differ from those who do not	No	Reminders, incentives; non-response analysis (Groves, 2006)
Measurement error	Questions measure poorly; recall; social desirability	No	Good questionnaire design and pretesting (E1)

A large sample cures only the first row. A survey of 5,000 self-selected respondents can be more biased than a well-run random sample of 400.

Key takeaways

1. Define the target population by who, where and when; name the frame and its gaps.
2. The sampling unit and the unit of analysis can differ.
3. Only sampling error falls as n rises; coverage, non-response and measurement error do not.

References cited

Groves, R. M. (2006). Nonresponse rates and nonresponse bias in household surveys. *Public Opinion Quarterly, 70*(5), 646–675.

Worksheet

Worksheet D1

Q1 For the study "Digital payment adoption among hawkers in George Town", define the target population, a feasible sampling frame, the sampling unit and the unit of analysis. Identify one coverage problem with your frame. (5 marks)

Model answer

Target population: licensed hawkers operating within the George Town municipal area in 2026. Frame: the city council's hawker licence register. Sampling unit: the licensed stall (from the register). Unit of analysis: the hawker (stall operator). Coverage problem: unlicensed hawkers are missing (undercoverage), and licences held for stalls no longer trading are included (overcoverage). Describe both and check eligibility at the start of each interview.

Q2 A student argues: "My online survey has 2,000 responses, so sampling error is tiny and my results represent all Malaysian youth." Evaluate the claim. (4 marks)

Model answer

A large n reduces sampling error only for a probability sample, where sampling error is defined. A self-selected online sample has undercoverage (youth without internet or outside the sharing networks) and self-selection bias, neither of which falls with n. The results describe the respondents and perhaps similar young people, not all Malaysian youth. Comparing the sample with national demographics and weighting can help, but cannot remove bias on unmeasured characteristics.

D2 Probability sampling: simple random, systematic, stratified

In a probability sample every member of the target population has a known, non-zero chance of selection. That known chance is what allows formal statistical inference: sampling distributions, standard errors and confidence intervals all assume random selection.

About 35 minutes | Live sampling simulator | Allocation calculator | Stata and Python code

After this section you can:

1. explain and carry out simple random, systematic and stratified random sampling;
2. allocate a stratified sample proportionately or disproportionately, and weight it back;
3. choose and justify a design for your own study.

1. Simple random sampling (SRS)

Every unit has an equal probability of selection, n/N, and every possible sample of size n is equally likely. Draw it with a random number generator applied to the sampling frame, or, for a small frame, by drawing paper slips out of a bowl.

- **Needs:** a complete list of the population.
- **Strengths:** simple; unbiased; the textbook formulas for standard errors apply directly.
- **Weaknesses:** small subgroups may be under-represented by chance; can be costly when units are spread out.

2. Systematic random sampling

Select every k-th unit from the frame after a random start between 1 and k, where $k = N/n$ (rounded down). With $N = 2{,}400$ and $n = 331$, $k = 7$; a random start of 2 selects units 2, 9, 16, 23 and so on.

- **Strengths:** easy in the field, such as every 7th customer leaving a shop or every 10th house on a street; spreads the sample evenly through the list.
- **Weakness:** vulnerable to bias when the list has a pattern that matches k, for example if every 10th house on a street is a corner house, or every 7th day is a Sunday.

✓ **Examiner:** *Report the interval and the random start, and say how the frame was ordered.*

3. Stratified random sampling

Partition the population into non-overlapping strata (for example urban/rural, ethnic group, industry, faculty) and draw a separate random sample within each. Stratify on characteristics that are related to your outcome, because that is what improves precision.

Proportionate allocation	Each stratum's sample size is proportional to its population size: $n_h = n \times N_h/N$. Every unit has the same selection probability, so no weights are needed.
Disproportionate allocation	Small strata are over-sampled to improve precision for subgroup estimates. Selection probabilities differ, so population estimates need design weights $w_h = N_h/n_h$.

Table D2.1. Proportionate and disproportionate stratified random sampling. Adapted from Sekaran and Bougie (2016, Table 13.1)

Job level	Number of elements	Proportionate sample (20% of elements)	Disproportionate sample
Top management	10	2	7
Middle-level management	30	6	15
Lower-level management	50	10	20
Supervisors	100	20	30
Clerks	500	100	60
Secretaries	20	4	10
Total	710	142	142

Both columns sample 142 people. The proportionate column gives top management only 2 people, too few to say anything about them; the disproportionate column gives 7, at the cost of needing weights for organisation-wide estimates. Chapter numbers differ between editions of Sekaran and Bougie.

4. Cluster and multistage sampling

When there is no list of individuals, or they are spread over a wide area, select *groups* first (villages, schools, enumeration blocks) and then units within them. National household surveys typically use multistage stratified designs, selecting areas first and then living quarters within them. Cluster samples are cheaper per interview but less precise than an SRS of the same size, because units in a cluster resemble each other. The design effect quantifies the loss (D4).

Choosing a design

Table D2.2. Probability designs compared

Design	Needs	Best when	Watch out for
Simple random	Full list	Population is fairly homogeneous; list available	Small subgroups under-represented by chance
Systematic	Ordered list, or a flow of units	Field selection (customers, houses)	Periodicity in the list
Stratified, proportionate	List with stratum information	Strata differ on the outcome	Too few units in small strata
Stratified, disproportionate	As above	Subgroup comparisons are a research question	Must weight for population estimates
Cluster / multistage	List of clusters only	Widely spread populations; no individual list	Design effect; clustered standard errors

Key takeaways

1. Probability sampling means known, non-zero selection probabilities, which makes inference possible.
2. Stratify on variables related to the outcome; allocate disproportionately when subgroups matter, then weight.
3. Report the frame, design, allocation, random seed or start, and weights.

References cited

Sekaran, U., & Bougie, R. (2016). *Research methods for business: A skill-building approach* (7th ed.). Wiley.

Example

Worked example: sampling 331 SMEs from a frame of 2,400

A hypothetical frame lists 2,400 SMEs in four northern states: 1,560 in services, 480 in manufacturing, 240 in agriculture and 120 in construction. The target is $n = 331$, the Krejcie–Morgan size for $N = 2{,}400$. The question is what share of SMEs accept e-wallet payments.

Table D2.3. Allocations for n = 331

Sector	N_h	Proportionate n_h	Disproportionate n_h	Weight N_h/n_h
Services	1,560	215	141	11.06
Manufacturing	480	66	70	6.86
Agriculture	240	33	60	4.00
Construction	120	17	60	2.00
Total	2,400	331	331	

With proportionate allocation, construction gets only 17 firms, too few for a separate estimate. The disproportionate design guarantees at least 60 per sector. Because acceptance differs by sector, from 55% in services to 15% in agriculture in this simulated frame, over-sampling small sectors distorts the overall estimate unless you weight:

Table D2.4. Share of SMEs accepting e-wallets (simulated frame)

Estimate	Value
True share in the frame	0.457
Disproportionate sample, unweighted	0.396
Disproportionate sample, weighted by N_h/n_h	0.467

01_sampling.do

```
use "$data/sme_sampling_frame.dta", clear
set seed 20261019                          // makes every draw reproducible: report it

* A. Simple random sampling
sample 331, count

* B. Systematic: interval k and a random start
sort sme_id
local k = floor(_N / 331)                  // k = 7
local start = runiformint(1, `k')
generate byte in_sys = (_n >= `start') & mod(_n - `start', `k') == 0

* C. Stratified, proportionate: n_h = n * N_h / N
bysort sector: generate N_h = _N
generate n_prop = round(331 * N_h / _N)
generate double u = runiform()
bysort sector (u): generate byte in_prop = _n <= n_prop

* D. Disproportionate, with design weights
generate double w_dis = N_h / n_dis
mean uses_ewallet_payments [pweight = w_dis] if in_dis
```

```
frame = load("sme_sampling_frame.csv")
N, n = len(frame), 331

srs = frame.sample(n=n, random_state=20261019)            # A. simple random

k = N // n                                                 # B. systematic, k = 7
start = int(rng.integers(1, k + 1))
sys_sample = frame.sort_values("sme_id").iloc[start - 1::k].head(n)

alloc = frame.groupby("sector").size().rename("N_h").to_frame()
alloc["n_prop"] = (n * alloc["N_h"] / N).round().astype(int)   # C. proportionate
prop = pd.concat([g.sample(n=alloc.loc[s, "n_prop"], random_state=1) for s, g in frame.groupby("sector")])

dis["w"] = dis["sector"].map(alloc["N_h"] / alloc["n_dis"])    # D. design weights
np.average(dis["uses_ewallet_payments"], weights=dis["w"])
```

Writeup

How it reads in the thesis

3.1.1 Sampling design

The sampling frame was the 2025 list of 2,400 SMEs registered with the state SME agencies of Kedah, Perlis, Pulau Pinang and Perak.[1] Because e-wallet acceptance was expected to differ by sector, and sectoral comparisons are a research question (RQ2), a stratified random sample was drawn with sector as the stratifying variable and disproportionate allocation, guaranteeing at least 60 firms per sector.[2] Within each sector, firms were selected by simple random sampling using Stata's random-number generator (seed 20261019).[3] Estimates for all SMEs use design weights equal to the inverse of each sector's sampling fraction (Table 3.2).[4]

1 *The frame, named, dated and sized.*

2 *The design, with a justification tied to a research question.*

3 *How selection was randomised, reproducibly.*

4 *The consequence for estimation: weights.*

Model write-up. *Hypothetical study.*

Worksheet

Worksheet D2

Q1 A frame lists 6,000 households: 3,600 urban and 2,400 rural. You need n = 361. (a) Give the proportionate allocation. (b) You want equal precision for urban and rural estimates: give an allocation and the design weights. (6 marks)

Model answer

(a) Urban: 361 × 3,600/6,000 = 216.6, so 217; rural: 361 × 2,400/6,000 = 144.4, so 144 (total 361). (b) Equal allocation: 181 urban and 180 rural. Weights: urban 3,600/181 = 19.89; rural 2,400/180 = 13.33. Each sampled urban household represents about 20 households and each rural household about 13.

Q2 You plan to survey every 8th customer leaving a supermarket between 9 a.m. and 9 p.m. on weekdays. Is this systematic sampling? What could bias it? (4 marks)

Model answer

It is systematic selection from a flow of customers, which works well in the field, but the population covered is only weekday shoppers at that supermarket. Biases: weekend shoppers and working people who shop in the evening or at weekends are under-represented; groups arriving together may all be skipped or all selected. It is a probability sample of customer visits on surveyed days, not of households. Choose a random start, sample different days and times, and state the target population precisely.

Q3 Use the simulator: draw ten proportionate stratified samples and ten simple random samples of n = 20. How does the number of agriculture firms vary across draws under each design, and why? (3 marks)

Model answer

Proportionate stratified sampling always gives 3 agriculture firms (20 × 30/200). SRS gives a varying number, typically 1 to 6, sometimes 0, because the stratum count is left to chance. Stratification fixes each stratum's share and so removes this source of sampling variation.

D3 Non-probability sampling: convenience, purposive, snowball

In non-probability sampling, selection probabilities are unknown, or zero for some units. These designs are used when no sampling frame exists, when the population is hidden, or when the aim is depth rather than statistical generalisation. Findings are hard to generalise, and Chapter 3 must say so honestly.

About 20 minutes | Four designs with economics examples

After this section you can:

1. explain convenience, purposive, snowball and quota sampling;
2. judge when a non-probability design is defensible;
3. describe one in Chapter 3 without over-claiming.

1. Convenience sampling

Recruiting whoever is available and willing: students in your class, shoppers at a mall, people who click a link shared on social media.

- **Use it for:** pilots, pretests, exploratory studies, testing an instrument.
- **Risk:** volunteers and the easily reached differ systematically from the rest. A bigger convenience sample is a bigger biased sample.

2. Purposive (judgemental) sampling

Deliberately selecting units that meet specific criteria, because they hold the information the study needs: female CEOs, judges, central bank economists, SME owners who stopped accepting cash, households flooded three times. Palinkas et al. (2015) describe variants:

- **Expert sampling**: people with specialist knowledge.
- **Maximum variation**: cases that differ widely, to capture the range of experience.
- **Typical case**: cases that represent the usual situation.
- **Critical or extreme case**: cases where a phenomenon is most visible.

Purposive sampling is the standard choice for interviews and case studies. State the selection criteria precisely so readers can judge them.

3. Snowball sampling

Existing participants recruit further participants from their networks. It is essential for stigmatised, hidden or illegal populations with no list: undocumented migrant workers, borrowers from unlicensed moneylenders, drug users, the homeless (Biernacki & Waldorf, 1981).

- **Risk:** samples reflect the starting “seeds” and their networks; well-connected people are over-represented.
- **Improvement:** start from several diverse seeds, limit referrals per person, and record who referred whom. Respondent-driven sampling formalises this to allow some population estimates (Heckathorn, 1997).

4. Quota sampling

Fill quotas that match known population proportions, for example 50% women, or ethnic shares from the census, but select people within each quota by convenience. The sample *looks* representative on the quota variables, yet selection within quotas is not random. Quota sampling is common in market research and opinion polling.

? **Examiner:** *You matched the census on gender and ethnicity. Does that make the sample representative of attitudes to e-wallets? Only on those two variables, not on anything else.*

When is a non-probability design defensible?

Table D3.1. Matching the design to the purpose

Purpose	Defensible?	What to write
Depth: why and how (interviews, FGDs)	Yes, usually purposive	Selection criteria; how many and why (saturation or information power)
Hidden population with no frame	Yes, snowball or respondent-driven	Seeds, referral chains, limits on generalisation
Testing relationships (theory testing)	Sometimes	Argue why the relationship should hold beyond the sample; compare the sample's profile with the population
Estimating a population proportion or mean	Weakly	Avoid design-based margins of error; any model-based adjustment needs stated assumptions (Baker et al., 2013)

Conventional margins of sampling error do not apply to non-probability samples (Baker et al., 2013). Do not report "±5% at 95% confidence" for a convenience sample, and do not cite Krejcie and Morgan as if it made the sample representative (D4).

Key takeaways

1. Convenience for pilots; purposive for depth; snowball for hidden populations; quota for a surface match.
2. Selection probabilities are unknown, so statistical generalisation is limited.
3. Describe the recruitment honestly and restrict your claims.

References cited

Baker, R., Brick, J. M., Bates, N. A., Battaglia, M., Couper, M. P., Dever, J. A., Gile, K. J., & Tourangeau, R. (2013). Summary report of the AAPOR task force on non-probability sampling. *Journal of Survey Statistics and Methodology, 1*(2), 90–143.

Biernacki, P., & Waldorf, D. (1981). Snowball sampling: Problems and techniques of chain referral sampling. *Sociological Methods & Research, 10*(2), 141–163.

Heckathorn, D. D. (1997). Respondent-driven sampling: A new approach to the study of hidden populations. *Social Problems, 44*(2), 174–199.

Palinkas, L. A., Horwitz, S. M., Green, C. A., Wisdom, J. P., Duan, N., & Hoagwood, K. (2015). Purposeful sampling for qualitative data collection and analysis in mixed method implementation research. *Administration and Policy in Mental Health and Mental Health Services Research, 42*(5), 533–544.

Example

Worked examples in economics

Convenience	**Mall-intercept survey of e-wallet use.** 300 shoppers approached at a Penang mall over two weekends. Fast and cheap, but it over-represents urban, mobile, higher-income shoppers. Suitable for piloting the questionnaire, not for estimating national adoption.
Purposive	**Interviews with SME owners who stopped accepting cash.** Twelve owners chosen because they made the switch, varied deliberately by sector and firm size (maximum variation). The study explains *why* firms switch, not how many do.
Snowball	**Borrowers from unlicensed moneylenders (Ah Long).** No list exists and the topic is sensitive. Start from five seeds introduced by community organisations in different districts; each participant may refer up to three others; record the referral chains. Report results as describing these networks.
Quota	**Attitudes to the sales and services tax.** 400 adults recruited online with quotas matching the census on gender, ethnicity and urban/rural residence. The sample matches on those variables only; attitudes may still differ from the population's.

Writeup

How it reads in the thesis

3.1.1 Sampling

Because no list of students who use e-wallets exists and the university did not release enrolment records, respondents were recruited by convenience: the questionnaire link was distributed through faculty student associations and course announcements.[1] This is a non-probability sample, so selection probabilities are unknown, conventional margins of sampling error do not apply, and the results describe the respondents rather than all students (Baker et al., 2013).[2] To gauge how far the sample departs from the population, Table 3.1 compares its composition by faculty, gender and year of study with enrolment statistics; the analysis focuses on relationships between variables rather than on population totals.[3]

1 *Why a non-probability design, and exactly how recruitment worked.*

2 *The consequence for inference, stated plainly and cited.*

3 *What was done to assess and limit the problem.*

Model write-up. *Honest description strengthens, not weakens, the chapter.*

Worksheet

Worksheet D3

Q1 Choose a non-probability design for each, and write the selection criteria: (a) understanding how Rohingya refugees in Malaysia earn a living; (b) how bank economists form inflation expectations; (c) piloting a questionnaire on household debt. (6 marks)

Model answer

(a) Snowball from several community-organisation seeds in different states; criteria: adult refugees who have earned income in Malaysia in the past year. (b) Purposive (expert) sampling; criteria: economists in research or treasury roles at licensed banks with at least three years' experience forecasting. (c) Convenience; criteria: adults with at least one household loan, recruited from the researcher's networks, outside the main sample.

Q2 Find the problems in this sentence: “A total of 384 respondents were selected using convenience sampling, as recommended by Krejcie and Morgan (1970), ensuring the sample is representative of the population with a 5% margin of error.” (4 marks)

Model answer

(1) Krejcie and Morgan give a sample size for a probability sample; they do not recommend convenience sampling. (2) A convenience sample cannot be called representative because of its size. (3) A design-based margin of error does not apply to a non-probability sample. (4) 384 respondents were not “selected”; they volunteered, so describe recruitment accurately. Rewrite to describe the recruitment and justify n by the analysis (power), with an honest statement of limits.

D4 How many respondents do we need?

Is it 30, the “magic number” of the Central Limit Theorem? 384, from the Krejcie–Morgan table? Or does it depend on statistical power? The short answer: a sample size should be justified by the inference you want to make, not by the size of the population or a rule of thumb.

About 45 minutes | Sample-size and power calculators | Stata and Python code

After this section you can:

1. explain why n = 30 is not a sample-size justification;
2. use and critique the Krejcie and Morgan (1970) table;
3. justify a sample size by precision or by power, with adjustments for non-response and design effects.

Is it 30? What the Central Limit Theorem does and does not say

The Central Limit Theorem says that the sampling distribution of a mean approaches a normal distribution as n grows, whatever the shape of the population. The rule of thumb “$n \geq 30$” is a rough guide to when that approximation is good enough for a t-test or a confidence interval on a mean, if the population is not too skewed. For heavily skewed variables such as income, 30 may not be enough even for that.

It says nothing about whether 30 observations are *enough to answer your question*:

- **Precision:** with 30 respondents, a 95% confidence interval for a proportion near 50% is about ±18 percentage points ($1.96\sqrt{0.25/30} = 0.179$).
- **Power:** with 30 per group, a two-sided t-test has only a 48% chance of detecting a medium effect ($d = 0.5$); detecting it with 80% probability needs 64 per group.
- **Regression:** with several predictors, 30 observations leave very few degrees of freedom.

✓ **Examiner:** *“The sample exceeds 30, so the CLT applies” justifies nothing about your sample size. Expect to be asked for a calculation.*

Two logics for sample size

Table D4.1. Precision-based and power-based sample sizes

	Precision (estimation)	Power (hypothesis testing)
Question	How accurately do I need to estimate a proportion or mean?	How likely am I to detect an effect of a given size, if it exists?
Inputs	Confidence level; margin of error d; variability (P or σ); population size N	Significance level α; power $1-\beta$; minimum effect size of interest; the planned test or model
Typical use	Descriptive surveys: "What share of SMEs accept e-wallets?"	Comparisons and regression: "Is literacy associated with saving?"
Tools	Krejcie–Morgan / Cochran formula	G*Power; Stata `power`; simulation

Lakens (2022) sets out both approaches, and others, in a single framework. It is the best single reading on sample-size justification.

The Krejcie and Morgan (1970) table

Krejcie and Morgan turned an existing formula for the sample size needed to estimate a population proportion into a lookup table:

$$s = \frac{\chi^2 NP(1-P)}{d^2(N-1) + \chi^2 P(1-P)}$$

s = required sample size; N = population size

χ^2 = chi-square value for 1 degree of freedom at the desired confidence level (3.841 at 95%)

P = population proportion, set to 0.50 because $P(1-P)$ is largest there, giving the most conservative size

d = margin of error as a proportion (0.05)

The table therefore builds in three choices: **95% confidence, ±5 percentage points, and** $P = 0.5$. Its entries agree with the formula to rounding precision. The calculator on the next tab reproduces every entry:

Table D4.2. Selected entries, recalculated from the formula

Population N	Table value s	Formula
100	80	79.51
1,000	278	277.73
9,000	368	368.46
10,000	370	369.97
50,000	381	381.22
1,000,000	384	384.00

As N grows, the finite-population correction vanishes and the formula approaches $n_0 = z^2 P(1-P)/d^2 = 1.96^2(0.25)/0.05^2 = 384.16$. That is why the required sample barely grows beyond a population of about 10,000. Sampling precision depends mainly on the sample size, not on the population size. Strictly, rounding down does not guarantee the target precision: for $N = 50{,}000$ a conservative requirement is 382, not 381. This is a minor point.

Five limitations of the table

1. **It assumes probability sampling.** The margin-of-error interpretation relies on random selection (usually SRS), a good frame and ignorable non-response. Collect 370 Google Form responses through WhatsApp groups and you have the same number with no comparable inferential justification. Raising a convenience sample from 370 to 1,000 does not remove selection bias (Baker et al., 2013).
2. **It fixes the margin of error at ±5 points.** For N = 10,000, ±3 points needs 965 and ±10 points needs 96. Population size alone does not settle the sample size; the precision you need matters far more.
3. **It does not calculate power.** The formula is about estimating one proportion. It ignores the effect size a test should detect. A sample of 370 split into two equal groups of 185 has only 48% power to detect a small difference ($d = 0.2$), 82% for $d = 0.3$ and 99.8% for $d = 0.5$. Detecting $d = 0.2$ with 80% power needs 394 per group, 788 in total.
4. **It ignores complex designs.** Clustered samples carry a design effect $DEFF = 1 + (m-1)\rho$, with m respondents per cluster and intraclass correlation ρ. With DEFF = 1.5, 384 clustered respondents are worth about $384/1.5 = 256$ independent ones.
5. **It gives completed, not invited, responses, and ignores subgroups and regression.** To obtain 370 completed questionnaires at a 60% response rate, invite $370/0.60 \approx 617$. If a key subgroup is 10% of the sample, it has only about 37 members. Regression precision depends on the variance of the regressors, the residual variance, collinearity and the coefficient of interest, none of which the table considers.

? **Examiner:** *Your target population is 10,000, so the table says 370. What margin of error did you need, and why that one?*

Power analysis

Power is the probability of rejecting a false null hypothesis. A priori power analysis fixes four quantities to find the fourth: significance level (usually 0.05), desired power (usually 0.80), the minimum effect size of substantive interest, and the sample size.

Table D4.3. Cohen's (1992) conventional effect sizes. Prefer an effect size grounded in your own literature or in what would matter for policy

Test	Effect size	Small	Medium	Large
Difference in two means	d	0.20	0.50	0.80
Correlation	r	0.10	0.30	0.50
Multiple regression	$f^2 = R^2/(1-R^2)$	0.02	0.15	0.35

- **Two groups:** $d = 0.2, 0.5, 0.8$ need 394, 64 and 26 per group for 80% power at $\alpha = 0.05$.
- **Regression, all slopes:** $f^2 = 0.15$ with 5 predictors needs N = 92 (Faul et al., 2009; Stata `power rsquared` gives the same).
- **One coefficient with controls:** a small partial effect ($f^2 = 0.02$) for one predictor alongside 7 controls needs N = 395. Detecting a small effect of your main variable usually drives the sample size.
- **Logit with a binary outcome:** a widely used guideline is at least 10 events (the rarer outcome) per estimated parameter (Peduzzi et al., 1996). With 8 parameters and 25% of respondents in the rarer category, you need at least $10 \times 8/0.25 = 320$ respondents.

Economics has a power problem: Ioannidis et al. (2017) estimate that the median statistical power in many areas of empirical economics is low, which inflates reported effects. A power analysis in Chapter 3 is good practice, not decoration.

Rules of thumb, and their limits

- **Green (1991):** $N \geq 50 + 8m$ to test the overall R^2, and $N \geq 104 + m$ for individual predictors (m = number of predictors). Simple, but it ignores effect size; power analysis is better.
- **Five or ten respondents per item** for factor analysis: not a guarantee. Required N depends on loadings, number of items per factor and model complexity.
- **Qualitative interviews:** justified by saturation (Guest et al., 2006) or information power (Malterud et al., 2016), not by statistical formulas.

Which method for which objective?

Table D4.4. Choose the sample-size method by the primary inferential objective

Primary research objective	More appropriate approach
Estimate a population percentage	Proportion-based precision calculation, with finite-population correction where appropriate
Estimate a population mean	Confidence-interval precision calculation using the anticipated standard deviation
Compare two groups	Power analysis for the planned difference-in-means or proportions test
Correlation analysis	Power or precision calculation for the correlation coefficient
Multiple linear regression	Power or precision analysis for the primary coefficient or joint hypothesis
Logistic regression	Outcome frequency (events per variable), coefficient precision, or simulation
Factor analysis or SEM	Model-specific precision, factor recovery, or Monte Carlo simulation
Difference-in-differences or field experiment	Minimum detectable effect and design-specific power analysis, including clustering and allocation (McKenzie, 2012)
Descriptive non-probability survey	Justify feasibility and analytical adequacy; acknowledge the limits of population inference

Key takeaways

1. “n ≥ 30” is about the normal approximation, not about precision or power.
2. Krejcie–Morgan is valid for estimating a proportion at ±5 points from a probability sample, and for nothing else.
3. For hypothesis-testing theses, justify n by a priori power for the main coefficient, then inflate for non-response and design effects.

References cited

Baker, R., Brick, J. M., Bates, N. A., Battaglia, M., Couper, M. P., Dever, J. A., Gile, K. J., & Tourangeau, R. (2013). Summary report of the AAPOR task force on non-probability sampling. *Journal of Survey Statistics and Methodology, 1*(2), 90–143.

Cohen, J. (1992). A power primer. *Psychological Bulletin, 112*(1), 155–159.

Faul, F., Erdfelder, E., Buchner, A., & Lang, A.-G. (2009). Statistical power analyses using G*Power 3.1: Tests for correlation and regression analyses. *Behavior Research Methods, 41*(4), 1149–1160.

Green, S. B. (1991). How many subjects does it take to do a regression analysis? *Multivariate Behavioral Research, 26*(3), 499–510.

Guest, G., Bunce, A., & Johnson, L. (2006). How many interviews are enough? *Field Methods, 18*(1), 59–82.

Ioannidis, J. P. A., Stanley, T. D., & Doucouliagos, H. (2017). The power of bias in economics research. *The Economic Journal, 127*(605), F236–F265.

Krejcie, R. V., & Morgan, D. W. (1970). Determining sample size for research activities. *Educational and Psychological Measurement, 30*(3), 607–610. https://doi.org/10.1177/001316447003000308

Lakens, D. (2022). Sample size justification. *Collabra: Psychology, 8*(1), 33267.

Malterud, K., Siersma, V. D., & Guassora, A. D. (2016). Sample size in qualitative interview studies: Guided by information power. *Qualitative Health Research, 26*(13), 1753–1760.

McKenzie, D. (2012). Beyond baseline and follow-up: The case for more T in experiments. *Journal of Development Economics, 99*(2), 210–221.

Peduzzi, P., Concato, J., Kemper, E., Holford, T. R., & Feinstein, A. R. (1996). A simulation study of the number of events per variable in logistic regression analysis. *Journal of Clinical Epidemiology, 49*(12), 1373–1379.

Example

Worked example: justifying n for the e-wallet survey

1. **Descriptive aim (RQ1):** estimate the share of the university's 10,000 undergraduates who use e-wallets within ±5 points at 95% confidence. Krejcie–Morgan: 370 completed questionnaires.
2. **Inferential aim (RQ2):** detect a small partial association ($f^2 = 0.02$) between perceived security and use, with 7 other predictors, at 80% power: 395 (computed with Stata's `power rsquared`). This exceeds 370, so 395 becomes the target.
3. **Non-response:** at an expected 60% response, invite $395 / 0.60 \approx 659$.
4. **Design:** stratified by faculty with proportionate allocation, so the design effect is at most 1 and no further inflation is needed.

02_sample_size.do

```
* Krejcie & Morgan for N = 10,000 (program defined in the do-file)
kmsize, n(10000)
display round(r(s))                                  // 370

* power for one tested coefficient with 7 controls, small partial f2 = 0.02
power rsquared `=0.10' `=0.10 + 0.02*(1 - 0.10)/(1 + 0.02)', ncontrol(7) ntested(1) power(0.8)   // N = 395

display ceil(395 / 0.60)                             // invitations: 659
```

```
print(round(km_size(10000)))                         # 370
print(reg_n_subset(0.02, tested=1, controls=7))      # 395
print(int(np.ceil(395 / 0.60)))                      # 659 invitations
```

Writeup

Two defensible statements

Choose the statement that matches your design, and fill it with your own numbers. The wording alone is not a justification; the calculation is.

> **For a probability-based descriptive survey.** The required sample size was determined using the finite-population formula presented by Krejcie and Morgan (1970), assuming a 95% confidence level, a margin of error of ±5 percentage points and a conservative population proportion of 0.50. The calculation assumed simple random sampling. The recruitment target was then adjusted for anticipated non-response.
>
> **For a hypothesis-testing study.** The target sample size was determined by an a priori power analysis for the primary hypothesis. The calculation specified the minimum effect size of substantive interest, the significance level, the desired power and the analytical model, and allowed for anticipated non-response and the characteristics of the sampling design.

Template statements. *Add the numbers: effect size and its source, α, power, model, resulting N, response-rate assumption, invitations.*

3.1.2 Sample size

The sample size was set to meet two requirements.[1] For the descriptive aim, estimating the proportion of e-wallet users within ±5 percentage points at 95% confidence in a population of about 10,000 requires 370 completed questionnaires (Krejcie & Morgan, 1970). For the main inferential aim, an a priori power analysis showed that detecting a small partial association ($f^2 = 0.02$) between perceived security and use, alongside seven other predictors, with 80% power at $\alpha = 0.05$ requires 395 observations (Faul et al., 2009).[2] The larger figure was adopted; anticipating a 60% response rate, 659 students were invited.[3]

1 *Separate the descriptive and inferential requirements.*

2 *Every input of the power analysis is stated.*

3 *The binding constraint, and the non-response inflation.*

Model write-up. *Hypothetical study.*

Worksheet

Worksheet D4

Q1 Use the calculator. For N = 5,000, find the completed sample for ±3, ±5 and ±7 points at 95% confidence. What do the results say about the claim that "sample size depends on population size"? (4 marks)

Model answer

About 880, 357 and 189 respectively (exact values 879.5, 356.8 and 188.6). Changing the margin of error from ±5 to ±3 points multiplies the sample by about 2.5, while changing N from 5,000 to 1,000,000 at ±5 points only moves it from 357 to 384. Precision, not population size, is the main driver.

Q2 You compare saving between treated and control households in a field experiment with 120 households per arm. What is the power to detect d = 0.3? How many per arm are needed for 80% power? (3 marks)

Model answer

Power is about 0.64 with 120 per arm; 176 per arm are needed for 80% power at $\alpha = 0.05$. If households were assigned by village, a design effect would increase this further (McKenzie, 2012).

Q3 A student writes: "Following Krejcie and Morgan (1970), the sample of 384 is adequate for structural equation modelling." Explain why this is not a justification and what would be. (4 marks)

Model answer

Krejcie–Morgan concerns the precision of a single proportion from a random sample; it says nothing about estimating an SEM, whose requirements depend on the number of indicators per factor, loadings, model complexity, missing data and the estimator. A justification would use a model-specific power or precision analysis, such as a Monte Carlo simulation of the planned model, or published evidence for models of similar size, plus a check that the sampling design supports the intended inference.

E1 Designing the questionnaire

The questionnaire is your data-collection instrument. Every variable in your model is only as good as the question that measured it. Good items are clear, neutral, single-barrelled and answerable; good questionnaires flow, protect respondents and are tested before fieldwork.

About 30 minutes | Fix-the-item exercises | Sample questionnaire outline

After this section you can:

1. distinguish constructs, items and questions, and choose suitable item formats;
2. spot and rewrite the seven kinds of problem item;
3. structure, pretest and translate a questionnaire.

Construct, item, question

Instrument	The data-collection tool as a whole: the questionnaire (Q), interview guide or observation sheet.
Construct	The concept you want to measure, often latent: perceived usefulness, financial literacy, trust.
Item	One statement or question with its response options. A construct is usually measured by several items.
Question	Often used loosely for item; strictly, an item phrased as a question (“How many days…?”) rather than a statement.

Item formats

Closed-ended items

- **Dichotomous**: yes/no, agree/disagree.
- **Multiple choice**: single answer or “select all that apply”. Options must be exhaustive (include “Other, please specify”) and mutually exclusive (income bands must not overlap).
- **Ranking**: respondents order alternatives. Beware cognitive load: rank no more than five or six.
- **Likert items**: degree of agreement on an ordered scale (E3).
- **Rating scales**: satisfaction, importance, frequency, 0–10 (E3).
- **Numeric**: age, income, days of use, with the unit and reference period stated.

Open-ended items

Free-text answers give rich information but are hard to code. Use them for (a) pilot studies, to surface categories you missed, and (b) “Other, please specify” options. Numeric answers (age, income) are open-ended too, but easy to analyse.

Items that should not be used

Seven kinds of problem item account for most measurement failures. Try rewriting each before you open the model answer on the *Fix the items* tab.

Table E1.1. Problem items

Problem	What goes wrong	Example
Leading	Suggests a preferred answer	"Do you agree that the new policy is beneficial?" ("beneficial" primes agreement)
Loaded	Embeds a contested assumption	"How often do you cheat on your exams?" "Why did the government fail on this policy?"
Double-barrelled	Asks two things at once	"Do you find Stata easy and enjoyable?"
Double negative	Two negatives overload the reader	"Do you disagree that researchers should not use AI?"
Socially desirable	Invites the respectable answer	"Do you always follow ethical digital practices?"
Ambiguous	Key words undefined	"Do you use social media often?" ("often" is undefined)
Jargon and acronyms	Respondents do not share your vocabulary	PTPTN and B40 are fine in Malaysia; "dirigisme" is not

✓ **Examiner:** *I will read your appendix questionnaire. A double-barrelled item in a key scale undermines the variable built from it.*

Structure and flow

1. **Introduction and consent**: who you are, the purpose, how long it takes, anonymity, voluntary participation, contact details.
2. **Screening questions** to confirm eligibility (for example, "Are you a full-time undergraduate?").
3. **Easy, engaging questions first**, then the core constructs, grouped by topic with a short heading for each section.
4. **Sensitive questions later** (income, debt, illegal behaviour), after trust is built.
5. **Demographics last**, except those needed for screening or skip logic.
6. **Close** with thanks and, if appropriate, an open comments box.

Keep response options in a consistent direction within a block (for example, 1 = strongly disagree on the left), state reference periods ("in the past 7 days"), and keep the questionnaire short. Every item should map to a variable in your codebook (B3). If it doesn't, delete it.

Testing before fieldwork

Expert review	Two to five experts judge relevance and clarity, which supports content validity (E4).
Cognitive interviews	A handful of target respondents think aloud while answering, revealing how they understand each question (Beatty & Willis, 2007).
Pilot	Usually 20–40 respondents outside the main sample: checks timing, skip logic, coding and preliminary reliability. If items change substantially, test them again.

Translation	For English–Malay (or other) versions: forward translation, independent back-translation, and reconciliation of differences (Brislin, 1970; Beaton et al., 2000).

Key takeaways

1. One idea per item, in neutral, specific, everyday words, with a stated reference period.
2. Order: consent, screening, easy, core, sensitive, demographics.
3. Expert review, cognitive interviews, pilot and back-translation before fieldwork.

References cited

Beaton, D. E., Bombardier, C., Guillemin, F., & Ferraz, M. B. (2000). Guidelines for the process of cross-cultural adaptation of self-report measures. *Spine, 25*(24), 3186–3191.

Beatty, P. C., & Willis, G. B. (2007). Research synthesis: The practice of cognitive interviewing. *Public Opinion Quarterly, 71*(2), 287–311.

Brislin, R. W. (1970). Back-translation for cross-cultural research. *Journal of Cross-Cultural Psychology, 1*(3), 185–216.

Krosnick, J. A. (1999). Survey research. *Annual Review of Psychology, 50*, 537–567.

Fix

Fix the items

Rewrite each item, then compare with the model rewrite.

Q1 **Leading.** “Do you agree that the new policy is beneficial?”

> **Model answer**
>
> “How would you rate the effects of the new policy?” with a balanced scale from “very negative” to “very positive”. The evaluative word “beneficial” is removed.

Q2 **Loaded.** “Why did the government fail on this policy?”

> **Model answer**
>
> “Has the policy succeeded or not? Why?” Ask about success first, without presupposing failure. Similarly, replace “How often do you cheat on your exams?” with a screening question (“Have you ever…?”), asked anonymously.

Q3 **Double-barrelled.** “Do you find Stata easy and enjoyable?”

Model answer

Split into two items: “Stata is easy to use” and “I enjoy using Stata”, each rated from strongly disagree to strongly agree. Easy and enjoyable are not the same thing.

Q4 **Double negative.** “Do you disagree that researchers should not use AI?”

Model answer

“To what extent do you agree or disagree with the following statement: Students should be permitted to use AI in their exam.”

Q5 **Socially desirable.** “Do you always follow ethical digital practices?”

Model answer

Ask about specific, recent behaviour: “In the past month, how often have you checked whether a website was secure before entering personal information?” (never / once or twice / several times / every time).

Q6 **Ambiguous.** “Do you use social media often?”

Model answer

“In the past seven days, on how many days did you use social media? (0, 1, 2, …, 7)”

Q7 **Overlapping options.** “Monthly income: RM 0–2,000 / RM 2,000–4,000 / RM 4,000–6,000 / above RM 6,000”

Model answer

Make the bands mutually exclusive and add a “prefer not to say” option: “below RM 2,000 / RM 2,000–3,999 / RM 4,000–5,999 / RM 6,000 or more / prefer not to say”. State whether income is individual or household, gross or net, and the reference month.

Example

Worked example: outline of the e-wallet questionnaire

Table E1.2. Questionnaire structure (hypothetical study)

Section	Content	Items	Design notes
Cover page	Purpose, 12 minutes, anonymous, voluntary, contact, consent	1	Continuing records consent
A. About you	Faculty, year of study (screening: full-time undergraduate)	2	Ineligible respondents are thanked and exit
B. E-wallet use	Used in past 30 days; days used in past 7 days; spending	3	Skip B2–B3 for non-users
C. Perceptions	Usefulness (4), security (4), ease of use (3)	11	Five-point agreement, same direction throughout
D. Money questions	Interest, inflation, diversification	3	Include a "do not know" option
E. Intentions	Intention to continue; satisfaction	2	Satisfaction for users only
F. Background	Gender, age, home state, urban/rural, monthly allowance	5	Sensitive allowance item last; "prefer not to say" available

Worksheet

Worksheet E1

Q1 **Write three items, in three different formats, that together measure "household debt burden". Give response options and state the reference period.** (6 marks)

Model answer

(1) Numeric: "What is your household's total monthly repayment on all loans and credit cards, in RM, last month?" (2) Multiple choice: "Which of these loans does your household currently have? Select all that apply: housing, vehicle, personal, PTPTN, credit card, other (please specify), none." (3) Rating: "In the past 12 months, how often has your household struggled to make loan repayments on time?" (never / rarely / sometimes / often / always). Together they capture amount, type and difficulty.

Q2 **Your pilot shows that 40% of respondents skip the income question. Suggest two design changes.** (4 marks)

Model answer

(1) Move income to the end and explain why it is asked and that answers are anonymous. (2) Offer income bands (B40/M40/T20 or RM ranges) instead of an exact amount, or follow a refusal with an unfolding bracket question ("Is it more or less than RM 4,000?"). Also check whether the question is ambiguous (individual or household; gross or net), which itself causes non-response.

E2 Adopting and adapting established questionnaires

You rarely need to write every item from scratch. Established instruments come with evidence of validity and reliability, and they let you compare your results with earlier studies. Using them well means choosing carefully, changing as little as possible, documenting every change, and testing again in your own sample.

About 25 minutes | Adaptation log template | Worked example from TAM

After this section you can:

1. distinguish adopting, adapting and developing an instrument;
2. evaluate an existing instrument's fit for your study;
3. document adaptations so an examiner can see exactly what changed and why.

Adopt, adapt or develop?

Adopt	Use the instrument as published: same items, same response format, same scoring. Its validity evidence carries over most directly, although you still check reliability in your sample.
Adapt	Modify it: change the context words, tense or target behaviour; translate it; drop or add items; change the response format. Every change weakens the inherited evidence, so validate again.
Develop	Write new items when no suitable instrument exists. This needs the full scale-development process: item generation, expert review, pretesting and factor analysis (Hinkin, 1998; Boateng et al., 2018).

Map each research question to an instrument

Start from the research questions and find an established instrument for each construct, recording its source, length and response format. Here is an example from a study of a teacher preparation programme and co-teaching:

Table E2.1. Research questions and established questionnaire instruments

Research question	Established questionnaire instrument
RQ1: Teacher efficacy = f(TPP programme)	Teachers' Sense of Efficacy Scale (TSES); Tschannen-Moran & Hoy (2001); 24 items; 9-point rating scale, from 1 (Nothing much I can do) to 9 (I can do a great deal)
RQ2: Student non-academic outcome = f(co-teaching): leadership, behaviour and motivation = f(co-teaching)	Student Leadership Practices Inventory (SLPI); Kouzes & Posner (2005); 30 items; 5-point rating scale, from 1 (Rarely/Never) to 5 (Very frequently)
	Motivated Self-Directed Learning & Collaborative Learning (MSDLCL); Choy et al. (2016); 28 items; 7-point rating scale, from 1 (Strongly disagree) to 7 (Strongly agree)
RQ3: Pedagogical skills = f(co-teaching)	Framework for Teaching (FFT); Danielson (1996); 22 items; 4-point rating scale, from 1 (Unsatisfactory) to 4 (Distinguished)

Notice that the four instruments use 9-, 5-, 7- and 4-point formats. Keep each instrument's original format unless you have a strong reason: the published validity and reliability evidence was obtained with that format. Different formats in different sections are fine if the instructions are clear.

✓ **Examiner:** *If you converted a 9-point scale to 5 points "for consistency", you adapted the instrument. Say so, and show its reliability and validity in your data.*

A procedure for adopting and adapting

1. **Define the construct** precisely before searching. "Financial literacy" as objective knowledge needs a test; as confidence it needs attitude items.
2. **Search** the literature on your construct: the original development papers, review articles, highly cited applications, measurement databases.
3. **Evaluate** each candidate: does its definition match yours? What validity and reliability evidence exists, and in which populations? What length and format? Is it free to use?
4. **Get permission** where needed. Some instruments are copyrighted or sold under licence; write to the authors or publisher and keep their reply for your appendix.
5. **Adapt minimally and systematically**: change only what the new context requires, keep the response format, and record every change in an adaptation log.
6. **Translate properly**: forward translation, independent back-translation, reconciliation (Brislin, 1970; Beaton et al., 2000).
7. **Review and pretest**: expert content review (content validity index), cognitive interviews, pilot (E1, E4).
8. **Re-establish evidence in your sample**: reliability (alpha, omega), factor structure, convergent and discriminant validity. A scale's reliability in someone else's sample is not evidence about yours.
9. **Report**: a sources table in Section 3.1 and the full adaptation log in an appendix.

? **Examiner:** *"The questionnaire is reliable because it was adopted from a published article." Reliability belongs to scores in a sample, not to a printed questionnaire. Show me yours.*

Key takeaways

1. Prefer established instruments; adopt where you can, adapt where you must, develop only when nothing fits.
2. Change as little as possible, keep the response format, and log every change.
3. Re-establish reliability and validity in your own sample.

References cited

Beaton, D. E., Bombardier, C., Guillemin, F., & Ferraz, M. B. (2000). Guidelines for the process of cross-cultural adaptation of self-report measures. *Spine, 25*(24), 3186–3191.

Boateng, G. O., Neilands, T. B., Frongillo, E. A., Melgar-Quiñonez, H. R., & Young, S. L. (2018). Best practices for developing and validating scales for health, social, and behavioral research: A primer. *Frontiers in Public Health, 6*, 149.

Brislin, R. W. (1970). Back-translation for cross-cultural research. *Journal of Cross-Cultural Psychology, 1*(3), 185–216.

Davis, F. D. (1989). Perceived usefulness, perceived ease of use, and user acceptance of information technology. *MIS Quarterly, 13*(3), 319–340.

Hinkin, T. R. (1998). A brief tutorial on the development of measures for use in survey questionnaires. *Organizational Research Methods, 1*(1), 104–121.

Tschannen-Moran, M., & Woolfolk Hoy, A. (2001). Teacher efficacy: Capturing an elusive construct. *Teaching and Teacher Education, 17*(7), 783–805.

Example

Worked example: adapting perceived usefulness from Davis (1989)

Davis's perceived-usefulness items were written for workers using a computer system at work. The e-wallet study needs them for students paying with a phone. The adaptation log records each decision.

Table E2.2. Adaptation log for perceived usefulness

Original item (Davis, 1989; [system] stands for the technology studied)	Adapted item	Change	Reason
Using [system] in my job would enable me to accomplish tasks more quickly.	Using an e-wallet enables me to complete payments more quickly.	Context (job → payments); tense (would → present)	Respondents are current users making personal payments
Using [system] in my job would increase my productivity.	Using an e-wallet makes managing my daily payments more efficient.	Context; "productivity" → "efficient"	"Productivity" has no clear meaning for personal payments (cognitive interviews)
Using [system] would enhance my effectiveness on the job.	Using an e-wallet improves the convenience of my payments.	Context; target outcome	Expert panel judged "convenience" the relevant benefit
I would find [system] useful in my job.	I find e-wallets useful for my everyday payments.	Context; tense	As above
Using [system] would improve my job performance.	(dropped)	Item removed	No meaningful equivalent for students; content coverage checked by expert review

Response format kept as a five-point agreement scale. The Malay version was back-translated. In the pilot (n = 30), the adapted four-item scale had $\alpha = 0.84$. In the main survey, $\alpha = 0.86$ and CR = 0.86 (E4).

Writeup

How it reads in the thesis

3.1.2 Instrument

Perceived usefulness and ease of use were measured with items adapted from Davis (1989), the instrument underlying the technology acceptance model, because it is the most widely validated measure of these constructs.[1] Adaptation was limited to replacing the workplace context with personal payments and changing the conditional tense to the present, because respondents are current users; one usefulness item with no student equivalent was dropped. Appendix B lists every original and adapted item.[2] The five-point agreement format was retained. The Malay version was produced by forward and independent back-translation, and three subject experts rated each adapted item's relevance (all item-level CVIs ≥ 0.80).[3] Reliability and validity in the present sample are reported in Section 3.3.

1 *Source and reason for choosing it.*

2 *What changed, why, and where the full log is.*

3 *Format, translation and expert review, with numbers.*

Model write-up. *Hypothetical study.*

Template: adaptation log

Template. One row per item (also in the ZIP as a Word file)

Construct	Original item and source	Adapted item	Type of change	Reason and evidence
construct	original wording (author, year)	new wording	context / tense / translation / dropped / added	why; expert or pilot evidence

Worksheet

Worksheet E2

Q1 You want to measure “trust in Islamic banks” among Malaysian depositors. Describe how you would decide whether to adopt, adapt or develop an instrument. (5 marks)

Model answer

Define the construct (trust in the integrity, competence and Shariah compliance of Islamic banks). Search the literature for validated bank-trust scales, especially any used with Islamic banking customers. Evaluate the fit of definition, validity evidence, population and language. If a validated scale fits closely and was used with Malaysian or similar depositors, adopt it. If it measures trust in conventional banks, adapt the items (adding Shariah-compliance content), get permission, back-translate, run an expert review and a pilot, and re-validate. Develop new items only if no scale covers the construct.

Q2 A student changes a validated 7-point scale to 5 points “to keep the questionnaire consistent” and cites the original alpha of 0.91 as evidence of reliability. Comment. (4 marks)

Model answer

Changing the response format is an adaptation, so the original evidence no longer transfers directly. The 0.91 was obtained in another sample with another format. Reliability must be estimated in the current sample. Keeping the original 7-point format is usually better; if changed, justify it, pilot the scale and report alpha (and omega) and factor structure from the current data.

E3 Likert scales and rating scales

The first misconception to correct: Likert scales and rating scales are not two mutually exclusive methods. A Likert-type item is one particular kind of rating scale. What matters is what you intend to measure, how respondents answer, and which analyses the answers can legitimately support.

About 25 minutes | Format examples | Which estimator for which outcome

After this section you can:

1. distinguish a Likert-type item, a Likert scale and other rating scales;
2. choose a response format to match each construct;
3. choose an estimator for an outcome measured on these scales.

What is the difference?

A rating scale is the broad category: respondents evaluate something using ordered categories or numerical scores. A Likert-type item is a more specific format that asks for degree of agreement or disagreement with a statement. Strictly, a Likert scale is several related Likert-type items combined (summed or averaged) to measure one underlying construct (Likert, 1932). A single agreement question is a Likert-type *item*, although many researchers loosely call it a Likert scale.

Example A: Likert-type item

"Using ChatGPT improves my academic productivity."

- ◯ 1 Strongly disagree
- ◯ 2 Disagree
- ◯ 3 Neither agree nor disagree
- ◯ 4 Agree
- ◯ 5 Strongly agree

Measures agreement with a proposition.

Example B: numerical rating scale

"How satisfied are you with your university's Wi-Fi service?"

Very dissatisfied  **7/10** Very satisfied

Directly evaluates satisfaction.

Both collect subjective evaluations, but the first operationalises agreement with a statement, while the second rates the thing itself.

Detailed comparison

Table E3.1. Likert-type items and other rating formats

Characteristic	Likert-type item	Other rating-scale formats
Main purpose	Agreement, attitudes, beliefs, perceptions	Satisfaction, intensity, frequency, quality, importance, performance
Typical wording	"I believe AI improves my learning."	"How useful is AI for your learning?"
Response format	Strongly disagree to strongly agree	Very poor to excellent; 0–10; never to always
Number of categories	Commonly 5 or 7	3, 5, 7, 10, 11, 100 or continuous
Measurement level	Each item is ordinarily ordinal	Ordinal unless interval properties can be justified
Multiple items needed?	Yes, for a summated Likert scale	Not necessarily; several ratings can also form a composite
Common uses	Technology acceptance, motivation, attitudes, trust	Satisfaction surveys, service quality, perceived risk, performance

A 1–10 rating is not automatically better than a five-point Likert item. More response categories do not by themselves produce interval-level measurement or improve validity (Sullivan & Artino, 2013; Norman, 2010).

When to use each

Agreement items for latent constructs. To measure perceived usefulness of AI, use several statements on the same five-point agreement format:

Table E3.2. A four-item perceived-usefulness scale

Code	Statement
PU1	Using generative AI improves the quality of my academic work.
PU2	Using generative AI helps me complete assignments more efficiently.
PU3	Using generative AI enhances my academic productivity.
PU4	Using generative AI makes it easier to understand difficult academic concepts.

If the items represent a common construct and support a defensible composite, one index is the mean:

$$PU_i = \frac{PU1_i + PU2_i + PU3_i + PU4_i}{4}$$

Before combining items, assess dimensionality and reliability (E4), rather than assuming that four similar statements make a valid scale.

Direct ratings for specific evaluations. "Overall, how satisfied are you with generative AI tools for your academic work?" (very dissatisfied to very satisfied) is direct, quick, and avoids an extra agreement statement. Agreement formats also invite **acquiescence bias**, the

tendency to agree whatever the content. An experiment by Revilla, Saris and Krosnick (2014) found that five categories performed better than seven or eleven for the agree–disagree scales they examined. That is evidence for that context, not a universal rule.

✓ **Examiner:** *Choose the format for the construct, construct by construct. One format for every question in the instrument is a habit, not a design decision.*

Advantages and disadvantages

Likert-type agreement format

Advantages	Disadvantages
Easy to understand	Acquiescence bias
Convenient for many constructs	Agreement may not reflect behaviour
Measures abstract constructs (trust, motivation)	Social desirability
Multiple items can raise reliability	Repetition encourages straight-lining
Supports multi-item indices	Ordinal categories complicate analysis

Other rating formats

Advantages	Disadvantages
Tailored to the construct	Scale points interpreted differently
Flexible anchors: frequency, intensity	Equal numerical steps may not be equal psychological steps
Single items reduce burden	One item may not capture a complex construct
Avoids some agreement biases	Response styles remain
Can capture fine distinctions	Too many categories imply false precision

Which estimator? It depends on the outcome, not the label

Your choice of econometric method should depend on the measurement properties of the dependent variable, not on whether the questionnaire calls it a Likert or a rating scale.

Table E3.3. Outcome, example and analytical options

Dependent variable	Example	Appropriate options
One five-point agreement item	Agreement that AI improves productivity	Ordered logit or probit
One five-point satisfaction rating	Satisfaction with university Wi-Fi	Ordered logit or probit
One 1–10 rating	Perceived AI usefulness	Ordered logit/probit; OLS with justification and robustness checks
Mean of several validated items	AI-dependence index	OLS can be considered; latent-variable models are alternatives
Binary	Uses ChatGPT at least weekly (yes/no)	Logit, probit, linear probability model
Observed frequency	Number of days using AI in a week	Count or binomial models

Is OLS acceptable for Likert data? The debate is long-running. Individual items are ordinal: moving from 1 to 2 need not equal moving from 4 to 5 (Jamieson, 2004). Norman (2010) shows that parametric methods are robust with such data, and Carifio and Perla (2008) distinguish single items from summated scales. Liddell and Kruschke (2018) show that treating ordinal outcomes as metric can produce wrong conclusions. A defensible rule: for a single five-category outcome, start with an ordered logit or probit. For a validated composite of several items, OLS is reasonable, ideally with a sensitivity check. For complex latent constructs, consider ordinal CFA or SEM.

Perceptions versus behaviour

Table E3.4. Three ways to ask about AI dependence

Format	Item	What it measures
Agreement	“I frequently rely on ChatGPT to complete my assignments.”	Self-reported agreement with a statement about reliance
Frequency rating	“During the past seven days, how often did you use ChatGPT for academic work?” (never / once / 2–3 times / 4–6 times / 7 or more)	Recalled frequency in ordered categories
Behavioural count	“On how many of the past seven days did you use ChatGPT for academic work?”	A self-reported count from 0 to 7 (recall error remains)

For frequency of use, the second and third formats beat the first. But even daily use is not dependence: dependence needs its own definition and, probably, a validated multi-item measure. Choose the measurement format from the construct and research question first, then choose the estimator.

Key takeaways

1. A Likert-type item is one kind of rating scale; a Likert scale combines several such items.
2. Match the format to each construct: agreement for latent attitudes, direct ratings, frequencies or counts for evaluations and behaviour.
3. The outcome's measurement properties, not its label, decide the estimator.

References cited

Carifio, J., & Perla, R. (2008). Resolving the 50-year debate around using and misusing Likert scales. *Medical Education, 42*(12), 1150–1152.

Jamieson, S. (2004). Likert scales: How to (ab)use them. *Medical Education, 38*(12), 1217–1218.

Liddell, T. M., & Kruschke, J. K. (2018). Analyzing ordinal data with metric models: What could possibly go wrong? *Journal of Experimental Social Psychology, 79*, 328–348.

Likert, R. (1932). A technique for the measurement of attitudes. *Archives of Psychology, 22*(140), 1–55.

Norman, G. (2010). Likert scales, levels of measurement and the “laws” of statistics. *Advances in Health Sciences Education, 15*(5), 625–632.

Revilla, M. A., Saris, W. E., & Krosnick, J. A. (2014). Choosing the number of categories in agree–disagree scales. *Sociological Methods & Research, 43*(1), 73–97.

Sullivan, G. M., & Artino, A. R., Jr. (2013). Analyzing and interpreting data from Likert-type scales. *Journal of Graduate Medical Education, 5*(4), 541–542.

Example

Worked example: choosing formats for an AI study

A thesis on students' use of generative AI measures eight constructs. Rather than one format throughout, each construct gets the format that fits it.

Table E3.5. Construct and suggested measurement

Construct	Suggested measurement
Attitude towards AI	Five-point agreement items
Perceived usefulness	Agreement items or direct usefulness ratings
Trust in AI	Validated multi-item measure
Satisfaction with AI	Direct five-point satisfaction rating
Frequency of AI use	Explicit frequency categories or counts
Time spent using AI	Hours or minutes over a defined recall period
Perceived importance of a policy	Direct importance rating
AI dependence	Validated multi-item construct, supplemented by behaviour questions

Questionnaire quality depends on whether items capture the construct, not only on the response scale. Scale development needs content validation, pretesting, tests of dimensionality, reliability and construct validity. Cronbach's alpha alone does not establish validity (E4).

Worksheet

Worksheet E3

Q1 For each outcome, name the estimator: (a) a single item "I intend to apply for a PTPTN loan" (1–5); (b) the mean of six validated financial-anxiety items (1–5); (c) "How many times did you use Touch 'n Go eWallet last week?" (3 marks)

Model answer

(a) Ordered logit or probit, with a test of the parallel-lines assumption. (b) OLS is reasonable for a validated composite, with an ordered-model or latent-variable check. (c) A count model (Poisson or negative binomial), or OLS if counts are spread widely, reported with robustness checks.

Q2 Rewrite "I use e-wallets frequently" (strongly disagree to strongly agree) into a behavioural count item, and explain what is gained. (3 marks)

Model answer

"On how many of the past seven days did you pay with an e-wallet? (0–7)". The count has a defined reference period, is less prone to acquiescence, is ratio-level, and means the same to every respondent, whereas "frequently" is interpreted differently by different people.

E4 Validity and reliability

Validity asks whether you are measuring the right thing, and whether your interpretations are justified. Reliability asks whether you are measuring it consistently. A measure can be reliable without being valid, so both have to be established, and for your questionnaire they are established with evidence from your own data.

About 50 minutes | Interactive validity–reliability tree | Three calculators | Full worked example with code

After this section you can:

1. explain every type of validity and reliability, with an economics example and a way to measure it;
2. assess a questionnaire's reliability (alpha, omega/CR) and validity (content, convergent, discriminant, criterion);
3. report the evidence in Chapter 3 without the common mistakes.

What validity and reliability mean

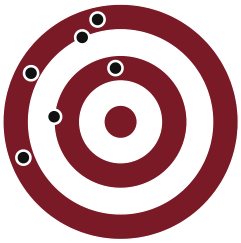
Validity	Are we measuring what we intend to measure, and are the interpretations and conclusions we draw justified? Modern measurement theory treats validity as the degree to which evidence and theory support particular interpretations of scores for particular purposes (Messick, 1995). So the rigorous question is not “is this questionnaire valid?” but “are these scores valid for this interpretation, in this population, for this purpose?”
Reliability	How consistent, stable or precise are the measurements under conditions where they should agree?



Reliable, not valid
Consistent but systematically off target



Low reliability
Scattered around the target: right on average, unreliable each time



Not reliable, not valid
Scattered and off target



Reliable and valid
Consistent and on target

Figure E4.1. *Reliability and validity as shots at a target. The centre is the true value of the construct; each dot is one measurement. Redrawn for this guide.*

The same idea with numbers: an object truly weighs 70 kg, and four scales each weigh it four times.

Table E4.1. Random and systematic measurement error

Scale	Readings (kg)	Verdict
A	70.0, 70.1, 69.9, 70.0	Accurate and consistent
B	75.0, 75.1, 74.9, 75.0	Consistent but systematically wrong: reliable, not valid
C	67, 73, 69, 72	Varies around the correct value: poorly reliable
D	75, 78, 73, 77	Both biased and variable

Random error (distraction, inconsistent interpretation, changing conditions) makes measurements fluctuate and mainly threatens *reliability*. **Systematic error** (consistently overstating income; a miscalibrated instrument; items that capture a different concept) pushes measurements one way and mainly threatens *validity*, possibly alongside high reliability. Both can happen at once.

Validity in the wider research context

Measurement validity is one of several kinds of validity a thesis must address.

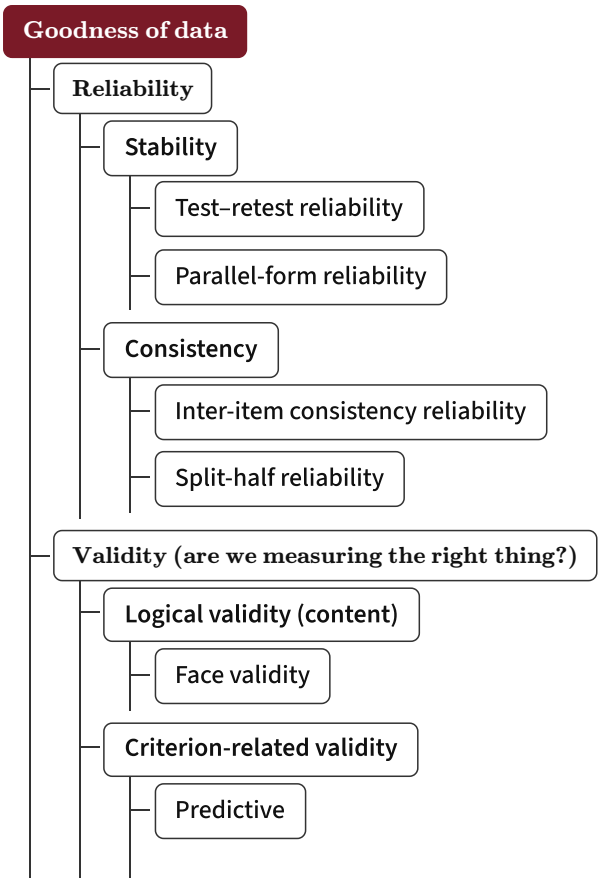
Table E4.2. Dimensions of validity

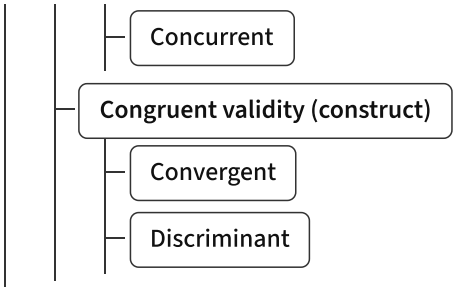
Dimension	Central question	Economics example
Measurement or construct validity	Do the variables measure the theoretical constructs?	Does the questionnaire capture financial literacy rather than financial confidence?
Internal validity	Is the estimated causal relationship credible?	Does training raise wages, or are abler workers more likely to be trained? (A2)
External validity	Do findings apply beyond the study setting?	Can a study of one university's students represent all Malaysian students?
Statistical conclusion validity	Are the statistical inferences justified?	Were standard errors appropriate and power sufficient? (D4)
Ecological validity	Do the conditions resemble real settings?	Do hypothetical purchase choices reflect actual purchases?

A study can have excellent measurement validity and reliability yet poor internal validity, if its identification strategy is weak. Similarly, measurement *reliability* is not the same as *reproducibility* (getting the same results from the same data and code) or *replicability* (finding the same result in new data). In qualitative research, quality is usually judged through trustworthiness: credibility, transferability, dependability and confirmability.

Every type of validity and reliability

The traditional classification (adapted from Sekaran and Bougie, 2016, Figure 12.1) splits the "goodness" of measures into reliability and validity. Select any box to see its meaning, an economics example, how to assess it, and a guideline.





Goodness of data

Before data are analysed, the measures that produced them must be shown to be good: **reliable** (consistent) and **valid** (measuring the intended concept). Reliability is necessary for validity but not sufficient.

Select a box on the left for detail.

Reliability

Question: do the measurements agree with themselves, over time (stability) and across items (consistency)?

A reliable measure of perceived financial insecurity gives similar scores when nothing has changed, and its items move together.

Reliability is a property of scores in a sample, not of a printed questionnaire. Report it for your data.

Stability

Question: does the measure give the same result when the construct has not changed?

Assessed by repeating the measurement (test–retest) or by using equivalent versions (parallel forms). Only meaningful for constructs expected to be stable over the interval.

Test–retest reliability

Meaning: the same respondents complete the same instrument twice, under comparable conditions.

Economics example: 50 students take a financial-knowledge test and retake it two weeks later, with no teaching in between. High agreement is evidence of temporal stability.

How to assess: intraclass correlation coefficient (ICC) for continuous scores (choose the form carefully: Shrout & Fleiss, 1979); weighted kappa for categories. Pearson's r measures association and can stay high even when everyone scores higher the second time.

Guideline: ICC below 0.50 poor, 0.50–0.75 moderate, 0.75–0.90 good, above 0.90 excellent (Koo & Li, 2016).

Choose the interval with care: too short and people remember answers; too long and the construct may genuinely change. Financial anxiety may change after a scholarship or a job loss, and that is real change, not unreliability.

Parallel-form reliability

Meaning: two equivalent versions of an instrument give comparable scores.

Economics example: two versions of a financial-literacy quiz with different numbers in the compound-interest and inflation questions, so that a pre-test and post-test around a workshop do not reward memory.

How to assess: correlation, or agreement, between scores on the two forms in the same respondents.

Equivalence is hard to achieve. Check that the forms have similar means and variances.

Consistency

Question: do the items meant to measure one construct hang together?

Assessed by inter-item consistency (Cronbach's alpha, McDonald's omega) and by split-half reliability. Internal consistency is not the same as stability over time.

Inter-item consistency reliability

Meaning: respondents answer the items of one scale consistently, because the items reflect the same construct.

Economics example: the four perceived-usefulness items in the e-wallet survey give α = 0.86 (worked example).

How to assess: Cronbach's alpha (Cronbach, 1951), $\alpha = \frac{k}{k-1}\left(1 - \frac{\sum \sigma_i^2}{\sigma_T^2}\right)$; McDonald's omega from a factor model, preferable when loadings differ (McNeish, 2018).

Guideline: 0.70 is the usual minimum for research use, 0.80+ good; above 0.95 may mean redundant items. These are conventions, not laws.

Alpha rises with the number of items even when items do not improve; it does not show the scale measures one dimension; and it is not evidence of validity (Sijtsma, 2009). Compute it per scale, never across the whole questionnaire.

Split-half reliability

Meaning: scores on two halves of a scale (for example odd and even items) agree.

Example: a ten-question knowledge test split into odd and even questions, with r = 0.70 between the halves.

How to assess: correlate the halves, then correct for halving the length with the Spearman–Brown formula, $r_{SB} = \frac{2r}{1+r}$: here $2(0.70)/1.70 = 0.82$.

The result depends on how you split. Alpha is the average of all possible split-half coefficients, which is one reason it is reported more often.

Validity

Question: are we measuring the right thing?

Traditional textbooks distinguish content, criterion-related and construct validity. Modern theory treats these as different sources of evidence for one validity argument (the AERA, APA and NCME *Standards*; Messick, 1995). See the five sources of evidence below the tree.

Logical (content) validity

Meaning: the items cover the relevant dimensions of the construct, and only those.

Economics example: financial literacy defined as knowledge of interest, inflation, diversification and debt. A questionnaire that asks only about interest omits half the domain (construct under-representation), however well its items are written.

How to assess: a panel of experts rates each item's relevance on a 4-point scale. Item-level CVI = share of experts rating it 3 or 4; scale-level CVI (S-CVI/Ave) = mean of the I-CVIs. Lawshe's (1975) content validity ratio is an alternative.

Guideline: I-CVI ≥ 0.78 with six or more experts; S-CVI/Ave ≥ 0.90 (Polit et al., 2007). Use the CVI calculator in the Calculators tab.

Expert agreement supports item content; it does not show respondents understand the items, or that the expected structure holds in the data.

Face validity

Meaning: the instrument *looks* as if it measures the construct, to respondents and lay reviewers.

Economics example: “Using an e-wallet makes my payments more efficient” looks like a usefulness item; “I enjoy shopping with my friends” does not.

How to assess: ask a few target respondents and reviewers whether items seem understandable, relevant and appropriate; cognitive interviews go further.

The weakest form of evidence: necessary for cooperation, never sufficient on its own.

Criterion-related validity

Question: do scores relate as expected to an independently measured outcome (the criterion)?

Split into concurrent (criterion measured at the same time) and predictive (criterion measured later). The evidence is only as good as the criterion: monthly spending is not a valid benchmark for financial literacy, because spending also reflects income.

Predictive validity

Meaning: scores predict a relevant future outcome.

Economics example: an intention-to-use scale collected in March predicts the number of e-wallet transactions recorded over the following three months.

How to assess: correlation or regression of the later criterion on the score; for a binary criterion, classification accuracy or AUC.

Prediction is not causation, and attrition between waves can bias the estimate.

Concurrent validity

Meaning: scores relate to a criterion measured at about the same time, or distinguish groups known to differ.

Economics example: self-reported e-wallet use compared with the same month's transaction records; a financial-knowledge test on which finance graduates outscore other graduates (known-groups comparison).

How to assess: correlation with the criterion; difference in means between known groups.

Congruent (construct) validity

Meaning: the instrument measures the theoretical construct it claims to, especially a latent one such as trust, confidence or perceived security (Cronbach & Meehl, 1955).

How to assess: factor analysis (EFA when the structure is uncertain, CFA to test a hypothesised structure), plus convergent and discriminant evidence.

Passing KMO and Bartlett's tests shows only that the data can be factor-analysed, not that the construct is valid.

Convergent validity

Meaning: measures that should be related are related: a scale's items load strongly on their factor, and the score correlates with other measures of the same construct.

Economics example: students who score high on perceived e-wallet security also score high on an established scale of trust in digital-payment security.

How to assess: standardised loadings; average variance extracted, $AVE = \overline{\lambda^2}$; composite reliability, $CR = (\sum \lambda)^2 / [(\sum \lambda)^2 + \sum(1 - \lambda^2)]$ (Fornell & Larcker, 1981).

Guideline: loadings ≥ 0.70, AVE ≥ 0.50, CR ≥ 0.70, as rules of thumb, not laws. CR is really a reliability coefficient.

Discriminant validity

Meaning: measures of constructs that should differ do differ (Campbell & Fiske, 1959).

Economics example: perceived usefulness (“e-wallets help me pay efficiently”) and ease of use (“learning to use an e-wallet is easy”) are related but distinct; their scales should not collapse into one factor.

How to assess: Fornell–Larcker criterion (√AVE of each construct exceeds its correlations with other constructs); heterotrait–monotrait ratio (HTMT), which detects problems the Fornell–Larcker criterion misses (Henseler et al., 2015); cross-loadings.

Guideline: HTMT below 0.85 (strict) or 0.90 (lenient).

The modern view: five sources of validity evidence

Table E4.3. Sources of validity evidence (Standards for Educational and Psychological Testing, 2014)

Source of evidence	How it applies to a questionnaire
Test content	Are the questions relevant and sufficiently comprehensive? (expert review, CVI)
Response processes	Do respondents understand and answer as intended? (cognitive interviews)
Internal structure	Do relationships among items fit the proposed dimensions? (EFA, CFA, reliability)
Relations to other variables	Are relationships with other measures, behaviours and criteria plausible? (convergent, discriminant, criterion evidence)
Consequences of testing	What are the intended and unintended consequences of using the scores?

Validation accumulates evidence rather than passing one statistical test. Response processes deserve special attention: a student may read “monthly income” as their parents' household income when you meant their own income from all sources. An expert would accept the item; only a cognitive interview reveals the misunderstanding.

? **Examiner:** *Your financial-confidence items have $\alpha = 0.95$. Do they measure financial literacy? High reliability does not make a scale valid for its intended construct.*

Which evidence for which kind of variable?

Table E4.4. Matching validity and reliability evidence to the variable

Variable or instrument	Validity evidence to prioritise	Reliability
Demographics (age, sex, faculty)	Clear definitions and coding; factual accuracy	Data verification
A factual measure (monthly income)	Reference period; comparison with records if possible	Repeatability if stable
Objective knowledge test	Content coverage; item difficulty; criterion evidence	KR-20 or alpha (with care); test–retest
Multi-item attitude scale	Content; response processes; factor structure; convergent and discriminant	Alpha, omega; test–retest as appropriate
Formative index (asset-based wealth index)	Content coverage; indicator justification; criterion evidence	Alpha generally inappropriate
Open-ended answers that are coded	Coding validity; meaningful categories	Inter-rater agreement (Cohen's kappa; Cohen, 1960)

Not every index should be judged by alpha. In a **reflective** model the construct causes its indicators, so they should correlate (perceived security drives answers to the security items). In a **formative** index the indicators define the construct and need not correlate: home ownership and vehicle ownership in a wealth index. Using alpha to validate a formative index is a conceptual error (MacKenzie et al., 2005).

Why reliability matters for your regression

If the true model is $Y_i = \alpha + \beta X_i^* + \varepsilon_i$ but you observe $X_i = X_i^* + u_i$ with classical measurement error, OLS converges to

$$\operatorname{plim} \hat{\beta} = \beta \, \frac{\operatorname{Var}(X^*)}{\operatorname{Var}(X^*) + \operatorname{Var}(u)}$$

The fraction is the reliability ratio. With reliability 0.70, a true coefficient of 0.50 is estimated near 0.35: attenuation bias. The result holds for classical error in a regressor; non-classical errors can bias in other directions, and error in the dependent variable has different consequences. Measurement reliability is not just a psychometric nicety; it changes your coefficients (Bound & Krueger, 1991, show how large reporting error is in survey earnings).

Common mistakes in theses

Table E4.5. What students write, why it fails, and what to do instead

Common statement or practice	Why it is a problem	Better approach
“Cronbach's alpha exceeds 0.70; therefore the questionnaire is valid.”	Alpha cannot establish validity.	Report reliability separately from validity evidence.
“The questionnaire was validated by three experts.”	Expert review is only part of the evidence.	Say what the experts assessed; add respondent and empirical evidence.
“The questionnaire is reliable because it was adopted from a published article.”	Reliability depends on the population and context.	Estimate reliability in the current sample.
One alpha for the entire questionnaire	Unrelated constructs and demographics are not one scale.	Compute coefficients for each scale or subscale.
Dropping every item that lowers alpha	Can destroy content coverage.	Weigh theoretical relevance and factor structure first.
“KMO and Bartlett's test confirm construct validity.”	They assess factorability only.	Evaluate factor structure and other evidence.
Treating a large sample as proof of validity	Size cannot fix poor concepts or biased items.	Improve measurement design and evaluate the instrument.
Assuming a validated scale transfers to a new language or population	Meanings and functioning may differ.	Check linguistic, cultural and empirical equivalence (measurement invariance).

Other threats to watch: **social desirability** (overstating saving discipline); **recall bias** (spending over six months; shorter reference periods help); and **common-method bias** when all variables come from one questionnaire at one time. Reduce it with careful design and separated measurement where possible; Harman's single-factor test alone does not show it is absent (Podsakoff et al., 2003). Finally, a valid questionnaire does not make an unrepresentative sample representative: sampling is a separate issue (D3).

Key takeaways

1. Reliability is consistency; validity is meaning. Reliable does not mean valid.
2. Collect several kinds of validity evidence (content, response process, structure, relations to other variables) and report reliability per scale, from your own data.
3. Report alpha and omega/CR, AVE and HTMT with their limits; never let one coefficient stand in for validity.

References cited

American Educational Research Association, American Psychological Association, & National Council on Measurement in Education. (2014). *Standards for educational and psychological testing*. AERA.

Bound, J., & Krueger, A. B. (1991). The extent of measurement error in longitudinal earnings data: Do two wrongs make a right? *Journal of Labor Economics, 9*(1), 1–24.

Campbell, D. T., & Fiske, D. W. (1959). Convergent and discriminant validation by the multitrait-multimethod matrix. *Psychological Bulletin, 56*(2), 81–105.

Cohen, J. (1960). A coefficient of agreement for nominal scales. *Educational and Psychological Measurement, 20*(1), 37–46.

Cronbach, L. J. (1951). Coefficient alpha and the internal structure of tests. *Psychometrika, 16*(3), 297–334.

Cronbach, L. J., & Meehl, P. E. (1955). Construct validity in psychological tests. *Psychological Bulletin, 52*(4), 281–302.

Fornell, C., & Larcker, D. F. (1981). Evaluating structural equation models with unobservable variables and measurement error. *Journal of Marketing Research, 18*(1), 39–50.

Henseler, J., Ringle, C. M., & Sarstedt, M. (2015). A new criterion for assessing discriminant validity in variance-based structural equation modeling. *Journal of the Academy of Marketing Science, 43*(1), 115–135.

Koo, T. K., & Li, M. Y. (2016). A guideline of selecting and reporting intraclass correlation coefficients for reliability research. *Journal of Chiropractic Medicine, 15*(2), 155–163.

Lawshe, C. H. (1975). A quantitative approach to content validity. *Personnel Psychology, 28*(4), 563–575.

MacKenzie, S. B., Podsakoff, P. M., & Jarvis, C. B. (2005). The problem of measurement model misspecification in behavioral and organizational research and some recommended solutions. *Journal of Applied Psychology, 90*(4), 710–730.

McNeish, D. (2018). Thanks coefficient alpha, we'll take it from here. *Psychological Methods, 23*(3), 412–433.

Messick, S. (1995). Validity of psychological assessment. *American Psychologist, 50*(9), 741–749.

Podsakoff, P. M., MacKenzie, S. B., Lee, J.-Y., & Podsakoff, N. P. (2003). Common method biases in behavioral research. *Journal of Applied Psychology, 88*(5), 879–903.

Polit, D. F., Beck, C. T., & Owen, S. V. (2007). Is the CVI an acceptable indicator of content validity? Appraisal and recommendations. *Research in Nursing & Health, 30*(4), 459–467.

Sekaran, U., & Bougie, R. (2016). *Research methods for business: A skill-building approach* (7th ed.). Wiley.

Shrout, P. E., & Fleiss, J. L. (1979). Intraclass correlations: Uses in assessing rater reliability. *Psychological Bulletin, 86*(2), 420–428.

Sijtsma, K. (2009). On the use, the misuse, and the very limited usefulness of Cronbach's alpha. *Psychometrika, 74*(1), 107–120.

Example

Worked example: assessing the e-wallet scales

Three reflective constructs, all five-point agreement items, 400 respondents (hypothetical). Computed with `04_reliability_validity.do` and its notebook.

Table E4.6. Reliability and convergent validity

Construct	Items	Loadings	Cronbach's α	CR (ω)	AVE	√AVE
Perceived usefulness (PU)	4	0.75–0.80	0.856	0.856	0.597	0.773
Perceived security (PS)	4	0.78–0.79	0.863	0.863	0.612	0.782
Ease of use (EOU)	3	0.70–0.75	0.775	0.775	0.535	0.731

Table E4.7. Discriminant validity: latent correlations (Fornell–Larcker) and HTMT

Pair	Latent correlation	Smaller √AVE of the pair	HTMT
PU – PS	0.447	0.773	0.449
PU – EOU	0.498	0.731	0.501
PS – EOU	0.376	0.731	0.379

Overall KMO = 0.86. Every √AVE exceeds the construct's correlations, and every HTMT is well below 0.85. All values are from simulated teaching data.

Reading the results. All three scales meet the conventional benchmarks for reliability (α and CR above 0.70) and convergent validity (AVE above 0.50), and the constructs are clearly distinct. These numbers alone would *not* establish validity: Chapter 3 should also report the

content review, the cognitive pretest and the adaptation from Davis (1989) (E2).

04_reliability_validity.do

```
use "$data/ewallet_survey_clean.dta", clear

* internal consistency, item by item
alpha PU1-PU4, item
alpha PS1-PS4, item
alpha EOU1-EOU3, item

* structure: do the 11 items form three factors?
factor PU1-PU4 PS1-PS4 EOU1-EOU3, ml factors(3)
rotate, promax
estat kmo

* CR and AVE from one-factor CFAs (program crave defined in the do-file)
crave PU1-PU4
display "CR = " r(cr) "   AVE = " r(ave)

* Fornell-Larcker: latent correlations from a joint CFA
sem (Usefulness -> PU1-PU4) (Security -> PS1-PS4) (EaseOfUse -> EOU1-EOU3), ///
    latent(Usefulness Security EaseOfUse) nocapslatent standardized nolog


from statsmodels.multivariate.factor import Factor

def cronbach_alpha(d):
    C = d.cov()
    k = C.shape[0]
    return k / (k - 1) * (1 - np.trace(C.values) / C.values.sum())

for name, cols in constructs.items():
    lam = np.abs(Factor(df[cols].dropna(), n_factor=1, method="ml").fit().loadings[:, 0])
    cr = lam.sum()**2 / (lam.sum()**2 + (1 - lam**2).sum())
    print(name, round(cronbach_alpha(df[cols]), 3), round(cr, 3), round((lam**2).mean(), 3))

# HTMT = mean heterotrait r / sqrt(mean monotrait r of i x of j)
Rall = df[items].corr()
```

Writeup

How it reads in the thesis

3.3.2 Reliability and validity of the measures

Content validity was assessed by three experts in consumer finance and survey methodology, who rated each adapted item's relevance; all item-level CVIs were at least 0.80. Cognitive interviews with five students led to the rewording of two items.[1] In the main sample, a confirmatory factor analysis supported the three-factor structure, with all standardised loadings between 0.70 and 0.80.[2] Reliability was acceptable for every scale: Cronbach's α was 0.86 (PU), 0.86 (PS) and 0.77 (EOU), and composite reliability took the same values. Average variance extracted exceeded 0.50 for all three constructs, supporting convergent validity.[3] Discriminant validity was supported by both the Fornell–Larcker criterion, with each √AVE above the construct's correlations, and the HTMT ratio, all below 0.51 against a threshold of 0.85 (Henseler et al., 2015).[4]

1 *Content and response-process evidence come first, before any statistics.*

2 *Internal structure.*

3 *Reliability per scale, and convergent validity, kept separate.*

4 *Discriminant validity by two criteria, with the source of the threshold.*

Model write-up. *Present the figures in a table; the prose interprets them.*

How much validation is enough?

Table E4.8. Proportionate expectations

Level	Suitable expectation
Undergraduate project	Clear constructs, literature-based items, supervisor review, pretest, pilot, basic reliability checks
Master's dissertation	Stronger theoretical justification, documented content review, factor analysis and reliability where applicable
PhD thesis or journal article	A validity argument proportionate to the instrument: documented provenance, measurement models, reliability evidence, robustness checks, transparent limitations

Worksheet

Worksheet E4

Q1 A student measures “financial literacy” with three items: “I am confident managing my finances”, “I am good at financial decisions”, “I understand how to manage money”. Alpha is 0.95. Evaluate the measure's reliability and validity. (5 marks)

Model answer

Reliability is very high, possibly too high (near-redundant items). But the items measure financial *confidence* (self-assessed competence), not financial literacy as objective knowledge, so content and construct validity are poor for the intended construct. A better measure would test knowledge of interest, inflation and diversification (Lusardi & Mitchell, 2014). The case shows that high reliability does not imply validity.

Q2 Using the CVI calculator: six experts rate five items; the numbers rating each item 3 or 4 are 6, 6, 5, 4, 6. Compute the I-CVIs, S-CVI/Ave and S-CVI/UA, and decide which item to revise. (4 marks)

Model answer

I-CVIs: 1.00, 1.00, 0.83, 0.67, 1.00. S-CVI/Ave = 4.50/5 = 0.90. S-CVI/UA = 3/5 = 0.60 (items with I-CVI = 1). Item 4 (0.67) is below the 0.78 guideline for six or more experts (Polit et al., 2007): revise or replace it, then ask the experts again.

Q3 Explain why calculating Cronbach's alpha for an asset-based wealth index (house, car, motorcycle, fridge, internet) is inappropriate, and say how you would support its validity instead. (4 marks)

Model answer

The index is formative: the assets jointly define wealth and need not be correlated (a household may own a car but no motorcycle), so low inter-item correlation is not a defect, and alpha assumes a reflective model. Support validity with content coverage (the assets that matter for wealth in Malaysia), justification of indicators and weights (for example, principal components), and criterion evidence such as correlation with reported income or expenditure.

Q4 A two-week test–retest of a financial-anxiety scale gives Pearson r = 0.85, but the mean rose by 0.6 points because exams started in week two. What does this tell you, and which statistic would be better? (3 marks)

Model answer

Pearson r shows that respondents kept their rank order, but it ignores the systematic shift in level, which here reflects a real change in anxiety (exams), not unreliability. An absolute-agreement ICC would be lower and more informative about agreement; better still, choose a retest interval with no expected change, since anxiety is not stable across exam periods.

F1 Worksheets and tutorials

Every section has a worksheet with model answers. This page collects them, adds an integrated tutorial that takes one study from research questions to model specification, and suggests a teaching sequence.

Worksheets

Worksheets by section

Table F1.1. Worksheets, with what each one practises

Worksheet	Practises	Questions
A1 Methodology, design, methods	Placing study elements at the right level	4
A2 Correlation and causation	Claims, threats and wording	4
B1 3.1 Data	Data description; item-to-variable map	3
B2 3.2 RQ–method table	Matching methods to questions	3
B3 3.3 Codebook	Operationalisation and coding	3
B4 3.4 Model specification	Equations, estimators, standard errors	3
B5 Putting it together	Limitations; the submission checklist	2
C1 Levels of measurement	Dummies; nominal variables in regression	3
C2 Data structures	Identifying structures; panel notation	3
C3 Primary data	Choosing methods; non-response	2
C4 Secondary data	Finding and citing datasets	2
C5 Data cleaning	Cleaning rules in Stata and Python	3
D1 Sampling concepts	Population, frame and errors	2
D2 Probability sampling	Allocation and weights	3
D3 Non-probability sampling	Designs and honest reporting	2
D4 How many respondents?	Precision and power calculations	3
E1 Fix the items and worksheet	Item writing	9
E2 Adopt and adapt	Choosing and adapting instruments	2
E3 Likert and rating scales	Formats and estimators	2
E4 Validity and reliability	Evaluating evidence; CVI	4

Word versions of all worksheets, with and without model answers, are in the offline ZIP (F4).

Tutorial

Integrated tutorial: from research questions to Chapter 3

The case

A PhD student plans a thesis on **"Remittances and children's schooling in rural Sabah."** The student wants to know (1) how many households receive remittances and how much; (2) whether receiving remittances is associated with higher education spending per child; and (3) whether any association is stronger for daughters than for sons. There is no list of households, but the district office keeps a list of the 120 villages in two districts, with their approximate number of households.

Work through the six steps. Each has a model answer.

Q1 **Step 1: Classify the research questions** as descriptive, correlational or causal, and say what wording Chapter 4 can use (A2).

Model answer

RQ1 is descriptive (proportions and means). RQ2 and RQ3 are correlational: one cross-sectional survey cannot rule out that families who value education both migrate and spend more. Wording: "associated with". A causal claim would need, for example, variation in remittances caused by shocks in migrants' destinations.

Q2 **Step 2: Design the sample** (D2, D4). Choose a design, explain how households are selected, and justify the size.

Model answer

Two-stage cluster sample: select 30 villages with probability proportional to size, list all households in each, then select 15 households per village by systematic sampling, giving 450 households. Size: for RQ2, detecting a small partial association ($f^2 = 0.02$) with 8 controls at 80% power needs about 400 observations; with an expected design effect of $1 + (15 - 1) \times 0.05 = 1.7$, the effective sample of 450 is about 265, so either more villages (about 45) or a stated acceptance of lower power. Inflate for 85% response.

Q3 **Step 3: Build the RQ–method table** (B2).

Model answer

RQ	Outcome (level)	Main variable	Method
RQ1 Incidence and size of remittances	Receives (binary); amount RM (ratio)	—	Weighted proportions and means, design-based SEs
RQ2 Remittances and education spending	Education spending per child (ratio; zeros possible)	Receives remittances; amount	OLS on log(1 + spending) or a two-part model; SEs clustered by village
RQ3 Daughters versus sons	Spending per child (child-level)	Remittances × girl	Child-level regression with household controls and an interaction; clustered SEs

---

Q4 **Step 4: Write three codebook rows** (B3): the remittance indicator, education spending, and the girl dummy.

Model answer

`remit` : household received money from a member living elsewhere in the past 12 months; item C1; nominal (binary); 1 = yes, 0 = no. `edu_spend` : spending on fees, books, uniforms, transport and tuition for the child in the past school year, RM; items D3–D7 summed; ratio. `girl` : child is female; household roster; nominal (binary); 1 = girl, 0 = boy.

Q5 **Step 5: Specify the model for RQ3** with correct subscripts (B4, C2).

Model answer

$spend_{ch} = \beta_0 + \beta_1 remit_h + \beta_2 girl_{ch} + \beta_3 (remit_h \times girl_{ch}) + \mathbf{x}'_{ch}\boldsymbol{\gamma} + \mathbf{w}'_h\boldsymbol{\delta} + u_{ch}$, for child c in household h. β_3 answers RQ3. Child controls $\mathbf{x}$: age, school level. Household controls $\mathbf{w}$: income, parents' education, distance to school. Standard errors clustered by village (the primary sampling unit).

Q6 **Step 6: Name two limitations** and their consequences (B5).

Model answer

(1) Cross-sectional data: remittance-receiving households may differ in unobserved ways (motivation, networks), so estimates are associations. (2) Recall of a year's education spending is error-prone; classical error in the outcome lowers precision but does not bias the coefficients, while systematic under-reporting by some groups would. Also acceptable: the two districts cannot represent all of Sabah.

Plan

A suggested six-week sequence

Table F1.2. One way to teach this material: a 2-hour lecture and a 1-hour tutorial each week

Week	Lecture	Tutorial	Students produce
1	Start, A1, A2	Worksheets A1–A2	Paradigm-and-design paragraph for their own study
2	C1, C2, C3, C4	Worksheets C1–C2; find a dataset (C4)	Data-structure decision and data sources
3	D1, D2, D3	Simulator; allocation exercise	Sampling design with justification
4	D4	Sample-size and power calculators; run `02_sample_size`	Sample-size justification
5	E1, E2, E3, E4	Fix the items; CVI; run `04_reliability_validity`	Draft instrument and adaptation log
6	B1–B5, C5	Integrated tutorial; run `03` and `06`	Draft Chapter 3 checked against the B5 checklist

F2 Glossary

Short definitions of the terms used across this guide, with a pointer to the section that explains each one. Search (Ctrl K) finds them too.

A C D E F H I L M N O P R S T V

A

Acquiescence bias — The tendency to agree with statements regardless of their content (E3).

Adapt (an instrument) — Modify an established instrument's wording, context, language or format; requires re-validation (E2).

Adopt (an instrument) — Use an established instrument unchanged (E2).

Attenuation bias — The shrinking of a regression coefficient towards zero caused by classical measurement error in a regressor (E4).

Average marginal effect (AME) — The average, over the sample, of the change in predicted probability for a one-unit change in a regressor; the usual way to report logit and probit results (B4).

Average variance extracted (AVE) — The mean squared standardised loading of a construct's items; ≥ 0.50 supports convergent validity (E4).

C

Cluster sampling — Selecting groups (villages, schools) and then units within them (D2).

Codebook — A table defining every variable's name, meaning, source item, level of measurement and coding (B3).

Common-method bias — Spurious correlation arising because all variables are measured with the same method at the same time (E4).

Composite reliability (CR) — A model-based reliability coefficient from standardised loadings, equal to omega for a single-factor model (E4).

Construct — The concept to be measured, often latent, such as trust or perceived usefulness (E1).

Content validity index (CVI) — The proportion of experts rating an item relevant; averaged across items for a scale-level index (E4).

Convenience sampling — Recruiting whoever is available and willing (D3).

Cronbach's alpha — A coefficient of internal-consistency reliability for a set of items (E4).

Cross-sectional data — Many units observed at one point in time (C2).

D

Design effect (DEFF) — The ratio of a complex design's variance to that of a simple random sample of the same size; $1 + (m - 1)\rho$ for equal clusters (D4).

Design weight — The inverse of a unit's selection probability, N_h / n_h in stratified sampling (D2).

Difference-in-differences (DiD) — Compares the change in outcomes for a treated group with the change for a comparison group (C2, B4).

Discriminant validity — Evidence that measures of different constructs are distinct (Fornell–Larcker, HTMT) (E4).

Dummy variable A 0/1 variable named after the category coded 1; its mean is the share coded 1 (C1).

E

Effect size The magnitude of a difference or association, such as Cohen's d or f^2, used in power analysis (D4).

Exclusion restriction The assumption that an instrument affects the outcome only through the endogenous regressor (A2).

External validity Whether findings apply beyond the study's setting and sample (E4).

F

Fixed effects Unit-specific intercepts in panel data that absorb all time-invariant characteristics (C2).

Focus group discussion (FGD) A moderated group discussion among 6–10 people sharing a relevant characteristic (C3).

Formative index A composite whose indicators define the construct and need not correlate; alpha is inappropriate (E4).

Fornell–Larcker criterion Discriminant validity holds if each construct's √AVE exceeds its correlations with other constructs (E4).

H

Hausman test Tests whether fixed- and random-effects estimates differ systematically; rejection favours fixed effects (C2).

HTMT Heterotrait–monotrait ratio of correlations; below 0.85 (or 0.90) supports discriminant validity (E4).

I

Identification strategy The design feature that lets an estimate be read causally, together with its assumption (A2).

Instrument (data collection) The tool used to collect data: questionnaire, interview guide, observation sheet (E1).

Internal validity Whether an estimated relationship can credibly be read as causal (A2, E4).

Interval scale Ordered values with equal intervals but no true zero (C1).

L

Likert scale Strictly, several agreement items combined to measure one construct; a single item is a Likert-type item (E3).

M

Margin of error Half the width of a confidence interval; ±5 percentage points in the Krejcie–Morgan table (D4).

Methodology The philosophy and logic that justify how knowledge is produced; it contains research design, which contains methods (A1).

Methods The specific techniques and tools of data collection and analysis (A1).

N

Nominal scale Categories without order (C1).

Non-probability sampling Designs in which selection probabilities are unknown or zero for some units (D3).

O

Omega (McDonald's ω) A factor-model-based reliability coefficient that allows unequal loadings (E4).

Omitted variable bias Bias from leaving out a variable that affects the outcome and is correlated with an included regressor (A2).

Operationalisation Turning a concept into a measurable, coded variable (B3).

Ordinal scale Ordered categories with unknown or unequal intervals (C1).

P

Panel data The same units observed repeatedly over time (C2).

Paradigm A set of assumptions about what exists and how it can be known: positivism, interpretivism, pragmatism (A1).

Parallel trends The DiD assumption that treated and comparison groups would have moved together without treatment (A2).

Pilot study A small trial of the instrument and procedures before fieldwork (E1).

Pooled cross-sections Independent samples from the same population at different times (C2).

Power The probability of detecting an effect of a given size when it exists; usually set at 0.80 (D4).

Probability sampling Every unit has a known, non-zero chance of selection (D2).

Purposive sampling Deliberately selecting units that meet criteria relevant to the research (D3).

R

Ratio scale Equal intervals and a true zero (C1).

Reflective measurement A model in which the latent construct causes its indicators, which should therefore correlate (E4).

Reliability The consistency or stability of measurement (E4).

Research design The overall plan of a study; sits between methodology and methods (A1).

Response rate Usable responses divided by eligible units invited (B1).

RQ–method table A table pairing each research question with its data, variables and method (B2).

S

Sampling frame The list or procedure from which the sample is drawn (D1).

Saturation The point at which new interviews add no new themes (C3).

Simple random sampling Every unit, and every sample of size n, has an equal chance of selection (D2).

Snowball sampling Participants recruit further participants; used for hidden populations (D3).

Stratified random sampling Random sampling within non-overlapping subgroups (strata) (D2).

Systematic sampling Selecting every k-th unit after a random start (D2).

T

Target population The precisely defined population the findings describe (D1).

Test–retest reliability	Agreement between scores from two administrations of the same instrument (E4).
Time-series data	One unit observed over many periods (C2).

V

Validity	The degree to which evidence supports the intended interpretation of scores, and more broadly the soundness of conclusions (E4).

F3 Reading list

Peer-reviewed journal articles on writing the Methodology chapter and the methods behind it. Every citation below is generated from the article's Crossref record. Where a legal open-access copy exists and loaded when checked, a “Free full text” link is given; otherwise use the DOI through your university library.

72 articles, each checked against the Crossref record on 2026-10-19. 24 have a free full-text copy that loaded when checked; the rest are reachable through your library via the DOI.

Methodology, paradigms and research design

Carter, S. M., & Little, M. (2007). Justifying Knowledge, Justifying Method, Taking Action: Epistemologies, Methodologies, and Methods in Qualitative Research. *Qualitative Health Research*, *17*(10), 1316-1328.

The clearest published statement of how epistemology, methodology and method nest — the backbone of tab A1.

doi:10.1177/1049732307306927

Johnson, R. B., & Onwuegbuzie, A. J. (2004). Mixed Methods Research: A Research Paradigm Whose Time Has Come. *Educational Researcher*, *33*(7), 14-26.

Defines mixed-methods research and its pragmatist rationale; useful if your design combines a survey with interviews or FGDs.

doi:10.3102/0013189x033007014

Morgan, D. L. (2007). Paradigms Lost and Pragmatism Regained. *Journal of Mixed Methods Research*, *1*(1), 48-76.

Explains paradigms in plain terms and argues for pragmatism as the working philosophy of applied research.

doi:10.1177/2345678906292462

Identification and model specification in economics

Angrist, J. D., & Pischke, J. (2010). The Credibility Revolution in Empirical Economics: How Better Research Design is Taking the Con out of Econometrics. *Journal of Economic Perspectives*, *24*(2), 3-30.

How design-based identification changed empirical economics; read before writing your model-specification justification.

doi:10.1257/jep.24.2.3 **Free full text (web)**

Imbens, G. W., & Wooldridge, J. M. (2009). Recent Developments in the Econometrics of Program Evaluation. *Journal of Economic Literature*, *47*(1), 5-86.

The standard survey of programme-evaluation estimators (matching, DiD, IV, RD) with their identifying assumptions.

doi:10.1257/jel.47.1.5 **Free full text (web)**

Kiel, K. A., & McClain, K. T. (1995). House Prices during Siting Decision Stages: The Case of an Incinerator from Rumor through Operation. *Journal of Environmental Economics and Management*, *28*(2), 241-255.

Source of the pooled cross-section incinerator example used in tab C2 (year81 × nearinc).

doi:10.1006/jeem.1995.1016

Lee, D. S., & Lemieux, T. (2010). Regression Discontinuity Designs in Economics. *Journal of Economic Literature*, *48*(2), 281-355.

The practitioner's guide to regression discontinuity — what to report and how to defend an RD design.

doi:10.1257/jel.48.2.281 **Free full text (PDF)**

Williams, R. (2006). Generalized Ordered Logit/Partial Proportional Odds Models for Ordinal Dependent Variables. *The Stata Journal: Promoting communications on statistics and Stata*, *6*(1), 58-82.

Methodological justification for the partial proportional odds model used in the RQ–method table example.

doi:10.1177/1536867x0600600104

Levels of measurement

Stevens, S. S. (1946). On the Theory of Scales of Measurement. *Science*, *103*(2684), 677-680.

The original nominal–ordinal–interval–ratio typology and the statistics each level permits.

doi:10.1126/science.103.2684.677

Likert and rating scales

Carifio, J., & Perla, R. (2008). Resolving the 50-year debate around using and misusing Likert scales. *Medical Education*, *42*(12), 1150-1152.

The reply to Jamieson: distinguishes Likert items from Likert scales and defends parametric analysis of summated scales.

doi:10.1111/j.1365-2923.2008.03172.x

Jamieson, S. (2004). Likert scales: how to (ab)use them. *Medical Education*, *38*(12), 1217-1218.

A short, much-cited warning about treating Likert items as interval data.

doi:10.1111/j.1365-2929.2004.02012.x

Liddell, T. M., & Kruschke, J. K. (2018). Analyzing ordinal data with metric models: What could possibly go wrong?. *Journal of Experimental Social Psychology*, *79*, 328-348.

Shows when treating ordinal outcomes as metric produces wrong conclusions — the case for ordered models.

doi:10.1016/j.jesp.2018.08.009 **Free full text (web)**

Norman, G. (2010). Likert scales, levels of measurement and the “laws” of statistics. *Advances in Health Sciences Education*, *15*(5), 625-632.

Shows parametric methods are robust with Likert data — the key reference when you run OLS on a scale score.

doi:10.1007/s10459-010-9222-y

Preston, C. C., & Colman, A. M. (2000). Optimal number of response categories in rating scales: reliability, validity, discriminating power, and respondent preferences. *Acta Psychologica*, *104*(1), 1-15.

Compares 2- to 11-point and 101-point rating scales on reliability, validity and respondent preference.

doi:10.1016/s0001-6918(99)00050-5

Revilla, M. A., Saris, W. E., & Krosnick, J. A. (2014). Choosing the Number of Categories in Agree–Disagree Scales. *Sociological Methods & Research*, *43*(1), 73-97.

Experimental evidence on how many response categories agree–disagree scales should have.

doi:10.1177/0049124113509605 **Free full text (web)**

Sullivan, G. M., & Artino, A. R. (2013). Analyzing and Interpreting Data From Likert-Type Scales. *Journal of Graduate Medical Education*, *5*(4), 541-542.

Two-page practical guide to analysing and reporting Likert-type data.

doi:10.4300/jgme-5-4-18 **Free full text (web)**

Sampling and sample size

Baker, R., Brick, J. M., Bates, N. A., Battaglia, M., Couper, M. P., Dever, J. A., Gile, K. J., & Tourangeau, R. (2013). Summary Report of the AAPOR Task Force on Non-probability Sampling. *Journal of Survey Statistics and Methodology*, *1*(2), 90-143.

Authoritative review of what can and cannot be inferred from non-probability samples, including online panels.

doi:10.1093/jssam/smt008

Biernacki, P., & Waldorf, D. (1981). Snowball Sampling: Problems and Techniques of Chain Referral Sampling. *Sociological Methods & Research*, *10*(2), 141-163.

The methodological classic on snowball sampling of hidden populations.

doi:10.1177/004912418101000205

Cohen, J. (1992). A power primer.. *Psychological Bulletin*, *112*(1), 155-159.

Defines small/medium/large effect sizes and gives sample sizes for 80% power across common tests.

doi:10.1037/0033-2909.112.1.155

Faul, F., Erdfelder, E., Lang, A., & Buchner, A. (2007). G*Power 3: A flexible statistical power analysis program for the social, behavioral, and biomedical sciences. *Behavior Research Methods*, *39*(2), 175-191.

The G*Power 3 paper — cite it when you report an a priori power analysis.

doi:10.3758/bf03193146 **Free full text (web)**

Faul, F., Erdfelder, E., Buchner, A., & Lang, A. (2009). Statistical power analyses using G*Power 3.1: Tests for correlation and regression analyses. *Behavior Research Methods*, *41*(4), 1149-1160.

Power analysis for correlation and regression in G*Power 3.1 (the f^2 approach used in tab D4).

doi:10.3758/brm.41.4.1149 **Free full text (web)**

Green, S. B. (1991). How Many Subjects Does It Take To Do A Regression Analysis. *Multivariate Behavioral Research*, *26*(3), 499-510.

Origin of the N ≥ 50 + 8m rule of thumb — and why power analysis does better.

doi:10.1207/s15327906mbr2603_7

Groves, R. M. (2006). Nonresponse Rates and Nonresponse Bias in Household Surveys. *Public Opinion Quarterly*, *70*(5), 646-675.

Why a high response rate does not guarantee low nonresponse bias.

doi:10.1093/poq/nfl033

Heckathorn, D. D. (1997). Respondent-Driven Sampling: A New Approach to the Study of Hidden Populations. *Social Problems*, *44*(2), 174-199.

How respondent-driven sampling improves on snowball sampling for hard-to-reach groups.

doi:10.1525/sp.1997.44.2.03x0221m

Ioannidis, J. P. A., Stanley, T. D., & Doucouliagos, H. (2017). The Power of Bias in Economics Research. *The Economic Journal*, *127*(605), F236-F265.

Finds most economics studies are underpowered — the economics case for power analysis.

doi:10.1111/ecoj.12461

Krejcie, R. V., & Morgan, D. W. (1970). Determining Sample Size for Research Activities. *Educational and Psychological Measurement*, *30*(3), 607-610.

The original sample-size table discussed in tab D4; read it with Lakens (2022).

doi:10.1177/001316447003000308

Lakens, D. (2022). Sample Size Justification. *Collabra: Psychology*, *8*(1), 33267.

The best modern guide to justifying a sample size: precision, power, resource constraints and heuristics.

doi:10.1525/collabra.33267 **Free full text (web)**

McKenzie, D. (2012). Beyond baseline and follow-up: The case for more T in experiments. *Journal of Development Economics*, *99*(2), 210-221.

Power and sample-size planning for field experiments with repeated rounds of data.

doi:10.1016/j.jdeveco.2012.01.002

Palinkas, L. A., Horwitz, S. M., Green, C. A., Wisdom, J. P., Duan, N., & Hoagwood, K. (2015). Purposeful Sampling for Qualitative Data Collection and Analysis in Mixed Method Implementation Research. *Administration and Policy in Mental Health and Mental Health Services Research*, *42*(5), 533-544.

Typology of purposive sampling strategies and when each fits.

doi:10.1007/s10488-013-0528-y **Free full text (web)**

Peduzzi, P., Concato, J., Kemper, E., Holford, T. R., & Feinstein, A. R. (1996). A simulation study of the number of events per variable in logistic regression analysis. *Journal of Clinical Epidemiology*, *49*(12), 1373-1379.

The events-per-variable guideline for logit models with a binary outcome.

doi:10.1016/s0895-4356(96)00236-3 **Free full text (web)**

Questionnaire design and surveys

Beaton, D. E., Bombardier, C., Guillemin, F., & Ferraz, M. B. (2000). Guidelines for the Process of Cross-Cultural Adaptation of Self-Report Measures. *Spine*, *25*(24), 3186-3191.

Step-by-step protocol for adapting an established instrument to a new language and culture.

doi:10.1097/00007632-200012150-00014

Beatty, P. C., & Willis, G. B. (2007). Research Synthesis: The Practice of Cognitive Interviewing. *Public Opinion Quarterly*, *71*(2), 287-311.

How to pretest questionnaire items with cognitive interviews before the pilot.

doi:10.1093/poq/nfm006

Bertrand, M., & Mullainathan, S. (2001). Do People Mean What They Say? Implications for Subjective Survey Data. *American Economic Review*, *91*(2), 67-72.

When subjective survey answers can and cannot be used as variables in economics.

doi:10.1257/aer.91.2.67

Brislin, R. W. (1970). Back-Translation for Cross-Cultural Research. *Journal of Cross-Cultural Psychology*, *1*(3), 185-216.

The back-translation procedure for English–Malay (or other) questionnaire versions.

doi:10.1177/135910457000100301

Krosnick, J. A. (1999). Survey research. *Annual Review of Psychology*, *50*(1), 537-567.

Evidence-based review of question wording, response options, acquiescence and satisficing.

doi:10.1146/annurev.psych.50.1.537

Lusardi, A., & Mitchell, O. S. (2014). The Economic Importance of Financial Literacy: Theory and Evidence. *Journal of Economic Literature*, *52*(1), 5-44.

How economists measure financial literacy (the 'Big Three' questions) — an example of an adopted instrument.

doi:10.1257/jel.52.1.5 **Free full text (web)**

Stantcheva, S. (2023). How to Run Surveys: A Guide to Creating Your Own Identifying Variation and Revealing the Invisible. *Annual Review of Economics*, *15*(1), 205-234.

Survey design written by and for economists: sampling, question design, attention checks, experiments in surveys.

doi:10.1146/annurev-economics-091622-010157

Validity and reliability

Boateng, G. O., Neilands, T. B., Frongillo, E. A., Melgar-Quiñonez, H. R., & Young, S. L. (2018). Best Practices for Developing and Validating Scales for Health, Social, and Behavioral Research: A Primer. *Frontiers in Public Health*, *6*, 149.

A nine-step primer for developing and validating scales — the practical workflow behind tab E4.

doi:10.3389/fpubh.2018.00149 **Free full text (PDF)**

Bound, J., & Krueger, A. B. (1991). The Extent of Measurement Error in Longitudinal Earnings Data: Do Two Wrongs Make a Right?. *Journal of Labor Economics*, *9*(1), 1-24.

Measures reporting error in survey earnings against administrative records — reliability in economics data.

doi:10.1086/298256

Campbell, D. T., & Fiske, D. W. (1959). Convergent and discriminant validation by the multitrait-multimethod matrix.. *Psychological Bulletin*, *56*(2), 81-105.

Introduced convergent and discriminant validity.

doi:10.1037/h0046016

Cohen, J. (1960). A Coefficient of Agreement for Nominal Scales. *Educational and Psychological Measurement*, *20*(1), 37-46.

The original kappa statistic for inter-rater reliability.

doi:10.1177/001316446002000104

Cronbach, L. J. (1951). Coefficient Alpha and the Internal Structure of Tests. *Psychometrika*, *16*(3), 297-334.

The original coefficient alpha paper.

doi:10.1007/bf02310555 **Free full text (web)**

Cronbach, L. J., & Meehl, P. E. (1955). Construct validity in psychological tests.. *Psychological Bulletin*, *52*(4), 281-302.

Where construct validity was defined.

doi:10.1037/h0040957

Fornell, C., & Larcker, D. F. (1981). Evaluating Structural Equation Models with Unobservable Variables and Measurement Error. *Journal of Marketing Research*, *18*(1), 39.

Source of average variance extracted (AVE) and the Fornell–Larcker discriminant-validity criterion.

doi:10.2307/3151312

Henseler, J., Ringle, C. M., & Sarstedt, M. (2015). A new criterion for assessing discriminant validity in variance-based structural equation modeling. *Journal of the Academy of Marketing Science*, *43*(1), 115-135.

Introduces the HTMT ratio and shows why Fornell–Larcker often misses discriminant-validity problems.

doi:10.1007/s11747-014-0403-8 **Free full text (web)**

Hinkin, T. R. (1998). A Brief Tutorial on the Development of Measures for Use in Survey Questionnaires. *Organizational Research Methods*, *1*(1), 104-121.

Step-by-step scale development: item generation, content validity, factor analysis, reliability.

doi:10.1177/109442819800100106

LAWSHE, C. H. (1975). A quantitative approach to content validity. *Personnel Psychology*, *28*(4), 563-575.

The content validity ratio (CVR) for expert panels.

doi:10.1111/j.1744-6570.1975.tb01393.x

MacKenzie, S. B., Podsakoff, P. M., & Jarvis, C. B. (2005). The Problem of Measurement Model Misspecification in Behavioral and Organizational Research and Some Recommended Solutions.. *Journal of Applied Psychology*, *90*(4), 710-730.

Reflective versus formative indicators, and the bias from getting this wrong.

doi:10.1037/0021-9010.90.4.710

McNeish, D. (2018). Thanks coefficient alpha, we'll take it from here.. *Psychological Methods*, *23*(3), 412-433.

Why and how to report omega and other reliability coefficients instead of alpha alone.

doi:10.1037/met0000144

Messick, S. (1995). Validity of psychological assessment: Validation of inferences from persons' responses and performances as scientific inquiry into score meaning.. *American Psychologist*, *50*(9), 741-749.

The unified view of validity as evidence for an interpretation — the basis of the modern standards.

doi:10.1037/0003-066x.50.9.741

Podsakoff, P. M., MacKenzie, S. B., Lee, J., & Podsakoff, N. P. (2003). Common method biases in behavioral research: A critical review of the literature and recommended remedies.. *Journal of Applied Psychology*, *88*(5), 879-903.

Common-method bias in single-source surveys and the procedural remedies to build into your questionnaire.

doi:10.1037/0021-9010.88.5.879

Polit, D. F., Beck, C. T., & Owen, S. V. (2007). Is the CVI an acceptable indicator of content validity? Appraisal and recommendations. *Research in Nursing & Health*, *30*(4), 459-467.

How to compute and interpret item- and scale-level content validity indices from expert ratings.

doi:10.1002/nur.20199

Shrout, P. E., & Fleiss, J. L. (1979). Intraclass correlations: Uses in assessing rater reliability.. *Psychological Bulletin*, *86*(2), 420-428.

Which intraclass correlation to use for test–retest and inter-rater reliability.

doi:10.1037/0033-2909.86.2.420

Sijtsma, K. (2009). On the Use, the Misuse, and the Very Limited Usefulness of Cronbach's Alpha. *Psychometrika*, *74*(1), 107-120.

A critique of how Cronbach's alpha is misused and misread.

doi:10.1007/s11336-008-9101-0 **Free full text (web)**

Tavakol, M., & Dennick, R. (2011). Making sense of Cronbach's alpha. *International Journal of Medical Education*, *2*, 53-55.

A short, readable explanation of what alpha does and does not tell you.

doi:10.5116/ijme.4dfb.8dfd **Free full text (PDF)**

Interviews and focus groups

Braun, V., & Clarke, V. (2006). Using thematic analysis in psychology. *Qualitative Research in Psychology*, *3*(2), 77-101.

The six-phase thematic analysis procedure for interview and FGD data.

doi:10.1191/1478088706qp063oa Free full text (web)

Guest, G., Bunce, A., & Johnson, L. (2006). How Many Interviews Are Enough?. *Field Methods*, *18*(1), 59-82.

Evidence on when interview data reach saturation.

doi:10.1177/1525822x05279903

Kallio, H., Pietilä, A., Johnson, M., & Kangasniemi, M. (2016). Systematic methodological review: developing a framework for a qualitative semi-structured interview guide. *Journal of Advanced Nursing*, *72*(12), 2954-2965.

A five-phase framework for building a semi-structured interview guide.

doi:10.1111/jan.13031

Malterud, K., Siersma, V. D., & Guassora, A. D. (2016). Sample Size in Qualitative Interview Studies. *Qualitative Health Research*, *26*(13), 1753-1760.

Justifying the number of interviews with 'information power' rather than saturation alone.

doi:10.1177/1049732315617444

Morgan, D. L. (1996). Focus Groups. *Annual Review of Sociology*, *22*(1), 129-152.

What focus groups are good for, how to run them, and how they combine with surveys.

doi:10.1146/annurev.soc.22.1.129

Secondary data and data quality

Deaton, A. (2005). Measuring Poverty in a Growing World (or Measuring Growth in a Poor World). *Review of Economics and Statistics*, *87*(1), 1-19.

Why national accounts and household surveys disagree — a lesson in choosing a data source.

doi:10.1162/0034653053327612

Einav, L., & Levin, J. (2014). Economics in the age of big data. *Science*, *346*(6210), 1243089.

New administrative and private-sector data sources and what they offer economic research.

doi:10.1126/science.1243089

Feenstra, R. C., Inklaar, R., & Timmer, M. P. (2015). The Next Generation of the Penn World Table. *American Economic Review*, *105*(10), 3150-3182.

Documentation of the Penn World Table — how to read and cite a major secondary macro dataset.

doi:10.1257/aer.20130954 Free full text (web)

Johnson, S., Larson, W., Papageorgiou, C., & Subramanian, A. (2013). Is newer better? Penn World Table Revisions and their impact on growth estimates. *Journal of Monetary Economics*, *60*(2), 255-274.

Shows that dataset revisions can change results — why you must report the exact data version.

doi:10.1016/j.jmoneco.2012.10.022

Meyer, B. D., Mok, W. K. C., & Sullivan, J. X. (2015). Household Surveys in Crisis. *Journal of Economic Perspectives*, *29*(4), 199-226.

Rising nonresponse and measurement error in the big household surveys economists rely on.

doi:10.1257/jep.29.4.199 **Free full text (web)**

Data cleaning, reproducibility and reporting

Broman, K. W., & Woo, K. H. (2018). Data Organization in Spreadsheets. *The American Statistician*, *72*(1), 2-10.

Practical rules for organising data in spreadsheets before analysis.

doi:10.1080/00031305.2017.1375989

Christensen, G., & Miguel, E. (2018). Transparency, Reproducibility, and the Credibility of Economics Research. *Journal of Economic Literature*, *56*(3), 920-980.

Why transparent, reproducible data handling (do-files, raw vs clean data) is now expected in economics.

doi:10.1257/jel.20171350

Leys, C., Ley, C., Klein, O., Bernard, P., & Licata, L. (2013). Detecting outliers: Do not use standard deviation around the mean, use absolute deviation around the median. *Journal of Experimental Social Psychology*, *49*(4), 764-766.

A robust rule for flagging outliers during cleaning.

doi:10.1016/j.jesp.2013.03.013 **Free full text (web)**

Schafer, J. L., & Graham, J. W. (2002). Missing data: Our view of the state of the art.. *Psychological Methods*, *7*(2), 147-177.

Missing-data mechanisms (MCAR, MAR, MNAR) and modern handling methods.

doi:10.1037/1082-989x.7.2.147

Van den Broeck, J., Argeseanu Cunningham, S., Eeckels, R., & Herbst, K. (2005). Data Cleaning: Detecting, Diagnosing, and Editing Data Abnormalities. *PLoS Medicine*, *2*(10), e267.

A framework for screening, diagnosing and editing data abnormalities.

doi:10.1371/journal.pmed.0020267 **Free full text (PDF)**

Wasserstein, R. L., & Lazar, N. A. (2016). The ASA Statement on p -Values: Context, Process, and Purpose. *The American Statistician*, *70*(2), 129-133.

The ASA statement on p-values — how to report and interpret significance in your results chapter.

doi:10.1080/00031305.2016.1154108 **Free full text (web)**

Wickham, H. (2014). Tidy Data. *Journal of Statistical Software*, *59*(10).

The tidy-data principles: one variable per column, one observation per row.

doi:10.18637/jss.v059.i10 **Free full text (web)**

Textbooks cited in the notes (Wooldridge; Sekaran and Bougie; Crotty) are not listed here because this list is limited to journal articles.

F4 Downloads

Keep everything. The offline ZIP contains this whole guide, which works without an internet connection, plus the lecture notes as a PDF, worksheets and templates in Word, the Stata do-files, the Jupyter notebooks and the practice datasets.

Everything in one file

The whole dashboard plus every file below, for offline use. Unzip it and double-click `index.html`; no internet connection needed.

Download the offline ZIP (3.9 MB)

Lecture notes as one PDF (2.7 MB)

Stata do-files and Jupyter notebooks

Run as they are: every file is tested on each build.

Stata do-files and Jupyter notebooks

File	Size
01_sampling.do	3.5 KB
01_sampling.ipynb	16.5 KB
02_sample_size.do	3.0 KB
02_sample_size.ipynb	59.6 KB
03_data_cleaning.do	4.4 KB
03_data_cleaning.ipynb	25.9 KB
04_reliability_validity.do	3.9 KB
04_reliability_validity.ipynb	21.4 KB
05_data_structures.do	3.1 KB
05_data_structures.ipynb	13.6 KB
06_model_specification.do	2.0 KB
06_model_specification.ipynb	15.8 KB

Practice datasets

Hypothetical data used in the worked examples (CSV and Stata .dta).

Practice datasets

File	Size
codebook_ewallet.xlsx	5.0 KB
ds_crosssection_students.csv	852 bytes
ds_crosssection_students.dta	7.2 KB
ds_panel_countries.csv	569 bytes
ds_panel_countries.dta	5.5 KB
ds_pooled_crosssection.csv	554 bytes
ds_pooled_crosssection.dta	4.8 KB
ds_timeseries_macro.csv	936 bytes
ds_timeseries_macro.dta	5.4 KB
ewallet_survey_clean.csv	63.6 KB
ewallet_survey_clean.dta	49.7 KB
ewallet_survey_raw.csv	34.5 KB
ewallet_survey_raw.dta	50.6 KB
panel_states_sim.csv	4.3 KB
panel_states_sim.dta	9.7 KB
pooled_incinerator_sim.csv	11.9 KB
pooled_incinerator_sim.dta	20.6 KB
sme_sampling_frame.csv	82.0 KB
sme_sampling_frame.dta	105.1 KB

Worksheets

Word versions of every worksheet, with and without model answers.

Worksheets

File	Size
Chapter3_Worksheets.docx	43.0 KB
Chapter3_Worksheets_with_model_answers.docx	52.2 KB

Templates

Starting points for your own Chapter 3 tables.

Templates

File	Size
Adaptation_log_template.docx	36.3 KB
Chapter3_outline_template.docx	36.9 KB
Codebook_template.xlsx	7.3 KB
RQ_method_table_template.docx	36.3 KB

Original lecture materials

The instructor's own notes and examples.

Original lecture materials

File	Size
datastructure_emtmodel.png	450.7 KB
questionnaire_adopt_adapt.png	84.0 KB
RQ_method_table_eg.docx	16.0 KB
SJJ_Chpt3_lectnotes.docx	366.9 KB
soon2018_VarOp_table.png	123.3 KB

Running the code

Stata Unzip, open Stata, change the working directory to `downloads/code` (File → Change Working Directory), then run the do-files in order: `do 03_data_cleaning.do` first, because 04 and 06 use its output. Built-in commands only; Stata 16 or later.

Python Open the notebooks in Jupyter, VS Code or Google Colab. The first cell installs any missing package. Run `03_data_cleaning.ipynb` before 04 and 06. On Colab the data load from this guide's GitHub repository automatically.

Tested Every do-file and notebook is run automatically each time this guide is rebuilt, and must finish without a single error before the ZIP is produced.

PhD Research Methods · Chapter 3: Methodology. Teaching notes, examples and worksheets for writing the Methodology chapter of an economics thesis. Built 19 October 2026. Works offline: get the whole dashboard from Downloads.